Lectures on Nonlinear Analysis

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1. Function Spaces and Differential Equations

1.1. Function Spaces. Let Ω be a bounded region in $\mathbb{R}^n$ with a smooth boundary. For $u : \Omega \to \mathbb{R}$, we denote

$$\partial^\alpha u = \frac{\partial^{\alpha_1+\cdots+\alpha_n} u}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}},$$

where $\alpha = (\alpha_1, \cdots, \alpha_n)$. We also denote $|\alpha| = \alpha_1 + \cdots + \alpha_n$.

Let $u \in L_{\text{loc}}(\Omega)$, i.e., u is integrable on every compact subset of Ω. If there exists $v \in L_{\text{loc}}(\Omega)$ such that

$$\int_\Omega u \partial^\alpha \phi \, \mathrm{d}x = (-1)^{|\alpha|} \int_\Omega v\phi \, \mathrm{d}x \tag{1.1}$$

holds for all $\phi \in C_0^\infty(\Omega)$, then v is called the α-th order weak derivative of u. If $u \in C^{|\alpha|}(\Omega)$, taking $v = \partial^\alpha u$, then by the integration by parts formula, equation (1.1) holds. Therefore, the weak derivative is a generalization of the partial derivative in mathematical analysis. Thus, we still denote the α-th order weak derivative of u as $v = \partial^\alpha u$. We also denote

$$W^k(\Omega) = \{u \in L_{\text{loc}}(\Omega) : \text{ for} |\alpha| \le k, u \text{has an } \alpha\text{-th order weak derivative}\}$$

Definition 1. *Let $k \in \mathbb{N}$ and $p \ge 1$. The linear space*

$$W^{k,p}(\Omega) = \left\{u \in W^k(\Omega) : \textit{for } |\alpha| \le k, \partial^\alpha u \in L^p(\Omega)\right\}$$

is endowed with the norm

$$\|u\|_{k,p} = \left(\sum_{|\alpha| \le k} \int_\Omega |\partial^\alpha u|^p \, \mathrm{d}x \right)^{1/p}, \tag{1.2}$$

which makes it a Banach space. $W^{k,p}(\Omega)$ is called a Sobolev space.

Clearly, $C_0^\infty(\Omega) \subset W^{k,p}(\Omega)$. The closure of $C_0^\infty(\Omega)$ in $W^{k,p}(\Omega)$ is denoted by $W_0^{k,p}(\Omega)$. This is a closed subspace of $W^{k,p}(\Omega)$, and it is also called a Sobolev space.

Remark 1. A

(1) The following norm is an equivalent norm on $W^{k,p}(\Omega)$:

$$\|u\| = \sum_{|\alpha| \le k} |\partial^\alpha u|_p,$$

where $|\cdot|_p$ is the L^p-norm, i.e., $|u|_p = \left(\int_\Omega |u|^p \, \mathrm{d}x\right)^{1/p}$. Alternatively, on $W_0^{1,p}(\Omega)$, the following equivalent norm can be used:

$$\|u\| = |\nabla u|_p = \left(\int_\Omega |\nabla u|^p \right)^{1/p}.$$

(2) When $p > 1$, $W^{k,p}(\Omega)$ is reflexive. Therefore, every bounded sequence in it has a weakly convergent subsequence.

(3) When $p = 2$, $W^{k,2}(\Omega)$ is a Hilbert space with respect to the norm (1.2), and the corresponding inner product is

$$\langle u, v \rangle = \sum_{|\alpha| \le k} \int_\Omega \partial^\alpha u \cdot \partial^\alpha v \, \mathrm{d}x.$$

This Hilbert space is also denoted by $H^k(\Omega)$, i.e., $H^k(\Omega) = W^{k,2}(\Omega)$. Also, $H_0^k(\Omega) = W_0^{k,2}(\Omega)$.

To introduce the most important property of Sobolev spaces—the embedding theorem—we introduce Hölder spaces. Let

$$C^k(\overline{\Omega}) = \left\{u \in C^k(\Omega) : \text{ for} |\alpha| \le k, \partial^\alpha u \text{is uniformly continuous on } \Omega\right\}$$

and endow it with the norm

$$\|u\|_{C^k(\overline{\Omega})} = \sum_{|\alpha|\le k} \sup_{\overline{\Omega}} |\partial^\alpha u|,$$

Then $C^k(\overline{\Omega})$ becomes a Banach space, called the space of k-times continuously differentiable functions. Due to uniform continuity, a function u in $C^k(\overline{\Omega})$ and its partial derivatives of all orders can be uniquely continuously extended to the boundary of Ω. Therefore, for a given u, the above norm is finite. This is the difference between $C^k(\overline{\Omega})$ and $C^k(\Omega)$.

If $0 < \alpha \le 1$, a function $u : \Omega \to \mathbb{R}$ is said to satisfy the Hölder condition with exponent α if there exists $K > 0$ such that for any $x, y \in \Omega$,

$$|u(x) - u(y)| \le K|x - y|^\alpha.$$

Let $k \ge 0$, $0 < \alpha \le 1$. Let

$$C^{k,\alpha}(\overline{\Omega}) = \left\{u \in C^k(\overline{\Omega}) \mid \partial^\alpha u \text{ satisfies the Hölder condition with exponent } \alpha\right\}$$

Equipped with the norm

$$\|u\|_{C^{k,\alpha}(\overline{\Omega})} = \|u\|_{C^k(\overline{\Omega})} + \max_{0\le|\alpha|\le k} \sup_{x,y\in\Omega,\, x\ne y} \frac{|\partial^\alpha u(x) - \partial^\alpha u(y)|}{|x - y|^\alpha}$$

it becomes a Banach space, called the space of $k + \alpha$ order Hölder continuous functions. When $k = 0$, we denote $C^\alpha(\overline{\Omega}) = C^{0,\alpha}(\overline{\Omega})$.

Theorem 1 (Sobolev Embedding Theorem)**.** *Let Ω be a bounded smooth domain in $\mathbb{R}^n$. Then the following continuous embeddings hold:*

$$W^{k,p}(\Omega) \hookrightarrow \begin{cases} L^q(\Omega), & 1 \le q \le \frac{np}{n-kp}, \quad kp < n, \\ L^q(\Omega), & 1 \le q < +\infty, \quad kp = n, \\ C^\alpha(\overline{\Omega}), & 0 < \alpha \le 1 - \frac{n}{kp}, \quad kp > n. \end{cases}$$

Theorem 2 (Rellich-Kondrachov Compact Embedding Theorem)**.** *Let Ω be a bounded smooth domain in $\mathbb{R}^n$, $1 \le p < n$. Let $p^* = \frac{np}{n-p}$. Then for any $q \in [1, p^*)$, the continuous embedding $W^{1,p}(\Omega) \hookrightarrow L^q(\Omega)$ is also compact. This p^* is called the critical exponent of the Sobolev embedding.*

In particular, when $p = 2$, for $1 \le q < 2^* = \frac{2n}{n-2}$, we have the compact embedding $H^1(\Omega) \hookrightarrow L^q(\Omega)$.

1.2. Estimates for Linear Elliptic Operators. A linear partial differential operator on a domain Ω

$$P(x, \partial) = \sum_{|\alpha|\le m} a_\alpha(x)\partial^\alpha, \qquad x \in \Omega$$

can be classified according to its principal part

$$P_m(x, \partial) = \sum_{|\alpha|=m} a_\alpha(x)\partial^\alpha$$

Let

$$p_m(x,\xi) = \sum_{|\alpha|=m} a_\alpha(x)\xi^\alpha, \qquad \xi = (\xi_1, \cdots, \xi_n) \in \mathbb{R}^n.$$

where $\xi^\alpha = \xi_1^{\alpha_1} \cdots \xi_n^{\alpha_n}$. If for all $x \in \Omega$, the polynomial $p_m(x,\xi)$ is definite with respect to ξ, then $P(x,\partial)$ is called elliptic in Ω, in which case m must be an even number. If there exists $c > 0$ such that

$$p_m(x,\xi) \geq c|\xi|^m,$$

then $P(x,\partial)$ is called uniformly elliptic in Ω.

Let $\mathcal{L}$ be a $2m$-th order uniformly elliptic operator with smooth coefficients on a bounded smooth domain Ω, and $f \in L^p(\Omega)$. We say that $u \in L^p(\Omega)$ is a weak solution to the boundary value problem

$$\begin{cases} \mathcal{L}u = f, \\ \partial^\alpha u|_{\partial\Omega} = 0, \quad |\alpha| \leq m-1. \end{cases} \tag{1.3}$$

if for all $\phi \in C_0^\infty(\Omega)$,

$$\int_\Omega u\mathcal{L}^*\phi \, \mathrm{d}x = \int_\Omega f\phi \, \mathrm{d}x.$$

Here $\mathcal{L}^*$ is the adjoint operator of $\mathcal{L}$ determined by integration by parts.

Theorem 3 (L^p-estimate)**.** *Let $1 < p < \infty$. If $u \in L^p(\Omega)$ is a solution to (1.3) in the above sense, then $u \in W^{2m,p}(\Omega)$ and there exist constants $c_1, c_2 > 0$ depending only on Ω and $\mathcal{L}$ such that*

$$\|u\|_{2m,p} \leq c_1|f|_p + c_2|u|_p.$$

Roughly speaking, the solution of $\mathcal{L}u = f$ has $2m$ orders higher weak differentiability than f: when $f \in L^p$, then $u \in W^{2m,p}$. $W^{2m,p}$.

Theorem 4 (Schauder Estimate)**.** *Let $f \in C^\alpha(\overline{\Omega})$. If $u \in C^\alpha(\overline{\Omega})$ is a solution of (1.3) in the above sense, then $u \in C^{2m,\alpha}(\overline{\Omega})$ and there exist constants $c_1, c_2 > 0$ depending only on Ω and $\mathcal{L}$ such that*

$$\|u\|_{C^{2m,\alpha}(\overline{\Omega})} \leq c_1\|f\|_{C^\alpha(\overline{\Omega})} + c_2\|u\|_{C^\alpha(\overline{\Omega})}.$$

Roughly speaking, the solution of $\mathcal{L}u = f$ has $2m$ higher order differentiability than f: when $f \in C^\alpha$, $u \in C^{2,\alpha}$.

2. Differential Calculus of Nonlinear Mappings

2.1. Continuity and Boundedness of Nonlinear Mappings. Let X and Y be Banach spaces, and U be a subset of X.

Definition 2. *A mapping $f : X \to Y$ is called*

(i) continuous at $x_0 \in X$, if $x_n \to x_0 \Rightarrow f(x_n) \to f(x_0)$,
(ii) continuous on $U \subset X$, if f is continuous at $\forall x_0 \in U$,
(iii) uniformly continuous on $U \subset X$, if $\forall$ January 24, 2026 > 0, $\exists\delta = \delta($ January 24, 2026$) > 0$ such that

$$\|x - x'\| < \delta \Rightarrow \|f(x) - f(x')\| < \text{January 24, 2026},$$

(iv) bounded on $U \subset X$, if f maps bounded sets in U to bounded sets in Y.
(iv) $f : U \to Y$ satisfies the local Lipschitz condition, if for any $x_0 \in U$, there exist $\rho > 0$, $K > 0$, such that for all $x, y \in B_\rho(x_0)$

$$\|f(x) - f(y)\| \leq K\|x - y\|.$$

For linear mappings, boundedness is equivalent to continuity. However, this equivalence does not hold for nonlinear mappings.

Example 1. The functional $f : \ell^2 \to \mathbb{R}$,

$$f(x) = \sum_{|x_i|>1} (|x_i| - 1)i$$

is continuous but unbounded. To see its unboundedness, Let $x_n = (0, \cdots, 0, 2, 0, \cdots)$, then $\|x_n\| = 2$ and $f(x_n) = n \to \infty$.

Exercise 1. Let X, Y be Banach spaces, U be a bounded convex subset of X, and $f : U \to Y$ be uniformly continuous. Then $f(U)$ is a bounded set in Y.

It is well known that a continuous function on a bounded closed set in $\mathbb{R}^n$ must attain its maximum and minimum values. Recall that the "compactness" of the bounded closed set plays a crucial role in the proof. Since bounded closed sets in infinite-dimensional spaces are not necessarily compact, this conclusion no longer holds. In this case, to attain extreme values, additional conditions need to be imposed on the functional.

Definition 3. *Let $M \subset X$. A function $f : M \to \mathbb{R}$ is said to be*

(i) weakly lower semicontinuous at $x_0 \in M$ if for any sequence $x_n \in M$ with $x_n \rightharpoonup x_0$, we have $\underline{\lim}_{n\to\infty} f(x_n) \geq f(x_0)$,

(ii) weakly upper semicontinuous at $x_0 \in M$ if for any sequence $x_n \in M$ with $x_n \rightharpoonup x_0$, we have $\overline{\lim}_{n\to\infty} f(x_n) \leq f(x_0)$,

(iii) weakly continuous at $x_0 \in M$ if for any sequence $x_n \in M$ with $x_n \rightharpoonup x_0$, we have $\lim_{n\to\infty} f(x_n) = f(x_0)$.

Remark 2. Weak continuity is actually a kind of compactness imposed on the functional. We will see that it plays a crucial role in solving extremum problems of functionals. Furthermore, a weakly continuous functional is obviously continuous.

Definition 4. *A functional $f : M \to \mathbb{R}$ is said to be coercive if $\lim_{x\in M, \|x\|\to\infty} f(x) = +\infty$.*

Theorem 5. *Let M be a weakly closed set in a reflexive Banach space X, and $f : M \to \mathbb{R}$ be coercive and weakly lower semicontinuous. Then there exists $x_0 \in M$ such that $f(x_0) = \inf_{x\in M} f(x)$.*

Proof. Let $a \in M$. Since f is coercive, there exists $R > 0$ such that for all $x \in M$ with $\|x\| > R$, Then $f(x) > f(a)$. Thus

$$-\infty \leq c := \inf_M f = \inf_{x\in M, \|x\|\leq R} f(x).$$

Let $\{x_n\} \subset M$, $\|x_n\| \leq R$ such that $f(x_n) \to c$. Since X is reflexive and M is weakly closed, $\{x_n\}$ has a subsequence $x_{n_k} \rightharpoonup x_0 \in M$. By weak lower semicontinuity,

$$f(x_0) \leq \underline{\lim}_{k\to\infty} f(x_{n_k}) = c.$$

Therefore, it must be $f(x_0) = c$. □

In the following, we will see the application of Theorem 5 in the study of the existence of solutions to the following nonlinear elliptic boundary value problem:

$$\begin{cases} -\mathrm{div}\,(|\nabla u|^{p-2}\nabla u) = g(x, u) \\ u|_{\partial\Omega} = 0 \end{cases} \tag{2.1}$$

Naturally, the question arises: what kind of functionals are weakly lower semicontinuous?

Theorem 6. *Let $f : X \to \mathbb{R}$ be continuous and convex, i.e., for $t \in [0,1]$ and $x, y \in X$, we have*

$$f(tx + (1-t)y) \le tf(x) + (1-t)f(y).$$

Then f is weakly lower semicontinuous.

Proof. Let $x_n \rightharpoonup x_0$. For any $c > \underline{\lim}_{n\to\infty} f(x_n)$, we can assume $f(x_n) \le c$, otherwise we can choose a subsequence to replace $\{x_n\}$.

By Mazur's theorem, $\exists \lambda_n^k \ge 0$, $\sum_{n=1}^{m_k} \lambda_n^k = 1$, such that

$$y_k = \sum_{n=1}^{m_k} \lambda_n^k x_n \to x_0.$$

Since f is convex,

$$f(y_k) = f\left(\sum_{n=1}^{m_k} \lambda_n^k x_n\right) \le \sum_{n=1}^{m_k} \lambda_n^k f(x_n) \le c.$$

Letting $k \to \infty$, we get $f(x_0) \le c$. Then letting $c \to \underline{\lim}_{n\to\infty} f(x_n)$ completes the proof. □

As an important example, we consider the following composite function operator. Let $\Omega \subset \mathbb{R}^n$ be measurable, $f : \Omega \times \mathbb{R} \to \mathbb{R}$. Define

$$(Fu)(x) = f(x, u(x)),$$

that is, F is a mapping between appropriate function spaces, which transforms the function u into a new function Fu as defined above. F is called... This is a composite function operator, or Nemitskii operator. It frequently appears in nonlinear analysis, for example, in the study of problem (2.1).

Now let's examine the boundedness and continuity of F, which, of course, depends on the choice of function space.

Proposition 1. *Let Ω be compact, and $f : \Omega \times \mathbb{R} \to \mathbb{R}$ be continuous. Then $F : C(\Omega) \to C(\Omega)$ is bounded and continuous.*

The proof is straightforward and is left as an exercise. The following case is what we are interested in. Let $\Omega \subset \mathbb{R}^n$ be measurable, and f satisfy the Caratheodory conditions, i.e.,

(i) For a.e. $x \in \Omega$, $f(x, \cdot) : \mathbb{R} \to \mathbb{R}$ is continuous,
(ii) For all $t \in \mathbb{R}$, $f(\cdot, t) : \Omega \to \mathbb{R}$ is measurable.

Lemma 1. *Let $f : \Omega \times \mathbb{R} \to \mathbb{R}$ satisfy the Caratheodory conditions, and u be a Lebesgue measurable function on Ω. Then Fu is also Lebesgue measurable.*

Proof. Since u is measurable on Ω, there exists a sequence of measurable simple functions $\{u_n\}$ such that $u_n \to u$ a.e. in Ω. Let

$$u_n(x) = \sum_{i=1}^{m} \lambda_i \chi_{\Omega_i}(x),$$

where $\{\Omega_i\}$ is a disjoint measurable partition of Ω, and χ_{Ω_i} is the characteristic function on Ω_i. Then

$$(Fu_n)(x) = f\left(x, \sum_{i=1}^{m} \lambda_i \chi_{\Omega_i}(x)\right) = \sum_{i=1}^{m} f(x, \lambda_i)\chi_{\Omega_i}(x).$$

By Caratheodory condition (ii), each term in the sum on the right is measurable, so Fu_n is measurable. Furthermore, by Caratheodory condition (i), $Fu_n \to Fu$ a.e. in Ω. Thus Fu is measurable. $\square$

Theorem 7. *Let $f : \Omega \times \mathbb{R} \to \mathbb{R}$ satisfy the Caratheodory conditions, and*

$$|f(x,t)| \le b(x) + a|t|^{p_1/p_2}, \tag{2.2}$$

where $a > 0$, $b \in L^{p_2}(\Omega)$, $p_1, p_2 \ge 1$. Then $F : L^{p_1}(\Omega) \to L^{p_2}(\Omega)$ is bounded and continuous.

Proof. From (2.2), we know that if $u \in L^{p_1}(\Omega)$, then $Fu \in L^{p_2}(\Omega)$, i.e., $F : L^{p_1} \to L^{p_2}$. The boundedness of F follows from Minkowski's inequality:

$$|Fu|_{p_2} \le |b|_{p_2} + a|u|_{p_1}^{p_1/p_2}.$$

Next, we prove that F is continuous. Suppose F is not continuous. Then there exists *January 24, 2026*$_0 > 0$ and a sequence $\{u_n\}_0^\infty \subset L^{p_1}$ such that

$$|u_n - u_0|_{p_1} \to 0, \qquad |Fu_n - Fu_0|_{p_2} \ge \textit{January 24, 2026}_0.$$

Let $f_n(x) = f(x, u_n(x))$ and $g_n(x) = b(x) + a|u_n(x)|^{p_1/p_2}$. Then we can choose a subsequence, still denoted by u_n, f_n, g_n, such that almost everywhere on Ω

$$u_n \to u_0, \qquad f_n \to f_0, \qquad g_n \to g_0, \qquad |f_n| \le g_n. \tag{2.3}$$

Since $||u_n|^{p_1} - |u_0|^{p_1}| \le |u_n|^{p_1} + |u_0|^{p_1}$, by Fatou's lemma

$$\int \varliminf_{n\to\infty} (|u_n|^{p_1} + |u_0|^{p_1} - ||u_n|^{p_1} - |u_0|^{p_1}|)$$
$$\le \varliminf_{n\to\infty} \int (|u_n|^{p_1} + |u_0|^{p_1} - ||u_n|^{p_1} - |u_0|^{p_1}|).$$

Right now

$$2\int |u_0|^{p_1} \le 2\int |u_0|^{p_1} - \varlimsup_{n\to\infty} \int ||u_n|^{p_1} - |u_0|^{p_1}|,$$
$$\lim_{n\to\infty} \int ||u_n|^{p_1} - |u_0|^{p_1}| = 0.$$

thereby

$$\int |g_n - g_0|^{p_2} = a^{p_2} \int ||u_n|^{p_1/p_2} - |u_0|^{p_1/p_2}|^{p_2}$$
$$\le a^{p_2} \int ||u_n|^{p_1} - |u_0|^{p_1}| \to 0.$$

It follows from this

$$\int |g_n|^{p_2} \to \int |g_0|^{p_2}.$$

again

$$|f_n - f_0|^{p_2} \le 2^{p_2}(|f_n|^{p_2} + |f_0|^{p_2}) \le 2^{p_2}(|g_n|^{p_2} + |g_0|^{p_2}),$$

Using Fatou's lemma,

$$\int \varliminf_{n\to\infty} (2^{p_2}(|g_n|^{p_2} + |g_0|^{p_2}) - |f_n - f_0|^{p_2})$$
$$\le \varliminf_{n\to\infty} \int (2^{p_2}(|g_n|^{p_2} + |g_0|^{p_2}) - |f_n - f_0|^{p_2}).$$

From (2.3), we get

$$2^{p_2+1}\int |g_0|^{p_2} \le 2^{p_2+1}\int |g_0|^{p_2} - \overline{\lim_{n\to\infty}}\int |f_n - f_0|^{p_2},$$

$$\lim_{n\to\infty}\int |f_n - f_0|^{p_2} = 0.$$

This contradicts $|Fu_n - Fu_0|_{p_2} \ge \varepsilon$. □

Exercise 2. Let $\Omega \subset \mathbb{R}^N$ be a bounded open region, $g \in C(\Omega \times \mathbb{R}, \mathbb{R})$, for some $\theta \in [1, p^*)$

$$|g(x,t)| \le C(1 + |t|^\theta).$$

Then the functional $f : W_0^{1,p}(\Omega) \to \mathbb{R}$,

$$f(u) = \int g(x,u)$$

is weakly continuous.

Theorem 7 can be further generalized. Let $\Omega \subset \mathbb{R}^n$ be measurable, we say that $f : \Omega \times \mathbb{R}^n \to \mathbb{R}^m$ satisfies the Caratheodory condition if

a) For $x \in \Omega$, $f(x,\cdot) : \mathbb{R}^n \to \mathbb{R}^m$ is continuous,
b) For all $y \in \mathbb{R}^n$, $f(\cdot, y) : \Omega \to \mathbb{R}^m$ is measurable.

Then we have

Theorem 8. *Let $f : \Omega \times \mathbb{R}^n \to \mathbb{R}^m$ satisfy the Caratheodory condition,*

$$|f(x,y)| \le b(x) + a(|y_1|^{p_1/q} + \cdots + |y_n|^{p_n/q}).$$

Then $F : \prod_{j=1}^n L^{p_j}(\Omega) \to (L^q(\Omega))^m$ is bounded and continuous. Where $p_1, \cdots, p_n, q \ge 1$, $a > 0$, $b \in L^q(\Omega)$, $(Fu)(x) = f(x, u(x))$.

2.2. Derivatives of Nonlinear Mappings. Recall from mathematical analysis that the (total) derivative $f'(x_0)$ of a function $f = (f^1, \dots, f^m) : \mathbb{R}^n \to \mathbb{R}^m$ at the point

$$x_0 = (x_0^1, \dots, x_0^n)$$

is the Jacobi matrix $A = \left(\left.\frac{\partial f^i}{\partial x^j}\right|_{x_0}\right)$, which satisfies: as $|h| \to 0$,

$$f(x_0 + h) - f(x_0) - Ah = o(|h|).$$

Viewing the matrix A as a (bounded) linear operator, this definition can be generalized to Banach spaces.

Definition 5. *Let U be an open set in X. A function $f : U \to Y$ is said to be Fréchet differentiable (or simply differentiable) at a point $x_0 \in U$ if there exists $A \in \mathcal{L}(X,Y)$ such that*

$$f(x_0 + h) - f(x_0) - Ah = o(\|h\|), \qquad \text{as } \|h\| \to 0.$$

In this case, we denote $A = f'(x_0)$, and it is called the Fréchet derivative of f at the point x_0.

The Fréchet derivative is the most important concept in nonlinear analysis, and will be simply referred to as the derivative from now on. Clearly, if $A \in \mathcal{L}(X,Y)$, then A is differentiable at every $x \in X$ and $A'(x) = A$. If f is Fréchet differentiable at x_0, then it is also continuous at that point. Furthermore, if f is differentiable at every point in the open set U, then a mapping $f' : U \to \mathcal{L}(X,Y)$ can be defined, called the derivative mapping of f.

When f'... When continuous, f is said to be continuously differentiable on U, denoted by $f \in C^1(U, Y)$.

Exercise 3. Let X be a Hilbert space, $f : X \to \mathbb{R}$, $f(u) = \|u\|^2$. Then $f \in C^1(X, \mathbb{R})$,

$$\langle f'(u), \phi \rangle = 2\langle u, \phi \rangle.$$

In particular, let $f : H_0^1(\Omega) \to \mathbb{R}$,

$$f(u) = \frac{1}{2} \int |\nabla u|^2,$$

then $f \in C^1(H_0^1(\Omega), \mathbb{R})$, and

$$\langle f'(u), \phi \rangle = \int \nabla u \nabla \phi.$$

Theorem 9 (Chain Rule)**.** *Let X, Y, Z be Banach spaces, $U \subset X$, $V \subset Y$ be open subsets. If $f : U \to V$ is differentiable at point x, and $g : V \to Z$ is differentiable at point $f(x)$. Then $g \circ f : U \to Z$ is differentiable at point x and*

$$(g \circ f)'(x) = g'(f(x)) \circ f'(x). \tag{2.4}$$

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Proof. $f(x + h) = f(x) + f'(x)h + o(h)$. From g Differentiability of $f(x)$ when $h \to 0$

$$\begin{aligned}
&(g \circ f)(x + h) = g(f(x + h)) \\
&= g(f(x)) + g'(f(x))(f'(x)h + o(h)) + o(f'(x)h + o(h)) \\
&= (g \circ f)(x) + (g'(f(x)) \circ f'(x))h + o(h).
\end{aligned}$$

Therefore $g \circ f$ is differentiable in x and (2.4) holds. □

Corollary 1. *Suppose $f : U \to Y$ is differentiable in x, then for $\forall h \in X$, $y^* \in Y^*$, function*

$$\varphi(t) = \langle y^*, f(x + th) \rangle$$

Differentiable at $t = 0$ and $\varphi'(0) = \langle y^*, f'(x)h \rangle$.

Theorem 10 (differential mean inequality)**.** *Let $f : U \to Y$ be on the line segment $[a, b] \subset U$ It's possible to go up, Then $\exists \xi \in (a, b)$ makes*

$$\|f(b) - f(a)\| \leq \|f'(\xi)\| \|b - a\|.$$

Proof. The method is to convert it into a numerical function through linear functional action. According to the Hahn-Banach theorem, $\exists y^* \in Y^*$ makes $\|y^*\| = 1$ and

$$\langle y^*, f(b) - f(a) \rangle = \|f(b) - f(a)\|.$$

Let $\varphi(t) = \langle y^*, f(a + t(b - a)) \rangle$. From the inference1, we know that φ is differentiable in $[0, 1]$. Then apply Lagrange's mean value theorem, $\exists \tau \in (0, 1)$ so that

$$\begin{aligned}
\|f(b) - f(a)\| &= \varphi(1) - \varphi(0) = \varphi'(\tau) \\
&= \langle y^*, f'(a + \tau(b - a))(b - a) \rangle \\
&\leq \|f'(\xi)\| \|b - a\|.
\end{aligned}$$

where $\xi = a + \tau(b - a) \in (a, b)$. □

Theorem 11 (Fundamental Theorem of Calculus)**.** *Let $f \in C^1(U, Y)$, $[x, x+h] \subset U$, then*

$$f(x+h) - f(x) = \int_0^1 f'(x+th)h \, \mathrm{d}t. \tag{2.5}$$

The right-hand side of the equation is the limit of Riemann sums, i.e., the so-called Riemann integral of an abstract function. Using linear functionals, we can also define the integral $\int_A f \, \mathrm{d}\mu$ of an abstract function $f : A \to Y$ on a measurable set $A \in \mathcal{A}$ of a measure space $(S, \mathcal{A}, \mu)$, such that for all $y^* \in Y^*$

$$\left\langle y^*, \int_A f \, \mathrm{d}\mu \right\rangle = \int_A \langle y^*, f \rangle \, \mathrm{d}\mu.$$

Proof. Let $y^* \in Y^*$ be arbitrary, then $\varphi(t) = \langle y^*, f(x+th) \rangle$ is continuously differentiable on $[0, 1]$. Thus

$$\begin{aligned}
\langle y^*, f(x+h) - f(x) \rangle = \varphi(1) - \varphi(0) &= \int_0^1 \varphi'(t) \, \mathrm{d}t \\
&= \int_0^1 \langle y^*, f'(x+th)h \rangle \mathrm{d}t \\
&= \left\langle y^*, \int_0^1 f'(x+th)h \, \mathrm{d}t \right\rangle.
\end{aligned}$$

Since y^* is arbitrary, (2.5) is proven. □

Remark 3. From the proof, it can be seen that (2.5) holds as long as $t \longmapsto f'(x+th)$ is continuous; it is not necessary to require $f \in C^1(U, Y)$.

Next, we consider the calculation of the Fréchet derivative. For this purpose, we introduce a weaker type of derivative—the Gateaux derivative, which is a generalization of the directional derivative in mathematical analysis.

Definition 6. *If the limit*

$$Df(x, h) = \lim_{t \to 0} \frac{f(x+th) - f(x)}{t}$$

exists, then f is said to be Gateaux differentiable at point x in the direction h, and $Df(x, h)$ is called the Gateaux differential of f at point x in the direction h.

Fixing x, $Df(x, h)$ is not necessarily linear with respect to h. If $Df(x, \cdot) \in \mathcal{L}(X, Y)$, then it is called the Gateaux derivative of f at point x, and is denoted by $Df(x)$, i.e., $Df(x)h = Df(x, h)$.

Clearly, if f is differentiable at point x_0, then f has a Gateaux derivative at x_0 and $Df(x_0) = f'(x_0)$.

Remark 4. By identifying $\mathcal{L}(\mathbb{R}, Y)$ with Y via $\phi \mapsto \phi(1)$, the Gateaux differential of f at x in the direction of h is actually the Fréchet derivative of the mapping $t \mapsto f(x+th)$ at $t = 0$, i.e.,

$$Df(x, h) = \left. \frac{\mathrm{d}}{\mathrm{d}t} \right|_{t=0} f(x+th). \tag{2.6}$$

For the Gateaux derivative, the chain rule no longer holds, but Corollary 1, the mean value inequality for derivatives, and the fundamental theorem of calculus are still valid.

Theorem 12. *Let $f : U \to Y$,*

a) If f is Gateaux differentiable at point $x \in U$ in the direction of h, and $y^* \in Y^*$. Then

$$\varphi : t \longmapsto \langle y^*, f(x+th)\rangle$$

is differentiable at $t = 0$ and $\varphi'(0) = \langle y^*, Df(x,h)\rangle$.

b) If f has a Gateaux derivative on the line segment $[a,b] \subset U$, then $\exists \xi \in (a,b)$ such that

$$\|f(b) - f(a)\| \le \|Df(\xi)\|\|b-a\|.$$

c) If f has a Gateaux derivative on $[x, x+h] \subset U$, and $t \longmapsto Df(x+th)$ is continuous, then

$$f(x+h) - f(x) = \int_0^1 Df(x+th)h \,\mathrm{d}t.$$

Part (i) can be proved by taking difference quotients and limits, while the proofs of (ii) and (iii) are exactly the same as in the case of the Fréchet derivative.

Theorem 13. *If $f : U \to Y$ has a Gateaux derivative everywhere in U, and*

$$Df : U \to \mathcal{L}(X, Y)$$

is continuous at $x_0 \in U$. Then f is Fréchet differentiable at x_0 and $f'(x_0) = Df(x_0)$. Therefore, if Df is continuous, then $f \in C^1(U, Y)$.

Let $\delta > 0$ Let $B_\delta(x_0) \subset U$. Let $F = f - Df(x_0)$, then $F : U \to Y$ has a Gateaux derivative everywhere on U and

$$DF(x) = Df(x) - Df(x_0).$$

Let $\|h\| < \delta$. Since Df is continuous at x_0, as $h \to 0$,

$$\begin{aligned}\|f(x_0+h) - f(x_0) - Df(x_0)h\| &= \|F(x_0+h) - F(x_0)\| \\ &\le \|DF(\xi)\|\|h\| \\ &= \|Df(\xi) - Df(x_0)\|\|h\| = o(h).\end{aligned}$$

Therefore, f is Fréchet differentiable at x_0 and $f'(x_0) = Df(x_0)$.

In practice, when calculating the Fréchet derivative, we often first use (2.6) to find the Gateaux derivative, and then use Theorem 13 to conclude that the Gateaux derivative is the Fréchet derivative. Below is an important example.

Example 2. Let $\Omega \subset \mathbb{R}^n$ be a bounded smooth open region, $1 < p < n$, $g \in C(\Omega \times \mathbb{R}, \mathbb{R})$,

$$|g(x,t)| \le C(1 + |t|^{p^*-1}).$$

Consider $f : W_0^{1,p}(\Omega) \to \mathbb{R}$,

$$f(u) = \int G(x,u), \qquad \text{where } G(x,t) = \int_0^t g(x,s)\,\mathrm{d}s.$$

Then $f \in C^1(W_0^{1,p}(\Omega), \mathbb{R})$ and for $\forall u, \phi \in W_0^{1,p}(\Omega)$,

$$\langle f'(u), \phi\rangle = \int g(x,u)\phi.$$

Proof. First find the Gateaux derivative. According to Lebesgue's controlled convergence theorem, we get

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(u+t\phi) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \int G(x, u+t\phi) = \int g(x,u)\phi,$$

Let $A : W_0^{1,p}(\Omega) \to \mathbb{R}$,

$$A\phi = \int g(x,u)\phi.$$

but

$$|A\phi| \le |g(\cdot,u)|_{p^*/(p^*-1)}|\phi|_{p^*} \le C\|\phi\|.$$

Therefore $A \in (W_0^{1,p}(\Omega))^*$, f has Gateaux derivative $Df(u) = A$ at point u.

Just prove $Df : W_0^{1,p}(\Omega) \to (W_0^{1,p}(\Omega))^*$ continuous. Because

$$\begin{aligned}\left|\int (g(x,u) - g(x,v))\phi\right| &\le |g(\cdot,u) - g(\cdot,v)|_{p^*/(p^*-1)}|\phi|_{p^*} \\ &\le C|g(\cdot,u) - g(\cdot,v)|_{p^*/(p^*-1)}\|\phi\|.\end{aligned}$$

Therefore

$$\begin{aligned}\|Df(u) - Df(v)\| &= \sup_{\|\phi\|=1}\left|\int (g(x,u) - g(x,v))\phi\right| \\ &\le C|g(\cdot,u) - g(\cdot,v)|_{p^*/(p^*-1)}.\end{aligned}$$

Therefore, if $\|u - v\| \to 0$, then $|u - v|_{p^*} \to 0$. Then from the continuity of Nemitskii operator, we get

$$|g(\cdot,u) - g(\cdot,v)|_{p^*/(p^*-1)} \to 0.$$

So $\|Df(u) - Df(v)\| \to 0$. That is, Df is in $\forall v \in W_0^{1,p}(\Omega)$ Continuous. □

Exercise 4. Suppose Ω is the same as above, $f : W_0^{1,p}(\Omega) \to \mathbb{R}$,

$$f(u) = \frac{1}{p}\int |\nabla u|^p.$$

Then $f \in C^1(W_0^{1,p}(\Omega), \mathbb{R})$ and

$$\langle f'(u), \phi\rangle = \int |\nabla u|^{p-2}\nabla u \cdot \nabla\phi.$$

When $p = 2$, this is exactly the result of Exercise 3.

From Example 2 and Exercise 4, we know that for Ω and g as described above, the functional $f : W_0^{1,p}(\Omega) \to \mathbb{R}$

$$f(u) = \frac{1}{p}\int |\nabla u|^p - \int G(x,u) \tag{2.7}$$

is C^1 and

$$\langle f'(u), \phi\rangle = \int |\nabla u|^{p-2}\nabla u \cdot \nabla\phi - \int g(x,u)\phi.$$

Recalling the definition of a weak solution to the boundary value problem (2.1), we know that u is a weak solution of (2.1) if and only if for all $\phi \in W_0^{1,p}(\Omega)$, $\langle f'(u), \phi\rangle = 0$. A u satisfying the latter is called a critical point of f. The critical point theory to be introduced later studies when a functional has critical points and estimates their number. This theory has been developed to a very perfect state, and its application to nonlinear differential equations... This is what is known as the variational method.

We now present a simple example to initially demonstrate the power of variational methods. We first study the relationship between extreme points and critical points.

Theorem 14. *Let $f : X \to \mathbb{R}$ be differentiable at a local extremum u_0. Then $f'(u_0) = 0$.*

Proof. For all $h \in X$, the function $\varphi : t \longmapsto f(u_0 + th)$ is differentiable at $t = 0$ and has a local extremum at $t = 0$. Therefore, by Fermat's theorem,

$$\langle f'(u_0), h\rangle = \varphi'(0) = 0,$$

i.e., $f'(u_0) = 0$. □

Example 3. Let $\Omega \subset \mathbb{R}^n$ be a bounded smooth open region, $1 \le \theta < p < n$. Let $g \in C(\Omega \times \mathbb{R}, \mathbb{R})$,

$$|g(x,t)| \le C(1 + |t|^{\theta-1}).$$

Then the boundary value problem (2.1) has a weak solution in $W_0^{1,p}(\Omega)$.

Proof. We only need to prove that the C^1-functional f given by (2.7) reaches its minimum value. We know from the conditions

$$|G(x,t)| \le C(1 + |t|^{\theta}).$$

Therefore

$$\begin{aligned} f(u) &= \frac{1}{p}\int |\nabla u|^p - \int G(x,u) \ge \frac{1}{p}\|u\|^p - C|u|_\theta^\theta - C|\Omega| \\ &\ge \frac{1}{p}\|u\|^p - C\|u\|^\theta - C|\Omega|. \end{aligned}$$

From $p > \theta$ we can see that f is forced.

Suppose $u_n \rightharpoonup u$, then obviously $\|u\| \le \varliminf_{n\to\infty} \|u_n\|$. We also know from the exercises2

$$\int G(x,u_n) \to \int G(x,u).$$

Therefore

$$\begin{aligned} \varliminf_{n\to\infty} f(u_n) &= \varliminf_{n\to\infty}\left(\frac{1}{p}\|u_n\|^p - \int G(x,u_n)\right) \\ &\ge \frac{1}{p}\|u\|^p - \int G(x,u) = f(u) \end{aligned}$$

Thus, f is weakly lower semicontinuous.

Therefore, by Theorem 5, f attains a minimum at some point $u_0 \in W_0^{1,p}(\Omega)$. □

Although the variational method is very powerful and effective, one should not be overly optimistic about it. It is not always possible to find a functional for every differential equation boundary value problem such that its critical points are solutions to the differential equation. If a differential equation possesses such a functional, it is said to have a variational structure, and it is called the Euler-Lagrange equation of this functional, while the functional is called its energy functional.

Exercise 5. Let $\Omega \subset \mathbb{R}^n$ be a bounded smooth open region, and $g \in C(\Omega \times \mathbb{R}, \mathbb{R})$. For $x \in \Omega$, uniformly, we have

$$\lambda = \lim_{|t|\to\infty} \frac{g(x,t)}{t} < \lambda_1,$$

where λ_1 is the first eigenvalue of $(-\Delta, H_0^1(\Omega))$. Then

$$\begin{cases} -\Delta u = g(x,u), \\ u|_{\partial\Omega} = 0. \end{cases} \tag{2.8}$$

has a solution in $H_0^1(\Omega)$.

We have seen how to find weak solutions of nonlinear differential equations. However, if we cannot show that the weak solution is sufficiently smooth to be a classical solution, this method is hardly satisfactory.

Below, we illustrate with example (2.8) that when the nonlinear term satisfies certain regularity conditions, the weak solution of the equation possesses corresponding smoothness. The tools used are the L^p-estimates and Schauder estimates for linear elliptic equations.

Theorem 15. *Let $\Omega \subset \mathbb{R}^n$ be a bounded open region with a smooth boundary, and $g : \Omega \times \mathbb{R} \to \mathbb{R}$ be locally Lipschitz continuous, and there exists $\theta \in (1, 2^* - 1)$ such that*

$$|g(x,t)| \le C(1 + |t|^\theta).$$

If $u \in H_0^1(\Omega)$ is a weak solution of (2.8), then $u \in C^{2,\alpha}(\Omega)$, and therefore it is a classical solution of (2.8).

Proof. We only consider the case $n > 2$. By the Sobolev embedding theorem, $u \in L^{2^*}$, and furthermore, by Theorem 7, $g(\cdot, u) \in L^{s_0}$, where $s_0 = 2^*/\theta$.

Using L^p-estimates, we have $u \in W^{2,s_0}$. Then, as above, applying the Sobolev embedding theorem and Theorem 7, we get $u \in L^{ns_0/(n-2s_0)}$, $g(\cdot, u) \in L^{s_1}$, where $s_1 = \frac{ns_0}{\theta(n-2s_0)}$. Therefore, by L^p-estimates again, we have $u \in W^{2,s_1}$.

Our goal is to prove that for some $\alpha \in (0, 1)$, there exists $u \in C^\alpha(\overline{\Omega})$. To this end, we inductively define

$$s_k = \frac{ns_{k-1}}{\theta(n - 2s_{k-1})}, \qquad k = 1, 2, \cdots.$$

As above, if $u \in W^{2,s_{k-1}}$ and $n > 2s_{k-1}$, then $u \in W^{2,s_k}$. By mathematical induction, we can prove that

$$\begin{aligned} \frac{s_k}{s_{k-1}} &= \frac{n}{\theta(n - 2s_{k-1})} > \frac{n}{\theta(n - 2s_0)} \\ &> \frac{n}{\theta\left(n - 2\frac{2^*}{2^*-1}\right)} = \frac{1}{\theta}(2^* - 1) > 1. \end{aligned}$$

Therefore, there exists k such that $n \le 2s_{k-1}$. If $n < 2s_{k-1}$, then from $u \in W^{2,s_{k-1}}(\Omega)$ and the Sobolev embedding theorem, we have $u \in C^\alpha(\overline{\Omega})$, where $0 < \alpha \le 1 - \frac{n}{2s_{k-1}}$; if $n = 2s_{k-1}$, then by the Sobolev embedding theorem, for all $q \in [1, \infty)$, we have $u \in L^q$. In particular, $u \in L^{(n+1)\theta/2}$, so $g(\cdot, u) \in L^{(n+1)/2}$, and $u \in W^{2,(n+1)/2}(\Omega)$. Using the Sobolev embedding theorem again, we obtain $u \in C^\alpha(\overline{\Omega})$, where $0 < \alpha \le 1 - \frac{n}{n+1}$.

This proves that for some $\alpha \in (0, 1)$, we have $u \in C^\alpha(\overline{\Omega})$. Since g is locally Lipschitz, we have $g(\cdot, u) \in C^\alpha(\overline{\Omega})$. Now, by Schauder estimates, we know that $u \in C^{2,\alpha}(\overline{\Omega})$. □

Remark 5.
(i) For the quasilinear equation (2.1), currently only a weak solution $u \in C^{1,\alpha}(\overline{\Omega})$ can be obtained.
(ii) If $g(x, t)$ is allowed to have critical growth, i.e., $\theta = 2^* - 1$, the above proof fails! Because in this case $\frac{s_k}{s_{k-1}} = 1$. However, the corresponding regularity result still holds.
(iii) The iterative method used to improve regularity in the proof of this theorem is called the bootstrap technique.

2.3. Higher-Order Derivatives and Taylor's Formula. If $f : U \to Y$ is differentiable on U, then we have the derivative map $f' : U \to \mathcal{L}(X, Y)$. If f' is differentiable at x_0, then the derivative of f' at x_0, $(f')'(x_0)$, is called the second-order derivative of f at x_0, denoted by $f''(x_0)$. Note that

$$f''(x_0) \in \mathcal{L}(X, \mathcal{L}(X, Y)).$$

Higher-order derivatives are defined inductively in this way: Suppose f has a $(k-1)$-th order derivative map

$$f^{(k-1)} : U \to \underbrace{\mathcal{L}(X, \mathcal{L}(X, \cdots, \mathcal{L}}_{k-1 \text{ times}}(X, Y) \cdots)).$$

If $f^{(k-1)}$ is differentiable at $x_0 \in U$, then $(f^{(k-1)})'(x_0)$ is called the k-th order derivative of f at x_0, denoted by $f^{(k)}(x_0)$.

We see that

$$f^{(k)}(x_0) \in \underbrace{\mathcal{L}(X, \mathcal{L}(X, \cdots, \mathcal{L}}_{k \text{ times}}(X, Y) \cdots)),$$

the latter being a space that is quite complicated to write down. To characterize this space, we need to introduce multilinear operators.

Let $(X_1, ||\cdot||_1), \ldots, (X_n, ||\cdot||_n)$ be Banach spaces. On the product space

$$X = \prod_{i=1}^{n} X_i = \{x = (x_1, \cdots, x_n) | \, x_i \in X_i, i = 1, \cdots n\}$$

equipped with the norm

$$\|x\| = \sum_{i=1}^{n} \|x_i\|_i,$$

then $(X, ||\cdot||)$ is a Banach space.

Definition 7. *An operator $A : X \to Y$ is called n-linear if, for any fixed $n-1$ variables, it is linear with respect to the remaining variable.*

An n-linear operator $A : X \to Y$ is called bounded if there exists $M > 0$ such that

$$\|A(x_1, \cdots, x_n)\| \leq M \prod_{i=1}^{n} \|x_i\|_i, \qquad \forall x = (x_1, \cdots x_n) \in X.$$

The set of all bounded n-linear operators from X to Y forms a linear space $\mathcal{L}(X_1, \ldots, X_n; Y)$. Equipped with the norm

$$\|A\| = \sup_{\|x_i\|_i = 1} \|A(x_1, \cdots, x_n)\|$$

$\mathcal{L}(X_1, \ldots, X_n; Y)$ becomes a Banach space.

Exercise 6. For n- linear operator $A : X \to Y$, bounded $\Longleftrightarrow$ is continuous.

Proposition 2. *Space $\mathcal{L}(X_1, \cdots, X_n; Y)$ and $\mathcal{L}(X_1, \mathcal{L}(X_2, \cdots, \mathcal{L}(X_n, Y) \cdots))$ Isometric isomorphism.*

Proof. Let $\Phi : \mathcal{L}(X_1, \cdots, X_n; Y) \to \mathcal{L}(X_1, \mathcal{L}(X_2, \cdots, \mathcal{L}(X_n, Y) \cdots))$,

$$(\cdots((\Phi(A)x_1)x_2) \cdots)x_n = A(x_1, \cdots, x_n).$$

This is obviously a linear bijection, and

$$\|\Phi(A)\| = \sup_{\|x_1\|_1 = 1} \|\Phi(A)x_1\| = \sup_{\|x_1\|_1 = 1} \sup_{\|x_2\|_2 = 1} \|(\Phi(A)x_1)x_2\| = \cdots$$

$$= \sup_{\|x_i\|_i=1} \|(\cdots((\Phi(A)x_1)x_2)\cdots)x_n\|$$
$$= \sup_{\|x_i\|=1} \|A(x_1,\cdots,x_n)\| = \|A\|.$$

□

In particular, $\mathcal{L}(\underbrace{X,\cdots,X}_{n \text{ times}};Y)$ is isometrically isomorphic to $\underbrace{\mathcal{L}(X,\mathcal{L}(X,\cdots,\mathcal{L}}_{n \text{ times}}(X,Y)\cdots))$ Thus, the k-th order derivative of $f:U\to Y$ at x_0, denoted by $f^{(k)}(x_0)$, can be considered as an element of

$$\mathcal{L}^k(X,Y) = \mathcal{L}(\underbrace{X,\cdots,X}_{k \text{ times}};Y)$$

.

If $f:U\to Y$ has a k-th order derivative everywhere on U, then there is a k-th order derivative map

$$f^{(k)}:U\to\mathcal{L}^k(X,Y).$$

If $f^{(k)}$ is continuous, then f is said to be k-times continuously differentiable on U, denoted by $f\in C^k(U,Y)$.

And if $f\in C^{k-1}(U,Y)$ and $f^{(k-1)}$ satisfies the local Lipschitz condition on U, then it is recorded as $f\in C^{k-0}(U,Y)$.

Exercise 7. If $f\in C^k(U,Y)$ then $f\in C^{k-0}(U,Y)$.

Lemma 2. *Let $f\in C^k(U,Y)$, $x_0\in U$, $h_1,\cdots,h_k\in X$. Then*

$$\left.\frac{\partial}{\partial t_1\cdots\partial t_k}\right|_{t_1=\cdots=t_k=0} f\left(x_0+\sum_{j=1}^k t_jh_j\right) = f^{(k)}(x_0)(h_1,\cdots,h_k).$$

Proof. It is obviously true when $k=1$. If it is true when $k=n$, then it is also true when $k=n+1$:

$$\left.\frac{\partial}{\partial t_1\cdots\partial t_{n+1}}\right|_{t_1=\cdots=t_{n+1}=0} f\left(x_0+\sum_{j=1}^{n+1} t_jh_j\right)$$
$$= \left.\frac{\partial}{\partial t_1}\right|_{t_1=0}\left(\left.\frac{\partial}{\partial t_2\cdots\partial t_{n+1}}\right|_{t_2=\cdots=t_{n+1}=0} f\left(x_0+\sum_{j=2}^{n+1} t_jh_j\right)\right)$$
$$= \left.\frac{\partial}{\partial t_1}\right|_{t_1=0} f^{(n)}(x_0+t_1h_1)(h_2,\cdots,h_{n+1})$$
$$= \lim_{t_1\to 0}\left(\frac{f^{(n)}(x_0+t_1h_1)-f^{(n)}(x_0)}{t_1}(h_2,\cdots,h_{n+1})\right)$$
$$= \left(\lim_{t_1\to 0}\frac{f^{(n)}(x_0+t_1h_1)-f^{(n)}(x_0)}{t_1}\right)(h_2,\cdots,h_{n+1})$$
$$= (f^{(n+1)}(x_0)h_1)(h_2,\cdots,h_{n+1}) = f^{(n+1)}(x_0)(h_1,\cdots,h_{n+1}).$$

□

Let $A \in \mathcal{L}^k(X, Y)$. If $\forall \sigma \in \mathcal{S}_k$ Yes

$$A(x_1, \cdots, x_k) = A(x_{\sigma(1)}, \cdots, x_{\sigma(k)}),$$

Then A is said to be symmetric.

Theorem 16. *Suppose $f \in C^k(U, Y)$, $x_0 \in U$. Then $f^{(k)}(x_0)$ is symmetric.*

Proof. for $\forall \sigma \in \mathcal{S}_k$, Using the above lemma and the symmetry of mixed partial derivatives of multivariate functions, $\forall y^* \in Y^*$ has

$$\begin{aligned}
&\langle y^*, f^{(k)}(x_0)(h_1, \cdots, h_k)\rangle \\
&= \left\langle y^*, \frac{\partial}{\partial t_1 \cdots \partial t_k}\bigg|_{t_1=\cdots=t_k=0} f\left(x_0 + \sum_{j=1}^k t_j h_j\right)\right\rangle \\
&= \frac{\partial}{\partial t_1 \cdots \partial t_k}\bigg|_{t_1=\cdots=t_k=0} \left\langle y^*, f\left(x_0 + \sum_{j=1}^k t_j h_j\right)\right\rangle \\
&= \frac{\partial}{\partial t_{\sigma(1)} \cdots \partial t_{\sigma(k)}}\bigg|_{t_1=\cdots=t_k=0} \left\langle y^*, f\left(x_0 + \sum_{j=1}^k t_j h_j\right)\right\rangle \\
&= \left\langle y^*, \frac{\partial}{\partial t_{\sigma(1)} \cdots \partial t_{\sigma(k)}}\bigg|_{t_1=\cdots=t_k=0} f\left(x_0 + \sum_{j=1}^k t_j h_j\right)\right\rangle \\
&= \langle y^*, f^{(k)}(x_0)(h_{\sigma(1)}, \cdots, h_{\sigma(k)})\rangle.
\end{aligned}$$

So the conclusion is established. □

Theorem 17 (Taylor formula)**.** *Suppose $f \in C^n(U, Y)$, $[x_0, x_0 + h] \subset U$. Then*

$$f(x_0 + h) = \sum_{k=0}^{n-1} \frac{1}{k!} f^{(k)}(x_0) h^{(k)} + \frac{1}{n!} f^{(n)}(\xi) h^{(n)},$$

Among them $\xi \in [x_0, x_0 + h]$, $h^{(k)} = (h, \cdots, h)$.

Proof. arbitrarily choose $y^* \in Y^*$, let $\varphi(t) = \langle y^*, f(x_0 + th)\rangle$. Then $\varphi^{(n)}$ is continuous in $[0, 1]$ and

$$\varphi^{(k)}(0) = \langle y^*, f^{(k)}(x_0) h^{(k)}\rangle, \qquad k = 1, \cdots, n.$$

From Taylor's theorem in mathematical analysis, $\exists \tau \in (0, 1)$,

$$\begin{aligned}
\langle y^*, f(x_0 + th)\rangle = \varphi(1) &= \sum_{k=0}^{n-1} \frac{1}{k!} \varphi^{(k)}(0) + \frac{1}{n!} \varphi^{(n)}(\tau) \\
&= \left\langle y^*, \sum_{k=0}^{n-1} \frac{1}{k!} f^{(k)}(x_0) h^{(k)} + \frac{1}{n!} f^{(n)}(\xi) h^{(n)}\right\rangle,
\end{aligned}$$

where $\xi = x_0 + \tau h$. □

Corollary 2. *Under the conditions of Theorem 17,*

$$f(x_0 + h) = \sum_{k=0}^{n-1} \frac{1}{k!} f^{(k)}(x_0) h^{(k)} + o(\|h\|^n), \text{ as } \|h\| \to 0.$$

2.4. Inverse Function Theorem. The inverse function theorem and the implicit function theorem still hold in Banach spaces. The key to the proof is the contraction mapping principle. Therefore, the requirement of completeness of the space is essential.

Theorem 18 (Inverse Function Theorem)**.** *Let U be an open set in X, $f \in C^k(U, Y)$, $k \geq 1$, $x_0 \in U$. If $f'(x_0) : X \to Y$ is a topological linear isomorphism, then f is a C^k-diffeomorphism from an open neighborhood $O \subset U$ of x_0 to an open neighborhood V of $f(x_0)$.*

Proof. First, consider the case where $X = Y$ and $f'(x_0) = \mathbf{1}_X$. Without loss of generality, let $x_0 = 0$ and $f(x_0) = 0$.

Let $g(x) = x - f(x)$, then $g'(0) = 0$. Since f' is continuous, there exists $r > 0$ such that for $\|x\| < 2r$, $\|g'(x)\| < \frac{1}{2}$. Furthermore, by the mean value inequality, $g(\overline{B}_r(0)) \subset \overline{B}_{r/2}(0)$.

We assert that for any $y \in \overline{B}_{r/2}(0)$, there exists a unique $x \in \overline{B}_r(0)$ such that $f(x) = y$. To this end, consider

$$g_y(x) = y + x - f(x) = y + g(x).$$

It is easy to see that $g_y(\overline{B}_r(0)) \subset \overline{B}_r(0)$, and for $x_1, x_2 \in \overline{B}_r(0)$ we have

$$\|g_y(x_1) - g_y(x_2)\| = \|g(x_1) - g(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\|$$

By the contraction mapping principle, g_y has a unique fixed point in $\overline{B}_r(0)$. The assertion is proved.

Therefore, f has a local inverse $\varphi = f^{-1} : \overline{B}_{r/2}(0) \to \overline{B}_r(0)$. From

$$\begin{aligned}\|x_1 - x_2\| &\leq \|f(x_1) - f(x_2)\| + \|g(x_1) - g(x_2)\| \\ &\leq \|f(x_1) - f(x_2)\| + \frac{1}{2}\|x_1 - x_2\|\end{aligned}$$

we get

$$\|x_1 - x_2\| \leq 2\|f(x_1) - f(x_2)\| \tag{2.9}$$

Therefore, φ is continuous.

For $y_{1,2} \in B_{r/2}(0)$ and $x_{1,2} = \varphi(y_{1,2})$, i.e., $y_{1,2} = f(x_{1,2})$, from the differentiability of f at x_2 and (2.9), we have

$$\begin{aligned}&\|\varphi(y_1) - \varphi(y_2) - f'(x_2)^{-1}(y_1 - y_2)\| \\ &= |x_1 - x_2 - f'(x_2)^{-1}(f(x_1) - f(x_2))| \\ &\leq |f'(x_2)^{-1}||f'(x_2)(x_1 - x_2 - f'(x_2)^{-1}(f(x_1) - f(x_2)))| \\ &\leq |f'(x_2)^{-1}||f(x_1) - f(x_2) - f'(x_2)(x_1 - x_2)| \\ &= o(|x_1 - x_2|) = o(|y_1 - y_2|).\end{aligned}$$

Therefore, φ is differentiable and for $y \in B_{r/2}(0)$ we have

$$\varphi'(y) = f'(\varphi(y))^{-1}.$$

From this and the C^∞-smoothness of the inverse of bounded linear operators, for $0 \leq l < k$, if $\varphi \in C^l$, then $\varphi' \in C^l$, i.e., $\varphi \in C^{l+1}$. Finally, we obtain $\varphi \in C^k$.

For a general f, since $f'(x_0) : X \to Y$ is C^∞-diffeomorphism. Applying the above conclusion to $F = f'(x_0)^{-1} \circ f$, we know that the conclusion holds for $f = f'(x_0) \circ F$. $\square$

Theorem 19 (Implicit Function Theorem)**.** *Let U be an open set in $X \times Y$, $f \in C^k(U, Z)$, $(x_0, y_0) \in U$. If*

(i) $f(x_0, y_0) = 0$,
(ii) $f'_y(x_0, y_0) : Y \to Z$ *is a topological linear isomorphism.*

Then there exists an open neighborhood O *of* x_0 *and* $\varphi \in C^k(O, Y)$ *such that*

(i) $\varphi(x_0) = y_0$,
(ii) For all $x \in O$, $f(x, \varphi(x)) = 0$.

Remark 6. $f'_y(x_0, y_0) : Y \to Z$ is the derivative of $f(x_0, \cdot)$ at y_0, called the partial derivative of f with respect to y at the point (x_0, y_0). If $f : U \to Z$ is differentiable at (x_0, y_0), then we have partial derivatives $f'_x(x_0, y_0)$, $f'_y(x_0, y_0)$ and

$$f'(x_0, y_0)(h, k) = f'_x(x_0, y_0)h + f'_y(x_0, y_0)k, \qquad \forall (h, k) \in X \times Y. \tag{2.10}$$

Conversely, if $f : U \to Z$ has partial derivatives with respect to x and y in a neighborhood of (x_0, y_0), and f'_x and f'_y are continuous at (x_0, y_0), then f is differentiable at (x_0, y_0) and (2.10) holds.

Proof. Let $F : U \to X \times Z$ be defined by $F(x, y) = (x, f(x, y))$. Then

$$F'(x_0, y_0) = \begin{pmatrix} \mathbf{1}_X & * \\ 0 & f'_y(x_0, y_0) \end{pmatrix}$$

is a topological linear isomorphism from $X \times Y$ to $X \times Z$. Also, $F(x_0, y_0) = (x_0, 0)$. By the inverse function theorem, there exists a neighborhood V of $(x_0, 0)$ in $X \times Z$ and a local inverse $\psi \in C^k(V, X \times Y)$ of F. From the definition of F, there exists $\phi \in C^k(V, Y)$ such that

$$\psi(x, z) = (x, \phi(x, z)).$$

From

$$(x, z) = F(\psi(x, z)) = (x, f(x, \phi(x, z))),$$

letting $\varphi = \phi(\cdot, 0)$,

$$O = \{x \in X \,|\, (x, 0) \in V\},$$

the theorem is satisfied. □

If only the existence of a continuous implicit function is required, the conditions on f can be weakened.

Theorem 20 (Implicit Function Theorem)**.** *Let* U

Let U be an open set in $X \times Y$, and $f : U \to Z$, with $(x_0, y_0) \in U$. If $f'_y : U \to \mathcal{L}(Y, Z)$ is continuous at (x_0, y_0) and

(i) $f(x_0, y_0) = 0$,
(ii) $f'_y(x_0, y_0) : Y \to Z$ is a topological linear isomorphism.

Then there exists an open neighborhood O of x_0 and $\varphi \in C(O, Y)$ such that

(i) $\varphi(x_0) = y_0$,
(ii) For all $x \in O$, $f(x, \varphi(x)) = 0$.

Proof. Similar to the inverse function theorem. By continuity, there exist $\delta, r > 0$ such that when $\|x - x_0\| < \delta$ and $\|y - y_0\| < r$,

$$|f'_y(x, y) - f'_y(x_0, y_0)| < \frac{1}{2M}, \text{ and } |f(x, y_0)| < \frac{r}{2M},$$

where $M = \|f'_y(x_0, y_0)^{-1}\|$. From this, let

$$\Phi(x, y) = y - f'_y($$

Let

$$\Phi(x, y) = y - \left[\frac{\partial f}{\partial y}(x_0, y_0)\right]^{-1} f(x, y),$$

Then it can be proven that for all $x \in B_\delta(x_0)$, $\Phi(x, \cdot) : \overline{B}_r(y_0) \to \overline{B}_r(y_0)$ is a contraction mapping, and thus has a fixed point, denoted by $\varphi(x)$. It remains to prove that φ is continuous. □

2.5. Ordinary Differential Equations in Banach Spaces. Let U be an open set in X, and $I = [-\alpha, \alpha]$. Let $\phi : I \times U \to X$ be continuous, and for all $x_0 \in U$, there exist $R > 0$ and $K > 0$ such that

$$|\phi(t, x) - \phi(t, y)| \le K\|x - y\|, \text{ for } t \in I, x, y \in B_R(x_0).$$

We say that ϕ satisfies a local Lipschitz condition with respect to x (uniformly for $t \in I$). We consider the ordinary differential equation

$$\dot{x} = \phi(t, x), x(0) = a \in U. \tag{2.11}$$

Lemma 3 (Gronwall's Inequality)**.** *Let $w : [0, T] \to [0, \infty)$ be continuous, and let $C, K \ge 0$ such that*

$$w(t) \le C + K \int_0^t w(s)\, ds.$$

Then for all $t \in [0, T]$, $w(t) \le Ce^{Kt}$.

Proof. Let

$$\phi(t) = C + K \int_0^t w(s)\, ds,$$

then

$$\frac{d}{dt}(\mathrm{e}^{-Kt}\phi(t)) = \mathrm{e}^{-Kt}(Kw(t) - K\phi(t)) \le 0.$$

Therefore, $\mathrm{e}^{-Kt}\phi(t) \le \phi(0) = C$, i.e., $w(t) \le \phi(t) \le C\mathrm{e}^{Kt}$. □

Theorem 21. *If ϕ satisfies the local Lipschitz condition with respect to x, then there exists $\varepsilon > 0$ such that (2.11) has a unique solution on $[-\varepsilon, \varepsilon]$.*

Proof. By assumption,

$$M = \max_{|t| \le \alpha, \|x - x_0\| \le R} \|\phi(t, x)\| < +\infty.$$

Let

$$h = \min\left\{\alpha, \frac{R}{1 + M}, \frac{1}{K}\right\},$$

Choose $\varepsilon \in (0, h)$. Let $T : C([-\varepsilon, \varepsilon], X) \to C([-\varepsilon, \varepsilon], X)$,

$$(Tx)(t) = a + \int_0^t \phi(s, x(s))\, \mathrm{d}s.$$

Note that the norm in $\mathcal{C} = C([-\varepsilon, \varepsilon], X)$ is defined as

$$\|x\| = \max_{|t| \le \varepsilon} \|x(t)\|.$$

According to this norm $\mathcal{C}$ becomes a Banach space.

Let D be a closed sphere in $\mathcal{C}$ with the constant map a as the center and R as the radius. Then for $x \in D$ we have

$$\max_{|t| \le \varepsilon} \|x(t) - a\| \le R.$$

thereby

$$\begin{aligned}\|Tx - a\| &= \max_{|t|\le\varepsilon} \|(Tx)(t) - a\| \\ &\le \max_{|t|\le\varepsilon} \left| \int_0^t \|\phi(s, x(s))\| \, \mathrm{d}s \right| \le M\varepsilon \le R.\end{aligned}$$

Therefore $T : D \to D$. And for $x, y \in D$,

$$\begin{aligned}\|Tx - Ty\| &= \max_{|t|\le\varepsilon} \left\| \int_0^t (\phi(s, x(s)) - \phi(s, y(s))) \, \mathrm{d}s \right\| \\ &\le \max_{|t|\le\varepsilon} \left| \int_0^t \|\phi(s, x(s)) - \phi(s, y(s))\| \, \mathrm{d}s \right| \\ &\le K\varepsilon\|x - y\|\end{aligned}$$

Therefore, $T : D \to D$ is a contraction mapping and has a unique fixed point. Thus, (2.11) has a unique solution on $[-\varepsilon, \varepsilon]$. □

From this, we know that on the closed interval $[-\varepsilon, \varepsilon]$, there exists a unique solution $x = x(t)$ to (2.11). Taking $a = x(\pm\varepsilon)$ as the initial value, solving (2.11) shows that this solution can be extended to a larger interval. Therefore, the maximal interval of existence for the solution of (2.11) must be an open interval (t_a^-, t_a^+). If $t_a^+ < +\infty$ and the limit

$$x^* = \lim_{t \to t_a^+ - 0} x(t)$$

exists, then $x^* \in \partial U$.

Let

$$\mathcal{O} = \left\{(t, a) \,|\, a \in U, t \in (t_a^-, t_a^+)\right\}.$$

Let $\eta(t, a)$ be the value of the solution to (2.11) with initial value a at time t. Then we obtain a mapping $\eta : \mathcal{O} \to X$. The following theorem shows that $\mathcal{O}$ is a neighborhood of $0 \times U$ in $\mathbb{R} \times U$ and η is continuous.

Theorem 22 (Continuity of the solution with respect to the initial value)**.** *Suppose ϕ satisfies the local Lipschitz condition with respect to x. If $a \in U, \beta \in (0, t_a^+)$. Then for all $\varepsilon > 0$, there exists $r > 0$ such that when $\|b - a\| < r$, we have $t_b^+ > \beta$ and*

$$\|\eta(t, b) - \eta(t, a)\| < \varepsilon, \qquad \forall t \in [0, \beta].$$

The proof is rather complex; see Guo Dajun [**?**, p. 450].

Remark 7. In the finite-dimensional space case, since bounded closed sets are compact, the local Lipschitz condition implies the global Lipschitz condition on the tubular neighborhood of the solution curve

$$\ell = \{\eta(t, a) \,|\, t \in [0, \beta]\}$$

$$A = \{x \in U \,|\, \mathrm{dist}\,(x, \ell) \le \varepsilon\}$$

Therefore, the proof is simpler; see Zhang Jinyan [**?**, Chapter 1, §3]. In the infinite-dimensional space case, the above tubular neighborhood A is necessarily not compact, and thus the proof is much more complex.

Now let $I = \mathbb{R}$. When can the solution of (2.11) be extended indefinitely to the right?

Theorem 23. *Suppose ϕ satisfies the local Lipschitz condition with respect to x. If there exists $c > 0$ such that*

$$\|\phi(t,x)\| \le c(1+\|x\|), \qquad (t,x) \in [0,+\infty) \times U. \tag{2.12}$$

Then the right maximal existence interval of the solution of (2.11) is $[0,+\infty)$.

Proof. Let the right maximal existence interval be $[0,T)$. If $T < +\infty$, from (2.12) and

$$x(t) = a + \int_0^a \phi(s,x(s))\,\mathrm{d}s$$

we get

$$\|x(t)\| \le \|a\| + cT + c\int_0^t \|x(s)\|\,\mathrm{d}s.$$

From lemma3 we know

$$\|x(t)\| \le (\|a\| + cT)\mathrm{e}^{cT}, \qquad \forall t \in [0,T).$$

So for $t, t' \in [0,T)$ we have

$$\begin{aligned}\|x(t) - x(t')\| &= \left\| \int_{t'}^t \phi(s,x(s))\,\mathrm{d}s \right\| \\ &\le c|t-t'| + c\left| \int_{t'}^t \|x(s)\|\,\mathrm{d}s \right| \\ &\le (c + c(\|a\| + cT)\mathrm{e}^{cT})|t-t'|.\end{aligned}$$

So the limit

$$x^* = \lim_{t \to T-0} x(t)$$

Exists. Taking $a = x^*$ as the initial value solution (2.11), we know that the solution $x = x(t)$ can be extended to the right, which is inconsistent with the maximal nature of T. □

3. Differentiable Manifolds

3.1. Differentiable Manifolds and Differentiable Maps. In topological spaces, we can discuss the continuity of mappings. Now we will add a new structure to topological spaces to discuss the differentiability of mappings. We call a homeomorphism from a topological space X to an open subset of a topological space Y a homeomorphism from X into Y.

Definition 8. *A Hausdorff space M is called an m-dimensional C^r-differentiable manifold, or simply an m-manifold, if M has an open cover $\mathcal{A}$, and for all $U \in \mathcal{A}$, there exists a homeomorphism φ_U from U to an open subset of $\mathbb{R}^m$ such that: when $U \cap V \neq \emptyset$,*

$$\varphi_V \circ \varphi_U^{-1} : \varphi_U(U \cap V) \to \varphi_V(U \cap V)$$

is a C^r-mapping. Such a pair (U, φ_U) is called a chart of M.

Let Ω be an open set of an m-manifold M, and ψ be a homeomorphism from Ω to an open subset of $\mathbb{R}^m$. If for any chart (U, φ_U) of M, when $U \cap \Omega \neq \emptyset$,

$$\psi \circ \varphi_U^{-1} : \varphi_U(U \cap \Omega) \to \psi(U \cap \Omega)$$

and its inverse are both C^r-mappings, then (Ω, ψ) is also called a chart of M. If $p \in \Omega$, then (Ω, ψ) is also called a chart of M at p. the chart at this point.

Remark 8. In the above definition, if we replace $\mathbb{R}^m$ with a Banach space X, we obtain a Banach manifold.

Example 4. $\mathbb{R}^m$ is clearly an m-manifold. Using stereographic projection, we can obtain a family of charts on S^{m-1}, making S^{m-1} an $(m-1)$-manifold. Furthermore, if S is an open subset of an m-manifold M, by restricting the charts of M to S in an obvious way, we obtain a family of charts on S, making S an m-manifold. S is called an open submanifold of M.

With a differential structure, we can define the differentiability of mappings between manifolds.

Definition 9. *Let M, N be C^r-manifolds, $k \leq r$. $f : M \to N$ is said to be k-times differentiable at $p \in M$ if there exists a chart (U, φ) containing p and a chart (V, ψ) containing $f(p)$, such that $f(U) \subset V$ and*

$$\psi \circ f \circ \varphi^{-1} : \varphi(U) \to \psi(V)$$

is k-times differentiable at the point $\varphi(p)$.

$f : M \to N$ is called a C^k-mapping, denoted by $f \in C^k(M, N)$, if for any charts (U, φ) of M and (V, ψ) of N, the mapping $\psi \circ f \circ \varphi^{-1} : \varphi(U) \to \psi(V)$ is C^k. $C^{k-0}(M, N)$ is defined similarly.

Remark 9. Due to the C^r-compatibility between charts, the k-th differentiability of f at point p is independent of the choice of charts.

3.2. Tangent Space and Tangent Map. Let M be a C^r-manifold, $p \in M$. A C^r-map γ from an interval containing 0 to M is called a C^r-curve through point p if $\gamma(0) = p$. The set of all C^r-curves through point p is denoted by Γ_p.

Let $\gamma_1, \gamma_2 \in \Gamma_p$. We say $\gamma_1 \sim \gamma_2$ if there exists a chart (U, φ) containing point p such that

$$(\varphi \circ \gamma_1)'(0) = (\varphi \circ \gamma_2)'(0)$$

This is independent of the choice of (U, φ). The relation "$\sim$" is an equivalence relation on Γ_p. The quotient set $\Gamma_p / \sim$ is called the tangent space of M at point p, denoted by T_pM. Its elements are called tangent vectors of M at point p.

Given a chart (U, φ) at p, we obtain a bijection

$$\varphi_* : T_p(M) \to \mathbb{R}^m, \qquad [\gamma] \mapsto (\varphi \circ \gamma)'(0).$$

Here $[\gamma]$ is the tangent vector represented by the curve γ passing through point p. We sometimes also denote $[\gamma]$ as $\dot{\gamma}(0)$ or $\gamma'(0)$. It is natural to use φ_*. We introduce the linear space structure of $\mathbb{R}^m$ into T_pM, that is, for $[\gamma], [\eta] \in T_pM$ and $a, b \in \mathbb{R}$, we define

$$\begin{aligned} a[\gamma] + b[\eta] &= \varphi_*^{-1}(a\varphi_*[\gamma] + b\varphi_*[\eta]) \\ &= \varphi_*^{-1}(a(\varphi \circ \gamma)'(0) + b(\varphi \circ \eta)'(0)). \end{aligned} \tag{3.1}$$

For this, we need to prove that the right-hand side is independent of the choice of (U, φ), i.e., for another chart (V, ψ), we have

$$\varphi_*^{-1}(a(\varphi \circ \gamma)'(0) + b(\varphi \circ \eta)'(0)) = \psi_*^{-1}(a(\psi \circ \gamma)'(0) + b(\psi \circ \eta)'(0)),$$

or equivalently,

$$(\psi_* \circ \varphi_*^{-1})(a(\varphi \circ \gamma)'(0) + b(\varphi \circ \eta)'(0)) = a(\psi \circ \gamma)'(0) + b(\psi \circ \eta)'(0). \tag{3.2}$$

Since $(\varphi \circ \gamma)'(0)$ can span $\mathbb{R}^m$, and

$$\begin{aligned} (\psi_* \circ \varphi_*^{-1})(\varphi \circ \gamma)'(0) &= \psi_*[\gamma] = (\psi \circ \gamma)'(0) \\ &= ((\psi \circ \varphi^{-1})'(\varphi(p)))(\varphi \circ \gamma)'(0), \end{aligned}$$

therefore $\psi_* \circ \varphi_*^{-1} : \mathbb{R}^m \to \mathbb{R}^m$ is precisely the linear map

$$(\psi \circ \varphi^{-1})'(\varphi(p)) : \mathbb{R}^m \to \mathbb{R}^m.$$

This immediately gives (3.2). Thus, the linear operation defined by (3.1) is well-defined. It makes T_pM a linear space, and $\varphi_* : T_pM \to \mathbb{R}^m$ is a linear isomorphism.

Let (U, φ) be a chart at p, and $\{x^i\}$ be the coordinate functions corresponding to (U, φ). Let e_i be the i-th standard basis vector of $\mathbb{R}^m$, Let

$$\left.\frac{\partial}{\partial x^i}\right|_p := \varphi_*^{-1}(e_i), \qquad i = 1, \cdots, m. \tag{3.3}$$

Then $\left\{ \left.\frac{\partial}{\partial x^i}\right|_p \right\}$ forms a basis for T_pM. Due to (3.3), the coordinates of the tangent vector $v \in T_pM$ with respect to the basis $\left\{ \left.\frac{\partial}{\partial x^i}\right|_p \right\}$ are exactly equal to the coordinates of $\varphi_*(v) \in \mathbb{R}^m$ with respect to the basis $\{e_i\}$.

Within the chart (U, φ), the curve $\gamma : (-$ *January 24, 2026,* *January 24, 2026*$) \to M$ passing through p has a local representation

$$(\varphi \circ \gamma)(t) = (x^1(t), \cdots, x^m(t)).$$

The tangent vector $\dot\gamma(0)$ represented by γ can be expressed as a linear combination of $\left.\frac{\partial}{\partial x^i}\right|_p$:

$$\begin{aligned}\dot\gamma(0) &= \varphi_*^{-1}((\varphi \circ \gamma)'(0)) \\ &= \varphi_*^{-1}(\dot x^1(0), \cdots, \dot x^m(0)) = \dot x^i(0) \left.\frac{\partial}{\partial x^i}\right|_p .\end{aligned}$$

The coefficients of the combination are precisely the derivatives of the components of the local representation of γ at $t = 0$.

Let $\mathcal{C}_p^r$ be the set of C^r-functions defined near p, $r \le k$. We can view $\dot\gamma(0) \in T_pM$ as a functional $X : \mathcal{C}_p^r \to \mathbb{R}$ on $\mathcal{C}_p^r$,

$$X(f) = \langle \dot\gamma(0), f \rangle = (f \circ \gamma)'(0), \qquad \forall f \in \mathcal{C}_p^r.$$

The definition of X is well-defined: it is independent of the choice of the curve γ representing the tangent vector $\dot\gamma(0)$, and it has the following properties: For $f, g \in \mathcal{C}_p^r$, $\lambda, \mu \in \mathbb{R}$,

(1) $X(\lambda f + \mu g) = \lambda X(f) + \mu X(g)$;
(2) $X(fg) = f(p)X(g) + g(p)X(f)$.

In the above equations, $\lambda f + \mu g$ and fg are considered as functions defined on the intersection of the domains of f and g. Many books define tangent vectors as mappings $X : \mathcal{C}_p^r \to \mathbb{R}$ that satisfy the above properties.

Let $\gamma_i(t) = \varphi^{-1}(\varphi(p) + te_i)$, then $\varphi_*(\dot\gamma_i(0)) = (\varphi \circ \gamma_i)'(0) = e_i$, hence

$$\dot\gamma_i(0) = \left.\frac{\partial}{\partial x^i}\right|_p .$$

Therefore

$$\left.\frac{\partial}{\partial x^i}\right|_p (f) = \langle \dot\gamma_i(0), f \rangle = (f \circ \gamma_i)'(0) = \left.\frac{\partial (f \circ \varphi^{-1})}{\partial x^i}\right|_{\varphi(p)} .$$

The rightmost expression is the partial derivative familiar from mathematical analysis. This is also the reason why we denote the tangent vector $\varphi_*^{-1}(e_i)$, represented by γ_i above, as $\left.\frac{\partial}{\partial x^i}\right|_p$.

In particular, for the coordinate functions $x^j = \pi^j \circ \varphi$ of the chart (U, φ), we have

$$\left.\frac{\partial}{\partial x^i}\right|_p (x^j) = \delta_i^j, \tag{3.4}$$

where $\pi^j : \mathbb{R}^m \to \mathbb{R}$ is the projection onto the j-th coordinate, $\pi^j(x^1, \cdots, x^m) = x^j$.

Now let (V, ψ) be another chart at point p, with corresponding coordinate functions $\{y^i\}$. The chart (V, ψ) gives another basis for T_pM, $\left\{\frac{\partial}{\partial y^i}|_p\right\}$. We are naturally interested in the transformation formula between the basis $\left\{\left.\frac{\partial}{\partial x^i}\right|_p\right\}$ and the basis $\left\{\frac{\partial}{\partial y^i}|_p\right\}$. Let

$$\left.\frac{\partial}{\partial y^j}\right|_p = a^i_j \left.\frac{\partial}{\partial x^i}\right|_p.$$

By (3.4), applying both sides of the above equation to the function x^k gives

$a_j^k = a_j^i$

$$\left.\frac{\partial}{\partial x^i}\right|_p (x^k) = \left.\frac{\partial}{\partial y^j}\right|_p (x^k) = \left.\frac{\partial x^k}{\partial y^j}\right|_{\psi(p)},$$

that is,

$$\left.\frac{\partial}{\partial y^j}\right|_p = \left.\frac{\partial x^k}{\partial y^j}\right|_{\psi(p)} \cdot \left.\frac{\partial}{\partial x^k}\right|_p.$$

Therefore, the basis transformation matrix of T_pM is precisely the Jacobian matrix of the coordinate transformation function.

Exercise 8. Find the coordinate transformation formula for a tangent vector $v \in T_pM$ under two sets of bases $\left\{\left.\frac{\partial}{\partial x^i}\right|_p\right\}$ and $\left\{\frac{\partial}{\partial y^i}|_p\right\}$.

Let $f : M \to N$ be a C^1-mapping, $p \in M$. Define

$$f_* = f_{*p} : T_pM \to T_{f(p)}N, \qquad f_*[\gamma] = [f \circ \gamma],$$

which is called the tangent map of f at p.

Exercise 9. $[f \circ \gamma]$ is independent of the choice of the curve γ representing $[\gamma]$, therefore the definition of f_* is well-defined.

Take a chart (U, φ) at p and a chart (V, ψ) at $f(p)$, such that $f(U) \subset V$. Then we have

$$\begin{aligned}\psi_*[f \circ \gamma] &= (\psi \circ f \circ \gamma)'(0) = ((\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ \gamma))'(0)\\ &= (\psi \circ f \circ \varphi^{-1})'(\varphi(p))(\varphi \circ \gamma)'(0)\\ &= (\psi \circ f \circ \varphi^{-1})'(\varphi(p))\varphi_*[\gamma].\end{aligned} \tag{3.5}$$

Therefore,

$$f_*[\gamma] = [f \circ \gamma] = (\psi_*^{-1} \circ (\psi \circ f \circ \varphi^{-1})'(\varphi(p)) \circ \varphi_*)[\gamma].$$

That is, $f_* : T_pM \to T_{f(p)}N$ is the composition of three linear mappings,

$$f_* = \psi_*^{-1} \circ (\psi \circ f \circ \varphi^{-1})'(\varphi(p)) \circ \varphi_*.$$

Therefore, f_* is linear. If we denote the coordinate functions of the charts (U, φ) and (V, ψ) as $\{x^i\}$ and $\{y^j\}$, respectively, then we have

$$f_*\left(\frac{\partial}{\partial x^i}\right) = \psi_*^{-1} \circ (\psi \circ f \circ \varphi^{-1})'(\varphi(p))(e_i) = \frac{\partial y^j}{\partial x^i}\frac{\partial}{\partial y^j}. \tag{3.6}$$

That is, the tangent map f_* in the bases $\left\{\frac{\partial}{\partial x^i}\right\}$ and $\left\{\frac{\partial}{\partial y^j}\right\}$ The matrix below is precisely the Jacobi matrix of its local representation $\psi \circ f \circ \varphi^{-1}$. From (3.6), for $v = v^i \frac{\partial}{\partial x^i} \in T_pM$, we have

$$f_*(v) = f_*\left(v^i \frac{\partial}{\partial x^i}\right) = v^i f_*\left(\frac{\partial}{\partial x^i}\right) = v^i \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j}.$$

3.3. Submanifolds, Regular Value Preimage Theorem.

Definition 10. *Let M be an m-dimensional C^r-manifold. $S \subset M$ is called an s-dimensional submanifold of M if for all $p \in S$, there exists a chart (U, φ) of M at p such that*

$$\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^s \times 0).$$

That is, every point of S has a chart (U, φ) that flattens $U \cap S$ onto the subspace $\mathbb{R}^s$ of $\mathbb{R}^m$, or in other words, there exists a local coordinate system $(x^1, \cdots, x^m)$, such that S is locally represented by the equations $x^{s+1} = \cdots = x^m = 0$.

Definition 11. *Let $f \in C^1(M, N)$, $p \in M$. If $f_{*p} : T_pM \to T_{f(p)}N$ is not surjective, then p is called a critical point of f. Let $f(p)$ be called the critical value of f. If $q \in N$ is not a critical value of f, then q is called a regular value of f.*

Example 5. Let $f \in C^1(M, \mathbb{R})$, and $p \in M$ be a local minimum of f. Then p is a critical point of f.

Theorem 24. *Let $f \in C^r(M, N)$, and $q \in N$ be a regular value of f such that $f^{-1}(q) \neq \emptyset$. Then $S = f^{-1}(q)$ is an $(m-n)$-dimensional C^r-submanifold of M. Here $m = \dim M$ and $n = \dim N$.*

Proof. For any $p \in f^{-1}(q)$, take charts (U, φ) and (V, ψ) at p and q respectively such that

$$f(U) \subset V, \quad \varphi(p) = 0, \quad \psi(q) = 0.$$

Let the local representation of f under these charts be

$$(y^1, \ldots, y^n) = (f^1(x^1, \ldots, x^m), \ldots, f^n(x^1, \ldots, x^m)).$$

Since $f_{*p} : T_pM \to T_{f(p)}N$ is surjective, the Jacobian matrix of the local representation is of full column rank. Without loss of generality, assume that

$$\left(\frac{\partial y^i}{\partial x^j}\right)_{i=1,\ldots,n;\, j=m-n+1,??\ldots,m}$$

is invertible. Applying the implicit function theorem to

$$F = \psi \circ f \circ \varphi^{-1} : \mathbb{R}^{m-n} \times \mathbb{R}^n \to \mathbb{R}^n$$

we obtain a C^r-map $g : \mathbb{R}^{m-n} \to \mathbb{R}^n$ such that $g(0) = 0$,

$$F(x, g(x)) = 0.$$

Define a differential local diffeomorphism near the origin in $\mathbb{R}^m = \mathbb{R}^{m-n} \times \mathbb{R}^n$ by

$$\beta(x, z) = (x, z - g(x)).$$

$$),$$

Let $\phi = \beta \circ \varphi$, then we get

$$\phi(f^{-1}(q) \cap \Omega) = \phi(\Omega) \cap (\mathbb{R}^{m-n} \times 0).$$

Here Ω is an open neighborhood of p. (Ω, ϕ) is a chart of M at p, making $f^{-1}(q)$ an $(m-n)$-dimensional C^r-submanifold of M. $\square$

Exercise 10. Under the conditions of Theorem 24, for $p \in f^{-1}(q)$, we have $T_p f^{-1}(q) = \ker(f_{*p})$.

3.4. Constrained Optimization. Let X be a Banach space, $f, g_1, \ldots, g_n \in C^1(X, \mathbb{R})$. Let

$$M = \{x \in X \mid g_i(x) = 0, i = 1, \ldots, n\}.$$

Also let $g : X \to \mathbb{R}^n$,

$$g(x) = (g_1(x), \ldots, g_n(x)).$$

If for all $x \in M$, $g_1'(x), \ldots, g_n'(x)$ are linearly independent, then 0 is a regular value of g (and vice versa). Therefore, $M = g^{-1}(0)$ is a submanifold of X. Let M be a submanifold of codimension n, and $f|_M : M \to \mathbb{R}$ be a C^1-mapping. Therefore, we can study its critical points.

Theorem 25. *Under the above conditions, $p \in M$ is a critical point of $f|_M$ if and only if there exist $\lambda_1, \cdots, \lambda_n \in \mathbb{R}$ such that*

$$f'(p) + \sum_{i=1}^{n} \lambda_i g_i'(p) = 0. \tag{3.7}$$

Proof. $\Longleftarrow$. For $v = \dot{\gamma}(0) \in T_p M$, where $\gamma : (-$ *January 24, 2026*, *January 24, 2026*$) \to M$, $\gamma(0) = p$. Since

$$\langle g_i'(p), v\rangle = (g_i \circ \gamma)'(0) = 0,$$

we have

$$(f|_M)_{*p} v = (f \circ \gamma)'(0) = \langle f'(p), v\rangle = -\sum_{i=1}^{n} \lambda_i \langle g_i'(p), v\rangle = 0.$$

That is, $(f|_M)_{*p} = 0$, so p is a critical point of $f|_M$.

$\Longrightarrow$. Let $X_1 = T_p M$, and X_2 be the complement of X_1 in X. Then

$$X_1 \subset \bigcap_{i=1}^{n} \ker g_i'(p). \tag{3.8}$$

Thus $g_1'(p), \cdots, g_n'(p)$ are linearly independent as linear functionals on X_2. Note that $\dim X_2 = n$, there exist $\lambda_1, \cdots, \lambda_n \in \mathbb{R}$ such that for all $u \in X_2$ we have

$$\left\langle f'(p) + \sum_{i=1}^{n} \lambda_i g_i'(p), u \right\rangle = 0. \tag{3.9}$$

Furthermore, since p is a critical point of $f|_M$ and by (3.8), for $v \in X_1$ we have

$$\left\langle f'(p) + \sum_{i=1}^{n} \lambda_i g_i'(p), v \right\rangle = \langle f'(p), v\rangle = (f|_M)_{*p} v = 0. \tag{3.10}$$

From (3.9) and (3.10), we know that (3.7) holds. □

Corollary 3. *If $p \in M$ is a local minimum point of $f|_M$, then there exist $\lambda_1, \cdots, \lambda_n \in \mathbb{R}$ such that (3.7) holds.*

As an application, we consider the boundary value problem

$$\begin{cases} -\Delta u + \lambda u = |u|^{p-2}u, \\ u|_{\partial\Omega} = 0. \end{cases}$$

where Ω is a bounded domain in $\mathbb{R}^n$ with a smooth boundary $\partial\Omega$, $p \in (2, 2^*)$. Let $(-\Delta, H_0^1(\Omega)$ The eigenvalues are $\lambda_1 < \lambda_2 \leq \cdots$.

Theorem 26. *If $\lambda > -\lambda_1$, then (3.4) has a non-zero solution.*

Proof. The solutions of (3.4) are precisely the critical points of the C^1-functional $F : H_0^1(\Omega) - \mathbb{R}$,

$$F(u) = \frac{1}{2}\int(|\nabla u|^2 + \lambda u^2) - \frac{1}{p}\int |u|^p$$

However, F is neither bounded above nor below, so the existence of critical points cannot be obtained by finding the extrema of F on $H_0^1(\Omega)$.

Let $f, g : H_0^1(\Omega) \to \mathbb{R}$,

$$f(u) = \frac{1}{2}\int(|\nabla u|^2 + \lambda u^2), \qquad g(u) = 1 - \frac{1}{p}\int |u|^p.$$

Then $f, g \in C^1(H_0^1(\Omega), \mathbb{R})$, and 0 is a regular value of g. Therefore, $M = g^{-1}(0)$ is a submanifold of $H_0^1(\Omega)$.

It is easy to prove that M is a weakly closed set in $H_0^1(\Omega)$, and when $\lambda > -\lambda_1$, f is coercive and weakly lower semi-continuous on M. Therefore, $f|_M$ attains a minimum at some point $u \in M$, and thus u is a critical point of $f|_M$. By Theorem 25, there exists $\mu \in \mathbb{R}$ such that for all $v \in H_0^1(\Omega)$

$$\int(\nabla u \nabla v + \lambda u v - \mu|u|^{p-2}uv) = \langle f'(u) + \mu g'(u), v\rangle = 0, \tag{3.11}$$

Taking $v = u$, we get

$$\int(|\nabla u|^2 + \lambda u^2) = \mu\int |u|^p.$$

Thus $\mu > 0$. From (3.11), we know that u is a solution to

$$\begin{cases} -\Delta u + \lambda u = \mu|u|^{p-2}u, \\ u|_{\partial\Omega} = 0 \end{cases} \tag{3.12}$$

It is easy to see that $u_* = \mu^{1/(p-2)}u$ is a solution to (3.4). □

Remark 10. (i) In the process of multiplying the solution of (3.12) by a constant to absorb the Lagrange multiplier μ, the homogeneity of the nonlinear term $|u|^{p-2}u$ plays a crucial role.

(ii) Since $|u| \in M$ when $u \in M$, and $f(u) = f(|u|)$. Therefore, we can require $u \geq 0$. By the maximum principle,

Then, after obtaining $u \in C^2(\Omega)$, we can further use the strong maximum principle to obtain $u > 0$. That is, when $\lambda > -\lambda_1$, equation (3.4) has a positive solution. Similarly, we can obtain the negative solution of (3.4).

Exercise 11. (1) Prove that the functional F has neither an upper bound nor a lower bound.

(2) If (3.4) has a positive or negative solution, then $\lambda > -\lambda_1$.

3.5. Manifolds with Boundary. We will study manifolds with boundary whose local model is

$$\mathbb{R}^m_+ = \{x = (x^1, \cdots, x^m) | \, x^i \in \mathbb{R}, x^1 \geq 0\}$$

For this purpose, we need to generalize the concept of C^r-mappings. Let $U \subset \mathbb{R}^m$, $f : U \to \mathbb{R}^n$. If there exists $F \in C^r(\mathbb{R}^m, \mathbb{R}^n)$ such that $F|_U = f$, then f is called a C^r-mapping.

Definition 12. *A Hausdorff space M is called an m-dimensional C^r-differentiable manifold with boundary, or simply an m-manifold with boundary, if M has an open cover $\mathcal{A}$, and for every $U \in \mathcal{A}$, there is a homeomorphism φ_U from U to $\mathbb{R}^m_+$ such that when $U \cap V \neq \emptyset$,*

$$\varphi_V \circ \varphi_U^{-1} : \varphi_U(U \cap V) \to \varphi_V(U \cap V)$$

is a C^r-mapping. Such a pair (U, φ_U) is called a chart of M.

Let Ω be an open set of an m-manifold with boundary M, and ψ be a homeomorphism from Ω to $\mathbb{R}^m_+$. If for any chart (U, φ_U) of M, when $U \cap \Omega \neq \emptyset$,

$$\psi \circ \varphi_U^{-1} : \varphi_U(U \cap \Omega) \to \psi(U \cap \Omega)$$

and its inverse are both C^r-mappings, then (Ω, ψ) is also called a chart of M. If $p \in \Omega$, then (Ω, ψ) is called a chart of M at p.

Let M be an m-dimensional manifold with boundary. If there exists a chart (U, φ) of M at p such that $\varphi(p) \in \partial\mathbb{R}^m_+$, then p is called a boundary point of M. The set of boundary points of M is denoted by ∂M, and is called the boundary of M.

Relative to manifolds with boundary, the manifold in Definition 8 is called a manifold without boundary. Let (U, φ) be a chart of a manifold M without boundary, then

$$\varphi = (\varphi^1, \cdots, \varphi^m)$$

is a homeomorphism from U to $\mathbb{R}^m$. Let

$$\tilde{\varphi} = (\mathrm{e}^{\varphi^1}, \varphi^2, \cdots, \varphi^m).$$

Then $(U, \tilde{\varphi})$ is still a chart of M. However, $\tilde{\varphi}$ is a homeomorphism from U to $\mathbb{R}^m_+$. Therefore, a manifold M without boundary can also be considered as a manifold with boundary $\partial M = \emptyset$.

The restriction of the charts of a manifold M with boundary to ∂M forms charts of ∂M, making it an $(m-1)$-dimensional manifold without boundary.

C^k-functions on manifolds with boundary... Mappings, tangent spaces, and tangent maps can all be defined as before.

An important class of manifolds with boundary are the level sets of differentiable functions.

Theorem 27. *Let M be a C^r-manifold without boundary. If $c \in \mathbb{R}$ is a regular value of $f \in C^r(M, \mathbb{R})$, then $f_c := f^{-1}(-\infty, c]$ is a C^r-manifold with boundary, and $\partial f_c = f^{-1}(c)$.*

Proof. For any $p \in f_c$. If $f(p) < c$, then there exists a chart (U, φ) of M at p such that $U \subset f_c$. Thus $\tilde{\varphi}$ is a homeomorphism from the open subset U of f_c to $\mathbb{R}^m_+$.

If $f(p) = c$, then by Theorem 24, $f^{-1}(c)$ is an $(m-1)$-dimensional C^k-submanifold of M, and there exists a chart (U, φ) of M at p such that

$$\varphi(U \cap f^{-1}(c)) = \varphi(U) \cap (0 \times \mathbb{R}^{m-1}).$$

Therefore, $\varphi(U \cap f^{-1}(c)) \subset \mathbb{R}^m_+$. Thus $\varphi|_{U \cap f_c}$ is a homeomorphism from the open subset $U \cap f_c$ of f_c to $\mathbb{R}^m_+$.

The above yields two types of open subsets of f_c. They cover f_c. These open subsets, together with their homeomorphisms into $\mathbb{R}^m_+$, constitute a family of charts for f_c, making f_c an m-dimensional C^r-manifold with boundary. □

Definition 13. *Let M be an m-manifold with boundary, and $S \subset M$. If for all $p \in S$, there exists a chart (U, φ) of M at point p such that*

$$\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^s_+ \times 0), \text{ or } \varphi(U \cap S) = \varphi(U) \cap (0 \times \mathbb{R}^s),$$

then S is called an s-dimensional submanifold of M.

Theorem 28. *Let M be a C^r-manifold with boundary, and N be a C^r-manifold without boundary, and $f \in C^r(M, N)$. If $q \in N$ is a regular value of f and $f|_{\partial M}$, and $f^{-1}(q) \neq \emptyset$, then $f^{-1}(q)$ is an $(m-n)$-dimensional submanifold of M and $\partial f^{-1}(q) = f^{-1}(q) \cap \partial M$. Here $m = \dim M$ and $n = \dim N$.*

Proof. For $p \in f^{-1}(q)$, if $p \in \partial M$, then by Theorem 24, there exists a chart (U, φ) of M at p such that

$$\varphi(U \cap f^{-1}(q)) = \varphi(U) \cap (0 \times \mathbb{R}^{m-n}).$$

If $p \in \partial M$, take charts (U, φ) and (V, ψ) of M and N at p and q respectively such that

$$f(U) \subset V, \quad \varphi(p) = 0, \quad \psi(q) = 0.$$

Under these charts, f has a local representation

$$\tilde{f} = \psi \circ f \circ \varphi^{-1} : \mathbb{R}^m_+ \to \mathbb{R}^n.$$

We still denote it by $\tilde{f}$ in $\mathbb{R}^m$. a C^r-extension on it. Let the components of $\tilde{f}$ be represented as

$$(y^1, \cdots, y^n) = (f^1(x^1, \cdots, x^m), \cdots, f^m(x^1, \cdots, x^m)).$$

Then the local representation of $f|_{\partial M} : \partial M \to N$ is

$$(y^1, \cdots, y^n) = (f^1(0, x^2, \cdots, x^m), \cdots, f^m(0, x^2, \cdots, x^m)).$$

Since $(f|_{\partial M})_{*p} : T_p(\partial M) \to T_q N$ is surjective, we can assume that at $\varphi(p) = 0$

$$\frac{\partial(f^1, \cdots, f^n)}{\partial(x^{m-n+1}, \cdots, x^m)} \neq 0.$$

Applying the implicit function theorem to $\tilde{f} : \mathbb{R}^{m-n} \times \mathbb{R}^n \to \mathbb{R}^n$, we obtain an open neighborhood W of 0 in $\mathbb{R}^{m-n}$ and $g \in C^r(W, \mathbb{R}^n)$ such that $g(0) = 0$,

$$\tilde{f}(x, g(x)) = 0.$$

Construct a C^r-diffeomorphism $\beta : W \times \mathbb{R}^n \to W \times \mathbb{R}^n$,

$$\beta(x, z) = (x, z - g(x)).$$

make

$$\Omega = \varphi^{-1}(W \times \mathbb{R}^n), \qquad \phi = \beta \circ (\varphi|_\Omega).$$

Then (Ω, ϕ) is the picture card of M at p, so that

$$\phi(\Omega \cap f^{-1}(q)) = \phi(\Omega) \cap (\mathbb{R}^{m-n}_+ \times 0).$$

Therefore, $f^{-1}(q)$ is the $(m-n)$-dimensional submanifold of M. It is clear that $\partial f^{-1}(q) = f^{-1}(q) \cap \partial M$. □

4. Topological Degree

Let Ω be an open set in a Banach space X, and $f : \overline{\Omega} \to X$ be continuous. For $p \in X$, we want to define an integer $d = \deg(f, \Omega, p)$, which provides some estimate of the number of solutions to the equation $f(x) = p$ in Ω. This integer d is called the topological degree of f in Ω with respect to the point p.

In order to obtain the topological degree of complex mappings by calculating the topological degree of simple mappings, we also hope that $\deg(f, \Omega, p)$ is invariant under small perturbations of f.

The most natural idea is to define $d = \deg(f, \Omega, p)$ as the number of solutions to the equation $f(x) = p$ in Ω. However, d defined in this way does not have the property of invariance under small perturbations, and is therefore difficult to calculate.

Example 6. Let $|\varepsilon| < 1$. When $\varepsilon > 0$, $\varepsilon = 0$, and $\varepsilon < 0$, the equation

$$f_\varepsilon(x) = \sin x + \varepsilon = 1$$

has 2, 1, and 0 solutions in $(0, \pi)$, respectively.

This example shows that $\deg(f, \Omega, p)$ cannot be simply defined as the absolute number of solutions to $f(x) = p$ in Ω. Therefore, we take a less ambitious approach and define $\deg(f, \Omega, p)$ as the algebraic number of solutions to $f(x) = p$ in Ω.

4.1. Brouwer Degree. In this section, we always assume that Ω is a bounded open set in $\mathbb{R}^n$, and $f : \overline{\Omega} \to \mathbb{R}^n$ is continuous. $p \notin f(\partial\Omega)$.

Definition 14. *If $f \in C(\overline{\Omega}, \mathbb{R}^n) \cap C^1(\Omega, \mathbb{R}^n)$, and $p \in \mathbb{R}^n \setminus f(\partial\Omega)$ is a regular value of f, then we define*

$$\deg(f, \Omega, p) = \sum_{x \in f^{-1}(p)} \operatorname{sgn} |f'(x)|.$$

Remark 11. Since f is continuous and Ω is bounded, $f^{-1}(p)$ is compact. Furthermore, since p is a regular value of f and by the inverse function theorem, the points in $f^{-1}(p)$ are discrete. Therefore, by the finite covering theorem, $f^{-1}(p)$ is a finite set. In particular, if $f^{-1}(p) = \emptyset$, we define $\deg(f, \Omega, p) = 0$.

Let there be a family of functions $\{\varphi_\varepsilon\}_{\varepsilon > 0} \subset C^\infty(\mathbb{R}^n, \mathbb{R})$ satisfying

$$\operatorname{supp} \varphi_\varepsilon \subset B_\varepsilon(0), \qquad \int_{\mathbb{R}^n} \varphi_\varepsilon \, \mathrm{d}x = 1.$$

If Ω is a bounded open set in $\mathbb{R}^n$, $f \in C(\overline{\Omega}, \mathbb{R}^n) \cap C^1(\Omega, \mathbb{R}^n)$, $p \notin f(\partial\Omega)$ is the set of regular values ??of f. Then there exists $\varepsilon_0 > 0$ such that for $\varepsilon \in (0, \varepsilon_0)$,

$$\deg(f, \Omega, p) = \int_\Omega \varphi_\varepsilon(f(x) - p) |f'(x)| \, dx.$$

The above defined $\deg(f, \Omega, p)$ is locally constant with respect to p.

Lemma 4. *Let $f \in C(\overline{\Omega}, \mathbb{R}^n) \cap C^1(\Omega, \mathbb{R}^n)$, and $p \notin f(\partial\Omega)$ be a regular value of f. Then there exists an open neighborhood V of p such that every point q in V is a regular value of f and*

$$\deg(f, \Omega, p) = \deg(f, \Omega, q).$$

Proof. If $f^{-1}(p) = \emptyset$, since $f(\overline{\Omega})$ is compact, there exists an open neighborhood V of p such that $V \cap f(\overline{\Omega}) = \emptyset$. The conclusion is obviously true.

If $f^{-1}(p) \neq \emptyset$, then we can assume $f^{-1}(p) = \{x_1, \dots, x_k\}$. By the inverse function theorem, there exist open neighborhoods $U_1, \dots, U_k$ of $x_1, \dots, x_k$. They are pairwise disjoint and each is diffeomorphic to an open neighborhood of p. Let W be an open neighborhood of p. Furthermore, $|f'(x)|$ does not change sign on each U_j.

Let

$$V = W \setminus f\left(\overline{\Omega} \setminus \bigcup_{j=1}^{k} U_j\right).$$

Since each point q in V has no preimage outside $\bigcup_{j=1}^{k} U_j$ and exactly one preimage inside each U_j, this V is the desired set. □

In order to extend the definition of $\deg(f, \Omega, p)$ to continuous mappings, we need the following famous Sard's theorem.

Theorem 29 (Sard, 1942)**.** *Let Ω be an open set in $\mathbb{R}^m$, and $f \in C^k(\Omega, \mathbb{R}^n)$. If $k > \max\{0, m-n\}$, then the set of critical values of f is a set of measure zero in $\mathbb{R}^n$.*

This theorem shows that if f is sufficiently smooth, then its regular values are numerous. Its proof is irrelevant to our discussion and is omitted.

Lemma 5 (Nagumo, 1952)**.** *If $h \in C([0,1] \times \overline{\Omega}, \mathbb{R}^n) \cap C^2(([0,1] \times \Omega, \mathbb{R}^n))$ satisfies*

(1) $p \notin h([0,1] \times \partial\Omega)$,

(2) p is a regular value of $h(0,\cdot)$ and $h(1,\cdot)$.

Then $\deg(h(0,\cdot), \Omega, p) = \deg(h(1,\cdot), \Omega, p)$.

Proof. Let

$$r = \min_{(t,x)\in[0,1]\times\partial\Omega} |h(t,x) - p| > 0.$$

By Lemma 4 and Sard's theorem, there exists a regular value $y \in B_r(p)$ of h. Let

$$\deg(h(j,\cdot), \Omega, p) = \deg(h(j,\cdot), \Omega, y), \qquad j = 0, 1. \tag{4.1}$$

By Theorem 28, $h^{-1}(y) \subset [0,1] \times \Omega$ is a compact 1-manifold with boundary. Thus

$$h^{-1}(y) = C_1 \cup \cdots \cup C_k,$$

where C_j are smooth compact curves.

Let $v(s) = (t(s), x(s))$ be the arc-length parameterization of C_j, satisfying

$$h'(v)\dot{v} = (h \circ v)'(s) = 0, \qquad |\dot{v}|^2 = 1,$$

Since $h'(v)\dot{v} = 0$, $\dot{v}$ is orthogonal to the row vectors of $h'(v)$. We can also require

$$\det\begin{pmatrix} \dot{v} \\ h'(v) \end{pmatrix} > 0.$$

Thus

$$\begin{aligned} \dot{t}\det\begin{pmatrix} \dot{v} \\ h'(v) \end{pmatrix} &= \begin{vmatrix} \dot{t}^2 & \dot{x} \\ (\partial_t h)\dot{t} & \partial_x h \end{vmatrix} \\ &= \begin{vmatrix} |\dot{v}|^2 & \dot{x} \\ h'(v)\dot{v} & \partial_x h \end{vmatrix} = |\dot{v}|^2 |\partial_x h|. \end{aligned}$$

That is, $\operatorname{sgn}\dot{t} = \operatorname{sgn}|\partial_x h|$.

Each curve C_j must be one of the following cases: as s increases,

(1) C_j is a closed curve contained in $(0,1)\times\Omega$,
(2) C_j connects two points on the same base of the cylinder $[0,1]\times\Omega$,
(3) C_j runs from the base $0\times\Omega$ to $1\times\Omega$,
(4) C_j runs from the base $1\times\Omega$ to $0\times\Omega$.

Let there be ℓ and k curves C_j satisfying 3 and 4 respectively. Then

$$\deg(h(0,\cdot),\Omega,y)=\ell-k=\deg(h(1,\cdot),\Omega,y).$$

Combining with (4.1), the lemma is proved. □

Lemma 6. *Let $f\in C(\overline{\Omega},\mathbb{R}^n)$, $p\in f(\partial\Omega)$. If $g_0,g_1\in C(\overline{\Omega},\mathbb{R}^n)\cap C^2(\Omega,\mathbb{R}^n)$ make*

$$\max_{x\in\partial\Omega}|f(x)-g_j(x)|<\min_{x\in\partial\Omega}|f(x)-p|,\qquad j=0,1,$$

And p is the regular value of g_0 and g_1. Then $\deg(g_0,\Omega,p)=\deg(g_1,\Omega,p)$.

Proof. define $h:[0,1]\times\overline{\Omega}\to\mathbb{R}^n$,

$$h(t,x)=(1-t)g_0(x)+tg_1(x).$$

For $(t,x)\in[0,1]\times\partial\Omega$ Yes

$$|h(t,x)-f(x)|<\min_{x\in\partial\Omega}|f(x)-p|.$$

Therefore

$$|h(t,x)-p|\geq|f(x)-p|-|h(t,x)-f(x)|>0,$$

and Lemma 5 can be applied to h. □

Definition 15. *Let $f\in C(\overline{\Omega},\mathbb{R}^n)$, $p\notin f(\partial\Omega)$. Choose $g\in C(\overline{\Omega},\mathbb{R}^n)\cap C^2(\Omega,\mathbb{R}^n)$ such that p is a regular value of g and*

$$\max_{x\in\partial\Omega}|f(x)-g(x)|<\min_{x\in\partial\Omega}|f(x)-p|.$$

We define

$$\deg(f,\Omega,p):=\deg(g,\Omega,p)$$

as the Brouwer degree of f in Ω with respect to the point p.

Remark 12. (1) By Lemma 6, $\deg(f,\Omega,p)$ is independent of the choice of g.
(2) By the Weierstrass approximation theorem and Sard's theorem, such a g exists. We can first find a $g\in C^2$ approximating f, and then make a small perturbation to it so that p is a regular value.
(3) Take any sequence of $g_k\in C(\overline{\Omega},\mathbb{R}^n)\cap C^2(\Omega,\mathbb{R}^n)$ with p as a regular value such that $g_k\rightrightarrows f$ on $\overline{\Omega}$. Then when k is sufficiently large, $p\notin g_k(\partial\Omega)$ and

$$\deg(f,\Omega,p)=\lim_{k\to\infty}\deg(g_k,\Omega,p).$$

4.2. Properties of Brouwer degrees.

Theorem 30 (Basic properties of Brouwer degree)**.** (*a*) *Normative:*

$$\deg(\boldsymbol{I}_{\overline{\Omega}},\Omega,p)=\begin{cases}1, & p\in\Omega,\\ 0, & p\notin\Omega.\end{cases}$$

(*b*) *Regional additivity: Disjoint subsets Ω_1,Ω_2 of Ω, if $p\notin f(\overline{\Omega}\setminus(\Omega_1\cup\Omega_2))$, then*

$$\deg(f,\Omega,p)=\deg(f,\Omega_1,p)+\deg(f,\Omega_2,p).$$

(*c*) *Homotopy invariance: Let* $h \in C([0,1] \times \overline{\Omega}, \mathbb{R}^n)$, $p \notin h([0,1] \times \partial\Omega)$. *Then*

$$\deg(h(0,\cdot), \Omega, p) = \deg(h(1,\cdot), \Omega, p).$$

Proof. (*a*) is obvious.

(*b*). Take a sequence of $g_k \in C(\overline{\Omega}, \mathbb{R}^n) \cap C^2(\Omega, \mathbb{R}^n)$ with p as regular valur such that $g_k \rightrightarrows f$ on $\overline{\Omega}$. Then $p \notin g_k(\overline{\Omega} \setminus (\Omega_1 \cup \Omega_2))$ for $k \gg 1$. So

$$\begin{aligned}
\deg(g_k, \Omega, p) &= \sum_{x \in g_k^{-1}(p)} \operatorname{sgn} |g_k'(x)| \\
&= \left(\sum_{x \in g_k^{-1}(p) \cap \Omega_1} + \sum_{x \in g_k^{-1}(p) \cap \Omega_2} \right) \operatorname{sgn} |g_k'(x)| \\
&= \deg(g_k, \Omega_1, p) + \deg(g_k, \Omega_2, p).
\end{aligned}$$

Let $k \to \infty$ be the proof.

(*c*). takes $g \in C([0,1] \times \overline{\Omega}, \mathbb{R}^n) \cap C^2([0,1] \times \Omega, \mathbb{R}^n)$ make

$$\max_{(t,x) \in [0,1] \times \partial\Omega} |h(t,x) - g(t,x)| < \min_{(t,x) \in [0,1] \times \partial\Omega} |h(t,x) - p|. \tag{4.2}$$

And p is regular value of $g(0,\cdot)$ and $g(1,\cdot)$.

For $\forall t \in (0,1)$ there is $g(t,\cdot) \in C(\overline{\Omega}, \mathbb{R}^n) \cap C^2(\Omega, \mathbb{R}^n)$, and by (4.2) we have

$$\max_{x \in \partial\Omega} |h(t,x) - g(t,x)| < \min_{x \in \partial\Omega} |h(t,x) - p|.$$

Therefore

$$\deg(h(t,\cdot), \Omega, p) = \deg(g(t,\cdot), \Omega, p), \qquad \forall t \in [0,1]. \tag{4.3}$$

When $(t,x) \in [0,1] \times \partial\Omega$ time,

$$\begin{aligned}
|g(t,x) - p| &\geq |h(t,x) - p| - |h(t,x) - g(t,x)| \\
&> |h(t,x) - p| - \min_{(t,x) \in [0,1] \times \partial\Omega} |h(t,x) - p| > 0
\end{aligned}$$

By Lemma 5, we have $\deg(g(0,\cdot), \Omega, p) = \deg(g(1,\cdot), \Omega, p)$. Combining this with (4.3), we obtain the desired conclusion. □

Exercise 12 (Connected Component Property)**.** Let $p : [0,1] \to \mathbb{R}^n$, $p[0,1] \cap f(\partial\Omega) = \emptyset$. Then $\deg(f, \Omega, p(t)) = \text{const}$.

Theorem 31. *Let* $f \in C(\overline{\Omega}, \mathbb{R}^n)$, $p \notin f(\partial\Omega)$.

(*d*) *Kronecker Existence: If* $\deg(f, \Omega, p) \neq 0$, *then* $p \in f(\Omega)$, *i.e., the equation* $f(x) = p$ *has a solution in* Ω.

(*e*) *Excision Property: If* K *is a closed subset of* $\overline{\Omega}$, $p \notin f(K)$, *then*

$$\deg(f, \Omega, p) = \deg(f, \Omega \setminus K, p).$$

Proof. (d) Let $\{g_k\} \subset C(\overline{\Omega}, \mathbb{R}^n) \cap C^2(\Omega, \mathbb{R}^n)$ be a sequence of functions with p as a regular value, such that $g_k \rightrightarrows f$ on $\overline{\Omega}$. Then for sufficiently large k,

$$\deg(f, \Omega, p) = \deg(g_k, \Omega, p).$$

If $\deg(f, \Omega, p) \neq 0$, then there exists $x_k \in \Omega$ such that $g_k(x_k) = p$. Without loss of generality, assume $x_k \to x \in \overline{\Omega}$. Then $f(x) = p$. Since $p \notin f(\partial\Omega)$, it follows that $p \in f(\Omega)$.

(e) This can be proved similarly to (b). □

Theorem 32 (Poincare-Bohl)**.** *Let $f, g \in C(\overline{\Omega}, \mathbb{R}^n)$, When $x \in \partial\Omega$ $p \notin [f(x), g(x)]$. Then* $\deg(f, \Omega, p) = \deg(g, \Omega, p)$.

It follows that $\deg(f, \Omega, p)$ is only the same as $f|_{\partial\Omega}$ is related, which is called the boundary value property.

Proof. Let

$$h(t, x) = (1-t)f(x) + tg(x),$$

Then $h \in C([0,1] \times \overline{\Omega}, \mathbb{R}^n)$ and $p \notin h([0,1] \times \partial\Omega)$. by homotopy invariance

$$\begin{aligned}\deg(f, \Omega, p) &= \deg(h(0, \cdot), \Omega, p) \\ &= \deg(h(1, \cdot), \Omega, p) = \deg(g, \Omega, p).\end{aligned}$$

□

Exercise 13.

(1) Let $f, g \in C(\overline{\Omega}, \mathbb{R}^n)$, when $x \in \partial\Omega$ $|g(x)| < |f(x)|$, Then

$$\deg(f + g, \Omega, 0) = \deg(f, \Omega, 0).$$

(2) Let $0 \in \Omega$, $f \in C(\overline{\Omega}, \mathbb{R}^n)$ satisfy

$$\langle f(x), x\rangle \geq 0, \qquad \forall x \in \partial\Omega.$$

Then f must have a zero in $\overline{\Omega}$.

(3) Let $f \in C(\mathbb{R}^n, \mathbb{R}^n)$,

$$\lim_{|x|\to\infty} \frac{\langle f(x), x\rangle}{|x|} = +\infty,$$

Then $f(\mathbb{R}^n) = \mathbb{R}^n$.

(4) Let $0 \in \Omega$, $f : \partial\Omega \to \mathbb{R}^n \setminus 0$ be continuous. If n is odd, then there exists $x \in \partial\Omega$ and $\lambda \neq 0$ such that $f(x) = \lambda x$.

(5) Let $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ satisfy
(i) $\forall x \in \mathbb{R}^n$, $|f'(x)| \neq 0$; (ii) $\lim_{|x|\to\infty} |f(x)| = +\infty$.
Then $f(\mathbb{R}^n) = \mathbb{R}^n$.

Finally, in order to establish the Leray-Schauder theory in infinite-dimensional spaces... Degree, We prove the following simplification theorem.

Theorem 33 (Simplification Theorem)**.** *Let $f \in C(\overline{\Omega}, \mathbb{R}^n)$, $m \leq n$, $(\boldsymbol{I}_{\overline{\Omega}} - f)(\overline{\Omega}) \subset \mathbb{R}^m$, $p \in \mathbb{R}^m \setminus f(\partial\Omega)$. Then* $\deg(f, \Omega, p) = \deg(f|_{\overline{\Omega}\cap\mathbb{R}^m}, \Omega \cap \mathbb{R}^m, p)$.

Remark 13. For $x \in \Omega \cap \mathbb{R}^m$, we have $x - f(x) \in \mathbb{R}^m$. Thus $f(x) \in \mathbb{R}^m$. Therefore, $\deg(f|_{\overline{\Omega}\cap\mathbb{R}^m}, \Omega \cap \mathbb{R}^m, p)$ is well-defined.

Proof. The proof proceeds in the following steps.

We have $f^{-1}(p) = f|^{-1}_{\Omega\cap\mathbb{R}^m}(p)$. Let $x \in f^{-1}(p)$, then $x - p = x - f(x) \in \mathbb{R}^m$, so $x \in \mathbb{R}^m$. Thus $f^{-1}(p) \subset f|^{-1}_{\Omega\cap\mathbb{R}^m}(p)$. The reverse inclusion is obvious.

Let $f \in C(\overline{\Omega}, \mathbb{R}^n) \cap C^2(\Omega, \mathbb{R}^n)$. If $x \in \Omega \cap \mathbb{R}^m$, then

$$f'(x) = \begin{pmatrix} f|'_{\Omega\cap\mathbb{R}^m}(x) & 0 \\ * & \mathbf{1}_{\mathbb{R}^{n-m}} \end{pmatrix}, \tag{4.4}$$

Therefore, when p is a regular value of f, p is also a regular value of $f|_{\Omega\cap\mathbb{R}^m}$ and for

$$x \in f^{-1}(p) = f|^{-1}_{\Omega\cap\mathbb{R}^m}(p)$$

we have $\operatorname{sgn}|f'(x)| = \operatorname{sgn}|f|'_{\Omega\cap\mathbb{R}^m}(x)|$. Thus

$$\deg(f,\Omega,p) = \sum_{x\in f^{-1}(p)} \operatorname{sgn}|f'(x)| = \sum_{x\in f|^{-1}_{\Omega\cap\mathbb{R}^m}(p)} \operatorname{sgn}|f|'_{\Omega\cap\mathbb{R}^m}(x)|$$
$$= \deg(f|_{\overline{\Omega}\cap\mathbb{R}^m}, \Omega\cap\mathbb{R}^m, p).$$

Proof as follows (4.4). Set $x \in \mathbb{R}^n = \mathbb{R}^m \oplus \mathbb{R}^{n-m}$ is written as $x = (x_1, x_2)$. Remember $F = \mathbf{1}_{\overline{\Omega}} - f$, then $F(\overline{\Omega}) \subset \mathbb{R}^m$. So $f(x) = x - F(x)$ can be written as

$$f(x_1, x_2) = (x_1 - F(x_1, x_2), x_2).$$

From this, for $x = (x_1, 0) \in \Omega\cap\mathbb{R}^m$,

$$\begin{aligned} f'(x) &= f'(x_1, 0) \\ &= \begin{pmatrix} \mathbf{1}_{\mathbb{R}^m} - \partial_1 F(x_1,0) & 0 \\ * & \mathbf{1}_{\mathbb{R}^{n-m}} \end{pmatrix} = \begin{pmatrix} f|'_{\Omega\cap\mathbb{R}^m}(x) & 0 \\ * & \mathbf{1}_{\mathbb{R}^{n-m}} \end{pmatrix}. \end{aligned}$$

For $f \in C(\overline{\Omega}, \mathbb{R}^n)$, Take $G_k \in C(\overline{\Omega}, \mathbb{R}^n) \cap C^2(\Omega, \mathbb{R}^n)$ Let p be the regular value of

$$g_k = \mathbf{1}_{\overline{\Omega}} - G_k \in C(\overline{\Omega}, \mathbb{R}^n) \cap C^2(\Omega, \mathbb{R}^n)$$

and $G_k \rightrightarrows F = 1_{\overline{\Omega}} - f$ on $\overline{\Omega}$. Then on $\overline{\Omega}$ we have

$$g_k \rightrightarrows f, \qquad g_k|_{\overline{\Omega}\cap\mathbb{R}^m} \rightrightarrows f|_{\overline{\Omega}\cap\mathbb{R}^m}.$$

From 2°, we know that p is also a regular value of g_k and

$$\begin{aligned} \deg(f,\Omega,p) &= \lim_{k\to\infty} \deg(g_k,\Omega,p) = \lim_{k\to\infty} \deg(g_k|_{\overline{\Omega}\cap\mathbb{R}^m}, \Omega\cap\mathbb{R}^m, p) \\ &= \deg(f|_{\overline{\Omega}\cap\mathbb{R}^m}, \Omega\cap\mathbb{R}^m, p). \end{aligned}$$

□

4.3. Brouwer Fixed-Point Theorem, Borsuk Theorem. Solving operator equations can be transformed into finding fixed points of a certain mapping. Therefore, fixed-point theorems are powerful tools for studying the existence of solutions to equations.

Theorem 34 (Brouwer)**.** *Let Ω be a bounded convex open set in $\mathbb{R}^n$, and $f \in C(\overline{\Omega}, \mathbb{R}^n)$. If $f(\partial\Omega) \subset \overline{\Omega}$, then f must have a fixed point on $\overline{\Omega}$.*

Proof. Without loss of generality, assume $0 \in \Omega$. If f has no fixed point on $\partial\Omega$, let $h(t,x) = x - tf(x)$. For $x \in \partial\Omega$, clearly $h(0,x) \neq 0$, $h(1,x) \neq 0$. If there exists $(t,x) \in (0,1)\times\partial\Omega$ such that $h(t,x) = 0$. Take $r > 0$ such that $B_r(0) \subset \Omega$. Since $f(x) \in \overline{\Omega}$, $\exists y \in \Omega$ such that

$$|y - f(x)| < \frac{(1-t)r}{t}.$$

Since Ω is convex and $B_r(0) \subset \Omega$, we have $B_{(1-t)r}(ty) \subset \Omega$. Therefore

$$x = tf(x) = ty + t(f(x) - y) \in B_{(1-t)r}(ty) \subset \Omega.$$

This contradicts $x \in \partial\Omega$. This proves that $0 \notin h([0,1]\times\partial\Omega)$.

By the homotopy invariance of the Brouwer degree,

$$\begin{aligned} \deg(\mathbf{1}_{\overline{\Omega}} - f, \Omega, 0) &= \deg(h(1,\cdot), \Omega, 0) \\ &= \deg(h(0,\cdot), \Omega, 0) = \deg(\mathbf{1}_{\overline{\Omega}}, \Omega, 0) = 1. \end{aligned}$$

By the Kronecker existence theorem, there exists $x \in \Omega$ such that $f(x) = x$. □

Theorem 35 (Borsuk)**.** *Let Ω be a bounded symmetric open set in $\mathbb{R}^n$ containing the origin, and $f : \overline{\Omega} \to \mathbb{R}^n$ be a continuous odd mapping. Then, when $0 \notin f(\partial\Omega)$, $\deg(f, \Omega, 0)$ is an odd number.*

Proof. Let $\varepsilon > 0$ such that $B_\varepsilon(0) \subset \Omega$. By Tietze's extension theorem, there exists $h \in C(\overline{\Omega}, \mathbb{R}^n)$ such that

$$h(x) = \begin{cases} x, & x \in \overline{B}_\varepsilon(0), \\ f(x), & \end{cases}$$

$$\begin{cases} h(x) = f(x), & x \in \Omega; \\ h(x) = -f(-x), & x \in \partial\Omega. \end{cases}$$

We can also assume that h is odd; otherwise, replace it with its odd extension.

By the boundary value property and additivity of the Brouwer degree,

$$\begin{aligned} \deg(f, \Omega, 0) &= \deg(h, \Omega, 0) \\ &= \deg(h, B_\varepsilon, 0) + \deg(h, \Omega \setminus \overline{B}_\varepsilon(0), 0). \end{aligned} \tag{4.5}$$

The first term is 1. We will prove that the second term is an even number, thus proving Borsuk's theorem. □

To prove that the second term of (4.5) is an even number, since the zeros of an odd mapping appear in pairs, we only need to find an odd C^2-mapping g with 0 as a regular value that sufficiently approximates h. For this, we need the following special case of the transversality theorem.

Lemma 7. *Let Ω be an open set in $\mathbb{R}^n$, $F \in C^2(\mathbb{R}^m \times \Omega, \mathbb{R}^n)$, and 0 be a regular value of F. Then for almost all $x \in \mathbb{R}^m$, 0 is a regular value of $f_x = F(x, \cdot) : \mathbb{R}^n \to \mathbb{R}^n$.*

Proof. $S = F^{-1}(0)$ is an m-dimensional C^2-submanifold of $\mathbb{R}^m \times \Omega$. Let $\pi : S \to \mathbb{R}^m$ be the projection $(x, y) \mapsto x$, then $\pi \in C^1$(Let S be a manifold in $\mathbb{R}^m \times \mathbb{R}^n$, and $\pi : S \to \mathbb{R}^m$ be the projection onto the first component. By Sard's theorem, almost all $x \in \mathbb{R}^m$ are regular values ??of π. Therefore, we only need to prove that if x is a regular value of π, then for all $y \in f_x^{-1}(0)$,

$$f_x'(y) = \partial_y F(x, y) : \mathbb{R}^n \to \mathbb{R}^n$$

is surjective, so that 0 is a regular value of f_x.

For $v \in \mathbb{R}^n$, since $F'(x, y) : \mathbb{R}^m \times \mathbb{R}^n \to \mathbb{R}^n$ is surjective, there exists $(h, k) \in \mathbb{R}^m \times \mathbb{R}^n$ such that

$$v = F'(x, y)(h, k) = \partial_x F(x, y)h + \partial_y F(x, y)k \tag{4.6}$$

Since x is a regular value of π, there exists a curve γ on S passing through the point (x, y) such that

$$(\pi \circ \gamma)'(0) = \pi_{*(x,y)}\dot{\gamma}(0) = h.$$

That is, for some $l \in \mathbb{R}^n$, $\dot{\gamma}(0) = (h, l)$. Therefore,

$$\begin{aligned} 0 = (F \circ \gamma)'(0)0) &= F_{*(x,y)}\dot{\gamma}(0) \\ &= F'(x, y)(h, l) = \partial_x F(x, y)h + \partial_y F(x, y)l. \end{aligned} \tag{4.7}$$

From (4.6) and (4.7) we know that $k - l \in \mathbb{R}^n$ makes

$$\partial_y F(x, y)(k - l) = v.$$

That is $f_x'(y) = \partial_y F(x, y)$ is surjective. □

Next, we construct the odd C^2-mapping g that we need. By Weierstrass's theorem, we can assume that $h \in C^2(\Omega \setminus \overline{B}_\varepsilon(0), \mathbb{R}^n)$. Let $F : \mathbb{R}^{n^2} \times \Omega \setminus \overline{B}_\varepsilon(0) \to \mathbb{R}^n$,

$$F(A, x) = h(x) + Ax.$$

Let $(A, x) \in F^{-1}(0)$. Then for $(B, y) \in \mathbb{R}^{n^2} \times \mathbb{R}^n$,

$$F'(A, x)(B, y) = h'(x)y + Ay + Bx.$$

For any $z \in \mathbb{R}^n$, since $x \neq 0$, we can choose

$$B = \frac{\langle x, \cdot \rangle}{|x|^2} z, \qquad y = 0,$$

then

$$F'(A, x)(B, y) = z.$$

That is, $F'(A, x)$ is surjective, and 0 is a regular value of F.

By Lemma 7, we can choose A with arbitrarily small norm such that 0 is a regular value of $g = F(A, \cdot)$. This g is the desired mapping.

Now let's look at a simple application of Borsuk's theorem.

Theorem 36. *Let Ω be an open set in $\mathbb{R}^n$, $f \in C(\Omega, \mathbb{R}^n)$.*

If f is locally injective, then f is an open mapping.

Proof. Let $x_0 \in \Omega$. We want to show that $f(x_0)$ is an interior point of $f(\Omega)$. By translation, we can assume $x_0 = 0$ and $f(x_0) = 0$.

Take an ε-neighborhood $B_\varepsilon(0)$ of 0 such that f is injective on $\overline{B}_\varepsilon(0)$. Let

$$h(t, x) = f\left(\frac{x}{1+t}\right) - f\left(\frac{-tx}{1+t}\right).$$

Then $h \in C([0, 1] \times \overline{B}_\varepsilon(0), \mathbb{R}^n)$. Since f is injective on $\overline{B}_\varepsilon(0)$, we have $0 \notin h([0, 1] \times \partial\overline{B}_\varepsilon(0))$. Therefore, by the homotopy invariance of the Brouwer degree and Borsuk's theorem,

$$\begin{aligned} \deg(f, B_\varepsilon(0), 0) &= \deg(h(0, \cdot), B_\varepsilon(0), 0) \\ &= \deg(h(1, \cdot), B_\varepsilon(0), 0) = \text{odd}. \end{aligned}$$

Thus, there exists $r > 0$ such that when $|p| < r$,

$$\deg(f, B_\varepsilon(0), p) = \text{odd}.$$

$$\deg(f, B_\varepsilon(0), p) = \deg(f, B_\varepsilon(0), 0) \neq 0.$$

Therefore, $f(x) = p$ has a solution in $B_\varepsilon(0) \subset \Omega$. That is, $B_r(0) \subset \Omega$, and 0 is an interior point of $f(\Omega)$. □

Exercise 14. Let $f \in C(\mathbb{R}^n, \mathbb{R}^n)$ be locally injective and coercive. Then $f(\mathbb{R}^n) = \mathbb{R}^n$. This problem is a generalization of Exercise 13 (5).

4.4. Leray-Schauder Degree. Many problems in analysis (such as differential equations and integral equations) are infinite-dimensional. It is natural to hope to define some kind of topological degree for continuous mappings on infinite-dimensional spaces, which possesses properties similar to the Brouwer degree. However, the following example shows that the Brouwer fixed-point theorem does not hold in infinite-dimensional spaces, so such a topological degree does not exist: because the proof of the Brouwer fixed-point theorem only uses the most basic properties of the Brouwer degree, such as homotopy invariance and Kronecker existence.

Example 7 (Kakutani, 1943)**.** Let B be the closed unit ball in ℓ^2, and $f : B \to B$ be defined as

$$f(x) = ((1 - \|x\|)^{1/2}, x_1, x_2, \cdots), \text{ where } x = (x_1, x_2, \cdots).$$

Then $f \in C(B, B)$, but f has no fixed point.

The reason for this phenomenon is that closed balls in infinite-dimensional spaces are not compact. Similar to extremal problems, we turn to considering mappings with some kind of compactness, to see if we can establish a satisfactory topological degree theory for them.

Definition 16. *Let X, Y be Banach spaces, $D \subset X$. A continuous mapping $F : D \to Y$ is called compact if for any bounded subset S of D, $\overline{F(S)}$ is a compact set in Y.*

Theorem 37. *Let D be a bounded closed set in X, and $F : D \to Y$ be compact. Then for $\forall \varepsilon > 0$, there exists a finite-dimensional subspace Y_n of Y and a continuous mapping $F_n : D \to Y_n$*

make

$$\sup_{x \in D} \|F(x) - F_n(x)\| \le \varepsilon.$$

Proof. Because $\overline{F(D)}$ is tight, there exists $y_1, \cdots, y_n \in Y$ makes

$$F(D) \subset \bigcup_{i=1}^{n} B_\varepsilon(y_i).$$

make

$$\psi_i(x) = \begin{cases} \varepsilon - \|F(x) - y_i\|, & F(x) \in B_\varepsilon(y_i), \\ 0, & F(x) \notin B_\varepsilon(y_i). \end{cases}$$

Let $F_n : D \to Y_n = \operatorname{span}\{y_1, \cdots, y_n\}$,

$$F_n(x) = \frac{1}{\sum_{i=1}^{n} \psi_i(x)} \sum_{j=1}^{n} \psi_j(x) y_j.$$

Then $F_n : D \to Y_n$ is continuous and for $\forall x \in D$,

$$\|F(x) - F_n(x)\| \le \frac{1}{\sum_{i=1}^{n} \psi_i(x)} \sum_{j=1}^{n} \psi_j(x) \|F(x) - y_i\| \le \varepsilon.$$

□

Lemma 8. *Let D be a bounded closed set in X, $F : D \to X$ be compact, $f = \mathbf{1}_D - F$, and S be a closed subset of D. Then $f(S)$ is a closed set.*

Proof. Let $x_n \in S$ such that $f(x_n) \to y$. Since F is compact and $\{x_n\}$ is bounded, we can assume that $F(x_n) \to z$. Then

$$x_n = f(x_n) + F(x_n) \to x = y + z \in S.$$

By the continuity of f, we have $y = f(x) \in f(S)$. □

Let Ω be a bounded open subset of X, $F : \overline{\Omega} \to X$ be compact, and $f = \mathbf{1}_{\overline{\Omega}} - F : \overline{\Omega} \to X$. Let $p \in X \setminus f(\partial\Omega)$. Since $f(\partial\Omega)$ is closed,

$$\varepsilon = \operatorname{dist}(p, f(\partial\Omega)) > 0.$$

By Theorem 37, we can choose a finite-dimensional subspace X_n of X and a continuous mapping $F_n : \overline{\Omega} \to X_n$ such that $p \in X_n$,

$$\sup_{x \in \overline{\Omega}} \|F(x) - F_n(x)\| < \varepsilon.$$

Let $\Omega_n = X_n \cap \Omega$, $f_n = \mathbf{1}_{\overline{\Omega}_n} - F_n : \overline{\Omega}_n \to X_n$. When $x \in \partial\Omega_n \subset \partial\Omega$,

$$\|f_n(x) - p\| \geq \|f(x) - p\| - \|F_n(x) - F(x)\| > 0, \tag{4.8}$$

that is, $p \notin f_n(\partial\Omega_n)$. Therefore, the Brouwer degree $\deg(f_n, \Omega_n, p)$ is well-defined.

Lemma 9. *The above* $\deg(f_n, \Omega_n, p)$ *is independent of* X_n *and* F_n.

The selection does not matter.

Proof. Suppose (X_n, F_n) and (X_k, F_k) all meet the above conditions. Note

$$f_n = \mathbf{1}_{\overline{\Omega}_n} - F_n, \qquad f_k = \mathbf{1}_{\overline{\Omega}_k} - F_k.$$

We want proof

$$\deg(f_n, \Omega_n, p) = \deg(f_k, \Omega_k, p). \tag{4.9}$$

Let $X_m = X_n \oplus X_k$,

$$f_{n,m} = f_n = \mathbf{1}_{\overline{\Omega}_m} - F_n, \qquad f_{k,m} = f_n = \mathbf{1}_{\overline{\Omega}_m} - F_k.$$

Then by the simplification theorem,

$$\begin{aligned} \deg(f_n, \Omega_n, p) &= \deg(f_{n,m}, \Omega_m, p), \\ \deg(f_k, \Omega_k, p) &= \deg(f_{k,m}, \Omega_m, p). \end{aligned} \tag{4.10}$$

Let $h : [0, 1] \times \overline{\Omega}_m \to X_m$,

$$\begin{aligned} h(t, x) &= (1 - t) f_{n,m}(x) + t f_{k,m}(x) \\ &= x - (1 - t) F_n(x) - t F_k(x). \end{aligned}$$

Then for $x \in \partial\Omega_m$ and $t \in [0, 1]$,

$$\begin{aligned} \|h(t, x) - p\| \geq \|f(x) - p\| - (1 - t)\|F(x) - F_n(x)\| \\ -t\|F(x) - F_k(x)\| > 0. \end{aligned}$$

By the homotopy invariance of the Brouwer degree,

$$\begin{aligned} \deg(f_{n,m}, \Omega_m, p) &= \deg(h(0, \cdot), \Omega_m, p) \\ &= \deg(h(1, \cdot), \Omega_m, p) = \deg(f_{k,m}, \Omega_m, p). \end{aligned}$$

From (4.10), we obtain (4.9). □

Definition 17. *The degree* $\deg(f_n, \Omega_n, p)$ *is called the Leray-Schauder degree of* $f : \overline{\Omega} \to X$ *on* Ω *with respect to the point* p*, denoted by* $\deg(f, \Omega, p)$.

This establishes a topological degree for a class of continuous mappings (i.e., compact perturbations of the identity mapping) on infinite-dimensional spaces. The properties of this degree are studied below.

Theorem 38. *The Leray-Schauder degree has the following properties.*

(*a*) *Normative:*

$$\deg(\boldsymbol{I}_{\overline{\Omega}}, \Omega, p) = \begin{cases} 1, & when\, p \in \Omega, \\ 0, & when\, p \notin \Omega. \end{cases}$$

(*b*) *Regional additivity: Disjoint subsets* Ω_1, Ω_2 *of* Ω, *if* $p \notin f(\overline{\Omega} \setminus (\Omega_1 \cup \Omega_2))$, *then*

$$\deg(f, \Omega, p) = \deg(f, \Omega_1, p) + \deg(f, \Omega_2, p).$$

(*c*) *Compact homotopy invariance: Let* $H : [0,1] \times \overline{\Omega} \to X$ *tight,* $h(t,x) = x - H(t,x)$. *If* $p \notin h([0,1] \times \partial\Omega)$, *then*

$$\deg(h(0,\cdot), \Omega, p) = \deg(h(1,\cdot), \Omega, p).$$

(*d*) *Kronecker existence: if* $\deg(f, \Omega, p) \neq 0$, *then* $p \in f(\Omega)$.

Proof. (*a*) is obvious.

(*b*). Because $\Omega_0 = \overline{\Omega} \setminus (\Omega_1 \cup \Omega_2)$ closed and $p \notin f(\Omega_0)$, so

$$\varepsilon = dist\,(p, f(\Omega_0)) > 0.$$

Take the finite-dimensional subspace X_n of X and the continuous map $F_n : \overline{\Omega} \to X_n$ makes $p \in X_n$,

$$\sup_{x \in \overline{\Omega}} \|F(x) - F_n(x)\| < \varepsilon.$$

Let $f_n(x) = x - F_n(x)$. Because

$$\varepsilon \leq \min\{\mathrm{dist}\,(p, f(\partial\Omega)), \mathrm{dist}\,(p, f(\partial\Omega_1)), \mathrm{dist}\,(p, f(\partial\Omega_2))\},$$

By definition there is

$$\begin{aligned} \deg(f, \Omega, p) &= \deg(f_n, \Omega \cap X_n, p), \\ \deg(f, \Omega_1, p) &= \deg(f_n, \Omega_1 \cap X_n, p), \\ \deg(f, \Omega_2, p) &= \deg(f_n, \Omega_2 \cap X_n, p). \end{aligned} \tag{4.11}$$

Similarly to (4.8), we know that $p \notin f_n(\Omega_0 \cap X_n)$. By the additivity of the Brouwer degree,

$$\deg(f_n, \Omega \cap X_n, p) = \deg(f_n, \Omega_1 \cap X_n, p) + \deg(f_n, \Omega_2 \cap X_n, p).$$

From this and (4.11), (b) is proved.

(c) Since $H : [0,1] \times \overline{\Omega} \to X$ is compact, similarly to Lemma 8, it can be proved that $h([0,1] \times \partial\Omega)$ is a closed set. Furthermore, since $p \notin h([0,1] \times \partial\Omega)$, we have

$$\varepsilon = \inf_{(t,x) \in [0,1] \times \partial\Omega} \|h(t,x) - p\| > 0.$$

By Theorem 37, there exists a finite-dimensional subspace X_n of X and a continuous mapping $H_n : [0,1] \times \overline{\Omega} \to X_n$ such that $p \in X_n$,

$$\sup_{(t,x) \in [0,1] \times \overline{\Omega}} \|H(t,x) - H_n(t,x)\| < \varepsilon.$$

Let $h_n(t,x) = x - H_n(t,x)$, Then for $(t,x) \in [0,1] \times \partial\Omega_n$ Yes

$$\begin{aligned} \|h_n(t,x) - p\| &\geq \|h(t,x) - p\| - \|h(t,x) - h_n(t,x)\| \\ &\geq \varepsilon - \|h(t,x) - h_n(t,x)\| > 0. \end{aligned}$$

According to the homotopy invariance of Brouwer degree, we have

$$\begin{aligned}\deg(h(0,\cdot),\Omega,p) &= \deg(h_n(0,\cdot),\Omega_n,p)\\ &= \deg(h_n(1,\cdot),\Omega_n,p) = \deg(h(1,\cdot),\Omega,p).\end{aligned}$$

(d). Take the finite-dimensional subspace X_n of X and the continuous map $F_n : \overline{\Omega} \to X_n$ makes $p \in X_n$,

$$\varepsilon_n = \sup_{x\in\overline{\Omega}} \|F(x) - F_n(x)\| \to 0, \qquad (n \to \infty). \tag{4.12}$$

Let $f_n = \mathbf{1}_{\overline{\Omega}} - F_n$. Then when n is large enough, So that $\varepsilon_n < \operatorname{dist}(p, f(\partial\Omega))$,

$$\deg(f_n,\Omega_n,p) = \deg(f,\Omega,p) \neq 0.$$

So $\exists x_n \in \Omega_n$ makes $x_n - F(x_n) = p$. Since F is compact, we might as well set $F(x_n) \to y$. By

$$\|F_n(x_n) - y\| \le \|F_n(x_n) - F(x_n)\| + \|F(x_n) - y\|$$

That is (4.12) that is, $F_n(x_n) \to y$. So $x_n \to x = y + p \in \overline{\Omega}$. By

$$y = \lim_{n\to\infty} F(x_n) = F(x)$$

That is to say, $f(x) = x - F(x) = p$. Then $p \notin f(\partial\Omega)$ then $p \in f(\Omega)$. □

4.5. The Leray-Schauder Fixed Point Theorem and its Applications. Using the Leray-Schauder degree, we prove the famous Leray-Schauder fixed point theorem, and use it to discuss the existence of solutions to nonlinear differential equations and integral equations. Below, let X be a Banach space.

Theorem 39 (Leray-Schauder)**.** *Let Ω be a bounded open set in X containing the origin, and let $F : \overline{\Omega} \to X$ be a compact mapping satisfying the Leray-Schauder boundary condition: for $u \in \partial\Omega$ and $\lambda > 0$, we have $F(u) \neq \lambda u$. Then F has a fixed point on $\overline{\Omega}$.*

Proof. Let $f = \mathbf{1}_{\overline{\Omega}} - F$. We only need to show that f has a zero on $\overline{\Omega}$. If f has no zeros on $\partial\Omega$, let $h : [0,1] \times \overline{\Omega} \to X$,

$$h(t,u) = u - tF(u).$$

From the Leray-Schauder boundary condition, we know that $0 \notin h([0,1] \times \partial\Omega)$. Therefore, by the compact homotopy invariance of the Leray-Schauder degree, we have

$$\deg(f,\Omega,0) = \deg(\mathbf{1}_{\overline{\Omega}},\Omega,0) = 1.$$

Thus, f indeed has a zero in $\overline{\Omega}$. □

Corollary 4. *Let Ω be a bounded convex open set in X, and let $F : \overline{\Omega} \to X$ be a compact mapping satisfying $F(\partial\Omega) \subset \overline{\Omega}$. Then F has a fixed point on $\overline{\Omega}$. There exists a fixed point.*

Proof. Without loss of generality, assume $0 \in \Omega$. Since Ω is convex, F satisfies the Leray-Schauder boundary condition. □

In particular, if B is a closed ball in X, and $F : B \to B$ is compact, then F has a fixed point.

The aforementioned fixed point theorems all seek fixed points of mappings on the closure of a bounded open set. In applications, it is necessary to first determine this bounded open set. Sometimes this is inconvenient. The following theorem directly seeks fixed points on X.

Theorem 40. *Let X be a Banach space, and $F : X \to X$ be compact. If*

$$S = \{u \in X \,|\, \exists \lambda \in (0,1) \text{such that } u = \lambda F(u)\}$$

is bounded, then F has a fixed point.

Proof. Let $S \subset B_R(0)$. Then for $u \in \partial B_R(0)$, $u \notin S$, so when $\lambda > 1$, $F(u) \neq \lambda u$. By Theorem 39, F has a fixed point on $\overline{B}_R(0)$. □

Example 8. Let Ω be a bounded open set in $\mathbb{R}^n$, and $k \in C(\overline{\Omega} \times \overline{\Omega} \times \mathbb{R}, \mathbb{R})$. Also, let $a > 0$, $b \in (0, |\Omega|^{-1})$ such that

$$|k(x,y,u)| \leq a + b|u|, \qquad \forall (x,y,u) \in \overline{\Omega} \times \overline{\Omega} \times \mathbb{R}.$$

Then the integral equation

$$\varphi(x) = \int_\Omega k(x,y,\varphi(y))\,\mathrm{d}y$$

There is a solution $\varphi \in C(\overline{\Omega})$.

Proof. Let $K : C(\overline{\Omega}) \to C(\overline{\Omega})$,

$$(K\varphi)(x) = \int_\Omega k(x,y,\varphi(y))\,\mathrm{d}y.$$

It can be proved that K is compact by the Arzela-Ascoli theorem.

If $(\lambda, \varphi) \in (0,1) \times C(\overline{\Omega})$ satisfies $\varphi = \lambda K\varphi$, but

$$\begin{aligned} \|\varphi\| \leq \|K\varphi\| &= \max_{x \in \overline{\Omega}} \left| \int_\Omega k(x,y,\varphi(y))\,\mathrm{d}y \right| \\ &\leq \max_{x \in \overline{\Omega}} \int_\Omega |k(x,y,\varphi(y))|\,\mathrm{d}y \leq a|\Omega| + b|\Omega|\|\varphi\|. \end{aligned}$$

That is $\|\varphi\| \leq a|\Omega|(1 - b|\Omega|)^{-1}$. According to the theorem40, K has a fixed point. □

Example 9. If $f \in C([a,b] \times \mathbb{R}^n, \mathbb{R}^n)$, there is a constant $M > 0$ such that

$$|f(t,x)| \leq M(1 + |x|).$$

but

$$\begin{cases} \dot{x} = f(t,x), \\ x(a) = x_0, \end{cases}$$

There is a solution $x \in C^1([a,b], \mathbb{R}^n)$.

Proof. Only require the integral equation

$$x(t) = x_0 + \int_a^t f(s, x(s))\,\mathrm{d}s$$

Continuous solution of , that is, $T : C([a,b], \mathbb{R}^n) \to C([a,b], \mathbb{R}^n)$,

$$(Tx)(t) = x_0 + \int_a^t f(s, x(s))\,\mathrm{d}s$$

The fixed point of . It can be proved that T is compact.

If $x \in C([a,b], \mathbb{R}^n)$ and $\lambda \in (0,1)$ so that $x = \lambda Tx$. Then

$$|x(t)| \leq |(Tx)(t)| \leq |x_0| + \int_a^t |f(s, x(s))|\,\mathrm{d}s$$

$$\leq |x_0| + M(b-a) + M\int_a^t |x(s)|\,\mathrm{d}s, \qquad t \in [a,b].$$

By Gronwall's inequality,

$$|x(t)| \leq C := (|x_0| + M(b-a))e^{M(b-a)}, \quad t \in [a,b].$$

Therefore, $\|x\| = \max_{t\in[a,b]} |x(t)| \leq C$. By Theorem 40, T has a fixed point. □

Let Ω be a bounded open set in $\mathbb{R}^n$ with smooth boundary $\partial\Omega$,

$$L = a^{ij}\partial_{ij} + b^i\partial_i$$

be a uniformly elliptic operator on Ω, i.e., $a^{ij}, b^i \in C^\infty(\overline{\Omega})$, and there exists $\lambda > 0$ such that

$$a^{ij}(x)\xi_i\xi_j \geq \lambda|\xi|^2, \quad \forall (x,\xi) \in \overline{\Omega}\times\mathbb{R}^n.$$

By the L^p-theory of elliptic equations, for $p > 1$, when $f \in L^p(\Omega)$,

$$Lu = f, \quad u|_{\partial\Omega} = 0.$$

$$u|_{\partial\Omega} = 0$$

There exists a unique solution $u \in W^{2,p}(\Omega)\cap W_0^{1,p}(\Omega)$. Furthermore, there exists a constant $c_0 > 0$ independent of f and u such that

$$\|u\|_{2,p} \leq c_0|f|_p.$$

If $p > n$, due to the compact embedding $W_0^{1,p}(\Omega) \to C(\overline{\Omega})$, there also exists a constant $c_1 > 0$ independent of f and u such that

$$\|u\|_{C(\overline{\Omega})} \leq c_1|f|_p. \tag{4.13}$$

Example 10. Let $g : \overline{\Omega}\times\mathbb{R} \to \mathbb{R}$ satisfy the Carathéodory conditions, and $p > n$. For some $a > 0$ and $b \in (0, c_1^{-1}|\Omega|^{-1/p})$, we have

$$|g(x,t)| \leq a + b|t|, \qquad \forall (x,t) \in \overline{\Omega}\times\mathbb{R}.$$

Then

$$Lu = g(x,u), \qquad u|_{\partial\Omega} = 0 \tag{4.14}$$

has a solution $u \in W^{2,p}(\Omega) \cap W_0^{1,p}(\Omega) \cap C(\overline{\Omega})$.

Proof. We construct a mapping $T : C(\overline{\Omega}) \to C(\overline{\Omega})$ as follows. When $u \in C(\overline{\Omega})$,

$$g(x,u) \in C(\overline{\Omega}) \subset L^p(\Omega).$$

As mentioned above,

$$Lv = g(x,u), \qquad v|_{\partial\Omega} = 0$$

has a unique solution in $W^{2,p}(\Omega) \cap W_0^{1,p}(\Omega) \cap C(\overline{\Omega})$. Let's denote it as $v = Tu$. By (4.13) The mapping $u \mapsto g(\cdot,u)$ is continuous from $C(\overline{\Omega})$ to $L^p(\Omega)$, so T is continuous. By the compactness of the Sobolev embedding $W_0^{1,p}(\Omega) \to C(\overline{\Omega})$, T is also compact.

If $(\lambda, u) \in (0,1)\times C(\overline{\Omega})$ such that $u = \lambda Tu$, then

$$\begin{aligned}\|u\|_{C(\overline{\Omega})} &\leq \|Tu\|_{C(\overline{\Omega})} \leq c_1|g(\cdot,u)|_p \\ &\leq c_1|a + b|u||_p \leq c_1(|a|_p + |bu|_p) \\ &\leq c_1(a|\Omega|^{1/p} + b|\Omega|^{1/p}\|u\|_{C(\overline{\Omega})}).\end{aligned}$$

Therefore,

$$\|u\|_{C(\overline{\Omega})} \leq \frac{ac_1|\Omega|^{1/p}}{1 - bc_1|\Omega|^{1/p}}.$$

By Theorem 40, T has a fixed point, which is the solution to (4.14). the solution. $\square$

Example 11. Let Ω be a bounded open set in $\mathbb{R}^n$ with a smooth boundary $\partial\Omega$, $b \in C^\infty(\mathbb{R}^n, \mathbb{R})$ and there exists a constant $C > 0$ such that

$$|b(p)| \le C(1 + |p|), \qquad \forall p \in \mathbb{R}^n. \tag{4.15}$$

Then for sufficiently large $\mu > 0$,

$$-\Delta u + b(\nabla u) + \mu u = 0, \qquad u|_{\partial\Omega} = 0 \tag{4.16}$$

has a solution $u \in H^2(\Omega) \cap H_0^1(\Omega)$.

Proof. For $u \in H_0^1(\Omega)$, from (4.15), we know that $f = -b(\nabla u) \in L^2(\Omega)$. Thus,

$$-\Delta w + \mu w = f, \quad w|_{\partial\Omega} = 0$$

has a unique weak solution $w \in H_0^1(\Omega)$. Let $Au = w$, then we obtain a mapping $A : H_0^1(\Omega) \to H_0^1(\Omega)$. From L^2-theory and (4.15), we also know that $w = Au \in H^2(\Omega)$ and there exists $C_1 > 0$ such that

$$\|Au\|_2 \le C_1|f|_2 \le C_2(1 + \|u\|). \tag{4.17}$$

Next, we prove that $A : H_0^1(\Omega) \to H_0^1(\Omega)$ is compact. If A is not continuous at some $u \in H_0^1(\Omega)$, then there exists $\varepsilon_0 > 0$ and $\{u_k\} \subset H_0^1(\Omega)$ such that

$$\|u_k - u\| \to 0, \quad \|Au_k - Au\| \ge \varepsilon_0. \tag{4.18}$$

Let $w_k = Au_k$. From (4.17),... Since $\sup_k \|w_k\|_2 < \infty$, and the embedding $H^2(\Omega) \to H^1(\Omega)$ is compact, there exists $w \in H^1(\Omega)$ such that

$$\|w_k - w\| \to 0. \tag{4.19}$$

For all $\phi \in H_0^1(\Omega)$, letting $k \to \infty$ in

$$\int \nabla w_k \nabla\phi + \mu \int w_k\phi = -\int b(\nabla u_k)\phi$$

we obtain

$$\int \nabla w \nabla\phi + \mu \int w\phi = -\int b(\nabla u)\phi.$$

That is, $w = Au$, and (4.19) becomes $Au_k \to Au$. This contradicts (4.18).

For a bounded sequence of points $\{u_k\}$ in $H_0^1(\Omega)$, from (4.17) we know that $\{Au_k\}$ is bounded in $H^2(\Omega)$. Since the embedding $H^2(\Omega) \to H^1(\Omega)$ is compact, there exists $w \in H^1(\Omega)$ such that

$$\|Au_k - Au\| \to 0.$$

However, $H_0^1(\Omega)$ is a closed subspace of $H^1(\Omega)$, so $w \in H_0^1(\Omega)$. That is, $\{Au_k\}$ converges in $H_0^1(\Omega)$.

If there is $(\lambda, u) \in (0, 1) \times H_0^1(\Omega)$ so that $u = \lambda Au$, that is

$$-\Delta u + \mu u = -\lambda b(\nabla u). \tag{4.20}$$

because

$$C\int |u| \le \frac{1}{2}\int u^2 + \frac{1}{2}\int C^2 = \frac{1}{2}\int u^2 + \frac{C^2}{2}|\Omega|,$$

$$C\int |\nabla u||u| \le \frac{1}{2}\int |\nabla u|^2 + \frac{C^2}{2}\int u^2,$$

Multiply both sides of (4.20) by u and integrate over Ω to get

$$\int |\nabla u|^2 + \mu \int u^2 = -\lambda \int b(\nabla u)u \le C \int (1 + |\nabla u|)|u|$$
$$\le \frac{1}{2}\int u^2 + \frac{C^2}{2}|\Omega| + \frac{1}{2}\int |\nabla u|^2 + \frac{C^2}{2}\int u^2.$$

Right now

$$\frac{1}{2}\int |\nabla u|^2 + \left(\mu - \frac{C^2+1}{2}\right)\int u^2 \le \frac{C^2}{2}|\Omega|.$$

Therefore, when $\mu \ge 2^{-1}(1 + C^2)$,

$$\|u\| = \left(\int |\nabla u|^2\right)^{1/2} \le C\sqrt{|\Omega|}.$$

By Theorem 40, A has a fixed point, i.e., (4.16) has a solution. □

5. Critical Point Theory

Let X be a Banach space, and $f \in C^1(X, \mathbb{R})$. We know what a critical point of f is, and we have also seen that solutions to nonlinear differential equations can be obtained by finding critical points of specific functionals.

For functionals bounded below, the method of minimizing sequences can be used to find their extremum points. These are the simplest critical points. To deal with functionals that are not bounded below, and to find critical points that are not extremum points, the minimizing sequence method is clearly insufficient. In this case, the main method for determining the existence of critical points is to examine the change in the topological structure of the level sets $f_c = f^{-1}(-\infty, c]$ as $c \in \mathbb{R}$ varies.

Let $\mathcal{K} = \{u \in X \mid f'(u) = 0\}$, $K_c = \mathcal{K} \cap f^{-1}(c)$, $\tilde{X} = X \setminus \mathcal{K}$.

5.1. Deformation Lemma.

Definition 18. *Let $f \in C^1(X, \mathbb{R})$. $V \in C^{1-0}(\tilde{X}, X)$ is called a pseudo-gradient vector field of f if*

(i) $\|V(u)\| \le 2\|f'(u)\|$,
(ii) $\langle f'(u), V(u)\rangle \ge \|f'(u)\|^2$.

Theorem 41 (Palais)**.** *Let $f \in C^1(X, \mathbb{R})$, then f*

There exists a pseudo-gradient vector field.

Proof. (1) For all $x \in \tilde{X}$, since $f'(x) \ne 0$, there exists $w \in X$ such that $\|w\| = 1$,

$$\langle f'(x), w\rangle > \frac{2}{3}\|f'(x)\|.$$

Let $v = \frac{3}{2}\|f'(x)\|w$, then

$$\|v\| < 2\|f'(x)\|, \text{ and } \langle f'(x), v\rangle > \|f'(x)\|^2.$$

Since f' is continuous, there exists an open neighborhood U_x of x such that for all $y \in U_x$,

$$\|v\| < 2\|f'(y)\|, \text{ and } \langle f'(y), v\rangle > \|f'(y)\|^2. \tag{5.1}$$

The family of sets $\{U_x \mid x \in \tilde{X}\}$ is an open cover of the paracompact space $\tilde{X}$, therefore there exists a locally finite open refinement cover $\{O_\alpha \mid \alpha \in \Lambda\}$. Let

$$\rho_\alpha(x) = \operatorname{dist}(x, X \setminus O_\alpha), \qquad \alpha \in \Lambda.$$

Then $\rho_\alpha \in C^{1-0}(\tilde{X}, \mathbb{R})$. For each $\alpha \in \Lambda$, there exists $x \in \tilde{X}$ such that $O_\alpha \subset U_x$.

Let v_α be the vector determined by $x \in \tilde{X}$ according to (1) satisfying (5.1), and let

$$\beta_\alpha(x) = \frac{\rho_\alpha(x)}{\sum_{\gamma\in\Lambda} \rho_\gamma(x)}.$$

Since $\{O_\alpha | \alpha \in \Lambda\}$ is a locally finite open cover of $\tilde{X}$, $\sum_{\gamma\in\Lambda} \rho_\gamma(x) > 0$ and for each $x \in \tilde{X}$ it is a finite sum. Let

$$v(x) = \sum_{\alpha\in\Lambda} \beta_\alpha(x) v_\alpha,$$

Then $v \in C^{1-0}(\tilde{X}, X)$ is a pseudo-gradient vector field of f:

$$\|v(x)\| \le \sum_{\alpha\in\Lambda} \beta_\alpha(x)\|v_\alpha\| \le 2\|f'(x)\|,$$

$$\langle f'(x), v(x)\rangle = \sum_{\alpha\in\Lambda} \langle f'(x), v_\alpha\rangle \beta_\alpha(x) \ge \|f'(x)\|^2.$$

□

Definition 19. *Let $c \in \mathbb{R}$. A sequence $\{u_n\} \subset X$ is called a $(PS)_c$ sequence for $f \in C^1(X, \mathbb{R})$ if $f(u_n) \to c$ and $f'(u_n) \to 0$. We say that f satisfies the $(PS)_c$ condition if every $(PS)_c$ sequence has a convergent subsequence. If f satisfies the $(PS)_c$ condition for all $c \in \mathbb{R}$, then we say that f satisfies the (PS) condition.*

Lemma 10. *If $f \in C^1(X, \mathbb{R})$ satisfies the $(PS)_c$ condition, then $\mathcal{K}_c$ is compact.*

Proof. Let $\{u_n\} \subset K_c$. Then $f(u_n) = c$ and $f'(u_n) = 0$. That is, $\{u_n\}$ is a $(PS)_c$ sequence of f, and thus it has a convergent subsequence. By the continuity of f and f', this subsequence converges to some $u \in K_c$. □

Exercise 15. If $f \in C^1(X, \mathbb{R})$ satisfies the (PS) condition, and $a, b \in \mathbb{R}$. If $f^{-1}[a, b] \cap K = \emptyset$, then there exists $\varepsilon > 0$ such that

$$\|f'(u)\| \ge \varepsilon, \qquad \forall x \in f^{-1}[a - \varepsilon, b + \varepsilon].$$

Theorem 42 (First Deformation Lemma)**.** *Let $c \in \mathbb{R}$, $\overline{\varepsilon} > 0$. If $f \in C^1(X, \mathbb{R})$ satisfies the $(PS)_c$ condition and $\mathcal{K}_c \subset N^\circ$, then there exist $\varepsilon \in (0, \overline{\varepsilon})$ and $\eta \in C([0, 1] \times X, X)$ such that*

(i) $\eta(0, \cdot) = \boldsymbol{1}_X$,
(ii) $\eta(t, u) = u$, $\forall (t, u) \in [0, 1] \times (X \setminus f^{-1}[c - \overline{\varepsilon}, c + \overline{\varepsilon}])$,
(iii) $\eta(1, f_{c+\varepsilon} \setminus N) \subset f_{c-\varepsilon}$,
(iv) For $\forall u \in X$, $f(\eta(\cdot, u)) : [0, 1] \to \mathbb{R}$ *decreases.*

Proof. Suppose $A \subset X$, let $\rho > 0$

$$A^\rho = \{u \in X | \operatorname{dist}(u, A) < \rho\}.$$

Since $\mathcal{K}_c$ is compact, With $\delta > 0$, $\mathcal{K}_c^{3\delta} \subset N$. Since f satisfies $(PS)_c$, there exists $\varepsilon \in (0, 2^{-1}\overline{\varepsilon})$, when

$$u \in A := f^{-1}[c - 2\varepsilon, c + 2\varepsilon] \setminus \mathcal{K}_c^\delta$$

When, $\|f'(u)\| \ge 8\varepsilon/\delta$.

Let $B = f^{-1}[c - \varepsilon, c + \varepsilon] \setminus \mathcal{K}_c^{2\delta}$,

$$\psi(u) = \frac{\operatorname{dist}(u, X \setminus A)}{\operatorname{dist}(u, X \setminus A) + \operatorname{dist}(u, B)},$$

$$\Phi(u) = \begin{cases} 0, & u \notin A, \\ -\psi(u)\|V(u)\|^{-2}V(u), & u \in A. \end{cases}$$

where V is the pseudo-gradient vector field of f. Then $\Phi \in C^{1-0}(X, X)$,

$$\|\Phi(u)\| \le \frac{1}{\|V(u)\|} \le \frac{1}{\|f'(u)\|} \le \frac{\delta}{8\varepsilon}, \qquad \forall u \in X. \tag{5.2}$$

Therefore, $\sigma : [0, +\infty) \times X \to X$ satisfies

$$\dot{\sigma}(t, u) = \Phi(\sigma(t, u)), \qquad \sigma(0, u) = u.$$

Then when $u \in X \setminus f^{-1}[Whenc - \overline{\varepsilon}, c + \overline{\varepsilon}] \subset X \setminus A$, $\Phi(u) = 0$, so due to the uniqueness of the solution to the ordinary differential equation, for $\forall t \in [0, +\infty)$ has $\sigma(t, u) = u$.

For $\forall u \in X$,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} f(\sigma(t, u)) &= \langle f'(\sigma(t, u)), \dot{\sigma}(t, u)\rangle = \langle f'(\sigma(t, u)), \Phi(\sigma(t, u))\rangle \\ &\le \begin{cases} -4^{-1}\psi(\sigma(t, u)), & \text{if } \sigma(t, u) \in A, \\ 0, & \text{if} \sigma(t, u) \notin A. \end{cases} \end{aligned} \tag{5.3}$$

That is $f(\sigma(\cdot, u)) : [0, +\infty) \to \mathbb{R}$ is decreasing.

Let $u \in f_{c+\varepsilon} \setminus \mathcal{K}_c^{3\delta}$. If $t \in [0, 8\varepsilon]$ makes $f(\sigma(t, u)) \le c - \varepsilon$, then $f(\sigma(8\varepsilon, u)) \le c - \varepsilon$; If there is no such t, Since $f(\sigma(\cdot, u))$ decreases,

$$\sigma(t, u) \in f^{-1}[c - \varepsilon, c + \varepsilon], \qquad \forall t \in [0, 8\varepsilon]. \tag{5.4}$$

Then from (5.2) we have

$$\begin{aligned} \|\sigma(t, u) - u\| &= \left\| \int_0^t \Phi(\sigma(\tau, u))\, \mathrm{d}\tau \right\| \\ &\le \int_0^{8\varepsilon} \|\Phi(\sigma(\tau, u))\|\, \mathrm{d}\tau \le \delta, \qquad \forall t \in [0, 8\varepsilon]. \end{aligned}$$

Therefore, from $u \notin \mathcal{K}_c^{3\delta}$ we know $\sigma((t, u) \notin \mathcal{K}_c^{2\delta}$. Combining this with (5.4), we know that $\sigma(t, u) \in B$. Therefore, from (5.3), we have

$$\begin{aligned} f(\sigma(8\varepsilon, u)) &= f(\sigma(0, u)) + \int_0^{8\varepsilon} \frac{\mathrm{d}}{\mathrm{d}t} f(\sigma(t, u))\, \mathrm{d}t \\ &\le f(u) - \frac{1}{4}\int_0^{8\varepsilon} \psi(\sigma(t, u))\, \mathrm{d}t \\ &= f(u) - \frac{1}{4} \cdot 8\varepsilon \le c - \varepsilon. \end{aligned}$$

This proves that $\sigma(8\varepsilon, f_{c+\varepsilon} \setminus \mathcal{K}_c^{3\delta}) \subset f_{c-\varepsilon}$. Since $\mathcal{K}_c^{3\delta} \subset N$, we have $\sigma(8\varepsilon, f_{c+\varepsilon} \setminus N) \subset f_{c-\varepsilon}$.

Let $\eta(t, u) = \sigma(8\varepsilon t, u)$. Then $\eta : [0, 1] \times X \to X$ satisfies the requirements of the theorem. □

Remark 14. (1) The above η is clearly defined on $[0, +\infty) \times X$, and for all $t \in [0, +\infty)$, $\eta(t, \cdot) : X \to X$ is a homeomorphism. This η is generated by a pseudo-gradient vector field, and is therefore called a pseudogradient flow.

(2) If f is even, i.e., $f(-u) = f(u)$, then $f' : X \to X^*$ is odd, and thus $\tilde{X} = -\tilde{X}$. If $V : \tilde{X} \to X$ is a pseudogradient vector field of f, let

$$\tilde{V}(u) = \frac{V(u) - V(-u)}{2}.$$

Then $\tilde{V} : \tilde{X} \to X$ is an odd pseudogradient vector field of f. The $\eta : [0,1] \times X \to X$ obtained from $\tilde{V}$ in the above manner, in addition to satisfying (i)–(iv), also has the symmetry property:

(v) $\eta(t, -u) = -\eta(t, u)$, $\forall (t, u) \in [0,1] \times X$.

The above deformation lemma is only a local result near level c. We can also prove the following global result, which shows that if $f^{-1}[a, b]$ contains no critical points of f, then f_a is a strong deformation retract of f_b.

Theorem 43. *Let $f \in C^1(X, \mathbb{R})$*

Satisfy(PS) condition, $a, b \in \mathbb{R}$. If $f^{-1}[a, b] \cap \mathcal{K} = \emptyset$, then there is $\eta \in C([0,1] \times f_b, f_b)$ so that

(i) $\eta(0, \cdot) = \mathbf{1}_{f_b}$, $\eta(1, f_b) \subset f_a$,

(ii) $\eta(t, x) = x$, $\forall (t, x) \in [0,1] \times f_a$.

Proof. Because f satisfies the (PS) condition, $\exists \varepsilon > 0$, When $u \in f^{-1}[a - \varepsilon, b + \varepsilon]$, $\|f'(u)\| \geq \varepsilon$.

Let V be the pseudo-gradient field of f, consider

$$\dot{\sigma}(t, u) = -\frac{V(\sigma(t, u))}{\|V(\sigma(t, u))\|^2}, \qquad \sigma(0, u) = u \in f^{-1}[a, b] \tag{5.5}$$

The maximum solution of $\sigma(\cdot, u) : [0, T_u) \to X$. When $\sigma(t, u) \in f^{-1}[a-\varepsilon, b+\varepsilon]$ sometimes

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} f(\sigma(t, u)) &= \langle f'(\sigma(t, u)), \dot{\sigma}(t, u) \rangle \\ &= -\frac{\langle f'(\sigma(t, u)), V(t, u) \rangle}{\|V(t, u)\|^2} \leq -\frac{1}{4}, \end{aligned} \tag{5.6}$$

That is $f(\sigma(\cdot, u)) : [0, T_u) \to \mathbb{R}$ is strictly decreasing. Thus

$$\sigma(t, f_b) \subset f_b, \qquad \forall t \in [0, T_u).$$

And because

$$f(\sigma(t, u)) - f(u) = \int_0^t \frac{\mathrm{d}}{\mathrm{d}\tau} f(\sigma(\tau, u))\, \mathrm{d}\tau \leq -\frac{t}{4}, \tag{5.7}$$

There must be $r \in [0, T_u)$ such that $f(\sigma(r, u)) = a$. In fact, if $\forall r \in [0, T_u)$ has $f(\sigma(r, u)) > a$, Then from (5.7) we know $T_u < +\infty$. Since

$$\|\sigma(t, u) - \sigma(s, u)\| = \left\| \int_s^t \dot{\sigma}(\tau, u)\, \mathrm{d}\tau \right\| \leq \left| \int_s^t \frac{d\tau}{\|V(\sigma(\tau, u))\|} \right| \leq \frac{1}{\varepsilon} |t - s|.$$

Therefore, the limit $u_* = \lim_{t \to T_u -} \sigma(t, u)$ exists and $u_* \in f^{-1}[a, b]$. Solving (5.5) with u_* as the initial value, we see that σ can be extended further, which contradicts the maximality of T_u.

Since $f(\sigma(\cdot, u))$ is strictly decreasing, the value $r = r_u$ satisfying $f(\sigma(r, u)) = a$ is unique. From (5.6), we have

$$\left. \frac{\partial}{\partial t} \right|_{t = r_u} f(\sigma(t, u)) \leq -\frac{1}{4},$$

By the implicit function theorem, $r = r_u$ is a continuous function of u. Note that when $u \in f^{-1}(a)$, $r_u = 1$.

Let $\eta : [0,1] \times f_b \to f_b$,

$$\eta(t,u) = \begin{cases} u, & u \in f_a, \\ \sigma(r_u t, u), & u \in f^{-1}[a,b]. \end{cases}$$

By the continuity of the solution of the ordinary differential equation with respect to the initial value and the continuity of r_u, we know that η is continuous on $[0,1] \times f_a$ and $[0,1] \times f^{-1}[a,b]$. Therefore, by the gluing lemma, $\eta \in C([0,1] \times f_b, f_b)$. Clearly, η satisfies (i) and (ii). □

If a is a critical value of f, Theorem 43 still holds, and we have the following generalization.

Theorem 44 (Second Deformation Lemma)**.** *Let $f \in C^1(X,\mathbb{R})$ satisfy the (PS) condition, and f has only a finite number of critical points. Let a be the unique critical value of f in $[a,b)$. Then there exists a continuous $\eta : [0,1] \times (f_b \setminus \mathcal{K}_b) \to f_b \setminus \mathcal{K}_b$ such that*

(i) $\eta(0,\cdot) = 1_{f_b \setminus \mathcal{K}_b}$,
(ii) $\eta(t,u) = u$, $\forall (t,u) \in [0,1] \times f_a$,
(iii) $\eta(1, f_b \setminus \mathcal{K}_b) \subset f_a$,
(iv) For all $u \in f_b \setminus \mathcal{K}_b$, $f(\eta(\cdot,u)) : [0,1] \to \mathbb{R}$ is decreasing.

This theorem is crucial for establishing Morse inequalities. Its proof can be found in [**?**] or [**?**, page 137], the latter requiring $f \in C^{2-0}(X,\mathbb{R})$.

5.2. Linking Theorem.

Definition 20. *Let S be a closed set in X, and Q be a submanifold of X with boundary ∂Q. We say that S and ∂Q are linked if*

(i) $S \cap \partial Q = \emptyset$,
(ii) $\varphi(Q) \cap S \neq \emptyset$ *for any continuous map* $\varphi : Q \to X$ *satisfying* $\varphi|_{\partial Q} = 1_{\partial Q}$.

Example 12. Let $S = \{u \in X \mid \|u\| = r\}$, $Q = [0,e]$, where $e \in X$, $\|e\| > r$. Then S and $\partial Q = \{0, e\}$ are linked.

Example 13. Let $X = X^- \oplus X^+$, $\dim X^- < \infty$, $R > 0$. Take $S = X^+$,

$$Q = \{u \in X^- \mid \|u\| \le R\}.$$

Then S and $\partial Q = \{u \in X^- \mid \|u\| = R\}$ are linked.

Proof. Clearly $S \cap \partial Q = \emptyset$. Let $\varphi \in C(Q,X)$ such that $\varphi|_{\partial Q} = \mathbf{1}_{\partial Q}$. Let $P : X \to X^-$ be the projection. Then $P\varphi|_{\partial Q} = \mathbf{1}_{\partial Q}$. Therefore, by the boundary value property of the Brouwer degree,

$$\deg(P\varphi, Q^\circ, 0) = \deg(\mathbf{1}_Q, Q^\circ, 0) = 1.$$

Thus, there exists $u \in Q^\circ$ such that $P\varphi(u) = 0. = 0$, i.e., $\varphi(u) \in X^+ = S$. This proves that $\varphi(Q) \cap S \neq \emptyset$. □

Theorem 45 (Linking Theorem)**.** *Let $f \in C^1(X,\mathbb{R})$ satisfy the (PS) condition, and let S and ∂Q be linked such that*

$$\alpha = \inf_S f > \sup_{\partial Q} f$$

Then

$$c = \inf_{\varphi \in \mathcal{A}} \sup_{u \in Q} f(\varphi(u)) \ge \alpha$$

is a critical value of f, where $\mathcal{A} = \{\varphi \in C(Q,X) \mid \varphi|_{\partial Q} = \mathbf{1}_{\partial Q}\}$.

Proof. Since for all $\varphi \in \mathcal{A}$, $\varphi(Q) \cap S \neq \emptyset$, it is clear that $c \geq \alpha$. Since f satisfies the (PS) condition, if c is not a critical value of f, then $\exists \varepsilon > 0$ such that $f^{-1}[c-\varepsilon, c+\varepsilon] \cap K = \emptyset$. Since $c \geq \alpha > \sup_{\partial Q} f$, we can also require $c - \varepsilon \geq \sup_{\partial Q} f$, i.e., $\partial Q \subset f_{c-\varepsilon}$.

By the deformation lemma, there exists a continuous map $\eta : [0,1] \times f_{c+\varepsilon} \to f_{c+\varepsilon}$ such that

$$\eta(1, f_{c+\varepsilon}) \subset f_{c-\varepsilon}, \qquad \eta(1,\cdot)|_{f_{c-\varepsilon}} = \mathbf{1}_{f_{c-\varepsilon}}. \tag{5.8}$$

Let $\varphi \in \mathcal{A}$ such that $\sup_{u \in Q} f(\varphi(u)) < c + \varepsilon$, i.e., $\varphi(Q) \subset f_{c+\varepsilon}$. Let $\psi(\cdot) = \eta(1, \varphi(\cdot))$. Then $\psi \in C(Q, X)$. When $u \in \partial Q \subset f_{c-\varepsilon}$, from (5.8) and noting that $\varphi(u) = u$, we get $\psi(u) = u$, i.e., $\psi|_{\partial Q} = \mathbf{1}_{\partial Q}$. Therefore, $\psi \in \mathcal{A}$. Thus

$$\psi(Q) = \eta(1, \varphi(Q)) \subset \eta(1, f_{c+\varepsilon}) \subset f_{c-\varepsilon},$$

which implies $\sup_{u \in Q} f(\psi(u)) \leq c - \varepsilon$. This contradicts the definition of c. □

Applying Theorem 45 to Examples 12 and 13, we obtain the most famous results in minimax theory.

Theorem 46 (Mountain Pass Lemma, Ambrosetti-Rabinowitz 1973)**.** *Let $f \in C^1(X, \mathbb{R})$ satisfy the (PS) condition. If there exist $r > 0$ and $e \in X$ such that $\|e\| > r$,*

$$\alpha = \inf_{\|u\|=r} f(u) > \max\{f(0), f(e)\},$$

then

$$c = \inf_{\gamma \in \mathcal{A}} \sup_{u \in Q} f(\gamma(u)) \geq \alpha$$

is a critical value of f, where $\mathcal{A} = \{\gamma \in C([0,1], X) | \, \gamma(0) = 0, \, \gamma(1) = e\}$.

Theorem 47 (Saddle Point Theorem, Rabinowitz 1978)**.** *Let $f \in C^1(X, \mathbb{R})$ satisfy the (PS) condition, $X = X^- \oplus X^+$. If $\dim X^- < \infty$ and there exists $R > 0$ such that*

$$\alpha = \inf_{u \in X^+} f > \sup_{u \in X^-, \|u\|=R} f(u).$$

Then f

There exists a critical value $c \geq \alpha$.

In some problems, the following form of linking is encountered.

Example 14. Let $X = X^- \oplus X^+$, $\dim X^- < \infty$, $e \in X^+$, $\|e\| = 1$. $R > \rho > 0$. Let

$$S = \left\{u \in X^+ | \, \|u\| = \rho\right\},$$
$$Q = \{u = u^- + te | \, u^- \in X^-, t \geq 0, \|u\| \leq R\}.$$

Then S and ∂Q are linked.

Proof. Clearly $S \cap \partial Q = \emptyset$. Let $\varphi \in C(Q, X)$ such that $\varphi|_{\partial Q} = \mathbf{1}_{\partial Q}$. Let $P : X \to X^-$ be the projection. Let $g : Q \to \mathbb{R}e \oplus X^-$,

$$g(u) = \|(\mathbf{1}_X - P)\varphi(u)\|e + P\varphi(u).$$

Then from $\varphi|_{\partial Q} = \mathbf{1}_{\partial Q}$, we have $g|_{\partial Q} = \mathbf{1}_{\partial Q}$. Therefore

$$\deg(g, Q^\circ, \rho e) = \deg(\mathbf{1}_Q, Q^\circ, \rho e) = 1.$$

Thus there exists $u \in Q$ such that $g(u) = \rho e$. Therefore $\varphi(u) \in X^+$, $\|\varphi(u)\| = \rho$. That is, $\varphi(u) \in S$, $\varphi(Q) \cap S \neq \emptyset$. □

When $X^- = \{0\}$, this example degenerates to the case of Example 12. Therefore, the critical point theorem corresponding to this example is a generalization of the mountain pass lemma, called the generalized mountain pass lemma.

5.3. Applications of Differential Equations. Let Ω be a bounded domain in $\mathbb{R}^m$ with a smooth boundary $\partial\Omega$. We study the existence of solutions to the semilinear elliptic Dirichlet problem

$$-\Delta u = g(x,u), \qquad u|_{\partial\Omega} = 0 \tag{5.9}$$

In studying (5.9), the following eigenvalue problem

$$-\Delta u = \lambda u, \qquad u|_{\partial\Omega} = 0 \tag{5.10}$$

is very important. We know that (5.10) has a sequence of eigenvalues

$$\lambda_1 < \lambda_2 \le \lambda_3 \le \cdots, \qquad \lim_{n\to\infty} \lambda_n = +\infty,$$

where each eigenvalue has finite multiplicity and is listed according to its multiplicity. Let ϕ_n be the eigenfunction corresponding to λ_n satisfying $|\phi_n|_2 = 1$. When $k \neq n$, ϕ_k and ϕ_n are orthogonal in both $H_0^1(\Omega)$ and $L^2(\Omega)$. We also know that $\lambda_1 > 0$ is simple, and ϕ_1 does not take zero values and does not change sign in Ω, so we can assume $\phi_1 > 0$.

Furthermore, let $X_0 = \{0\}$, $X_n = \operatorname{span}\{\phi_1, \cdots, \phi_n\}$. Then

$$\lambda_n = \inf_{u \in X_{n-1}^{\perp} \setminus 0} \frac{\int |\nabla u|^2}{\int |u|^2}, \qquad n = 1, 2, \cdots. \tag{5.11}$$

Where $X_{n-1}^{\perp}$ is the orthogonal complement of X_{n-1} in $H_0^1(\Omega)$. From (5.11) we also know that for $n = 0, 1, 2, \cdots$,

$$\begin{aligned} \int |\nabla u|^2 &\le \lambda_n \int |u|^2, \quad \text{when} u \in X_n, \\ \int |\nabla u|^2 &\ge \lambda_{n+1} \int |u|^2, \quad \text{when} u \in X_n^{\perp}. \end{aligned} \tag{5.12}$$

Example 15. Let $g \in C(\Omega \times \mathbb{R}, \mathbb{R})$ satisfies

(g_1) there exists $\theta \in (2, 2^*)$ makes $|g(x,t)| \le C(1 + |t|^{\theta-1})$.

(g_2) there exists $\mu > 2$, $R > 0$, such that

$$0 < G(x,t) := \int_0^t g(x,s)\mathrm{d}s \le \frac{1}{\mu} g(x,t)t \qquad \text{for } |t| \ge R.$$

(g_0) $\lambda = \lim_{t\to 0} \frac{G(x,t)}{t^2} < \frac{\lambda_1}{2}$.

Then equation (5.9) has two non-zero weak solutions $u \ge 0$, $v \le 0$.

Proof. make

$$g_+(x,t) = \begin{cases} g(x,t), & t > 0, \\ 0, & t \le 0. \end{cases}$$

but

$$-\Delta u = g_+(x,u), \qquad u|_{\partial\Omega} = 0$$

The solution of is exactly C^1- functional $f_+ : H_0^1(\Omega) \to \mathbb{R}$,

$$f_+(u) = \frac{1}{2}\int |\nabla u|^2 - \int G_+(x,u)$$

critical point. Where $G_+(x,t) = \int_0^t g_+(x,s)\,\mathrm{d}s$.

Take $\varepsilon \in (\lambda, 2^{-1}\lambda_1)$. From (g_1), (g_0), There exists $C_\varepsilon > 0$,

$$|G_+(x,t)| \le \varepsilon |t|^2 + C_\varepsilon |t|^\theta.$$

So from (5.12) and continuous embedding $H_0^1(\Omega) \to L^\theta(\Omega)$,

$$\begin{aligned} f_+(u) &\geq \frac{1}{2}\int |\nabla u|^2 - \varepsilon \int |u|^2 - C_\varepsilon \int |u|^\theta \\ &\geq \left(\frac{1}{2} - \frac{\varepsilon}{\lambda_1}\right)\int |\nabla u|^2 - C_\varepsilon \int |u|^\theta \\ &\geq \left(\frac{1}{2} - \frac{\varepsilon}{\lambda_1}\right)\|u\|^2 - C_\varepsilon \|u\|^\theta. \end{aligned} \tag{5.13}$$

From $\theta > 2$ we know $\exists r > 0$, when $\|u\| = r$ when $f_+(u) > 0 = f_+(0)$.

And from (g_2), there exists $C > 0$ so that

$$G_+(x,t) \geq C(|t|^\mu - 1), \qquad \forall t > 0.$$

For $s > 0$, since $\mu > 2$, when $s \to +\infty$

$$\begin{aligned} f_+(s\phi_1) &= \frac{s^2}{2}\int |\nabla \phi_1|^2 - \int G_+(x, s\phi_1) \\ &\leq \frac{s^2}{2}\|\phi_1\|^2 - \frac{s^\mu C}{2}|\phi_1|_\mu^\mu - C|\Omega| \to -\infty. \end{aligned}$$

Therefore, for sufficiently large $s > 0$, take $e = s\phi_1$ and we have

$$\alpha = \inf_{\|u\|=r} f_+(u) > \max\{f_+(0), f_+(e)\}.$$

We will prove that f_+ satisfies the (PS) condition. Then, by the mountain pass lemma, f_+ has a critical value $c \geq \alpha > f_+(0)$, with a corresponding critical point $u \neq 0$. From

$$\|u_-\|^2 = \int \nabla u \nabla u_- - \int g_+(x,u)u_- = \langle f_+'(u), u_-\rangle = 0$$

we have $u_- = 0$, so $u \geq 0$. Thus u is a non-zero solution of (5.9). Similarly, we can obtain another non-zero solution $v \leq 0$ of (5.9).

Now we verify that f_+ satisfies the (PS) condition. Suppose $f_+(u_n) \to c$, $f'(u_n) \to 0$. Let

$$M = \sup_{(x,t)\in\Omega\times[-R,R]} |\mu G_+(x,u_n) - g_+(x,u_n)u_n|.$$

Then for large n, by (g_2), we have

$$\begin{aligned} c + 1 + \|u_n\| &\geq f_+(u_n) - \frac{1}{\mu}\langle f_+'(u_n), u_n\rangle \\ &= \left(\frac{1}{2} - \frac{1}{\mu}\right)\int |\nabla u|^2 - \frac{1}{\mu}\int (\mu G_+(x,u_n) - g_+(x,u_n)u_n) \\ &\geq \left(\frac{1}{2} - \frac{1}{\mu}\right)\|u_n\|^2 - \frac{1}{\mu}\int_{|u_n|\leq R} (\mu G_+(x,u_n) - g_+(x,u_n)u_n) \\ &\geq \left(\frac{1}{2} - \frac{1}{\mu}\right)\|u_n\|^2 - \frac{1}{\mu}M|\Omega|, \end{aligned}$$

Therefore $\{u_n\}$ is bounded. Because $H_0^1(\Omega)$ is reflexive, $\{u_n\}$ has a weakly convergent subsequence. It might as well be set in $H_0^1(\Omega)$ $u_n \rightharpoonup u$. Because

$$p := \frac{2^*(\theta - 1)}{2^* - 1} < 2^*,$$

Embed $H_0^1(\Omega) \to L^p(\Omega)$ is compact. Thus in $L^p(\Omega)$ in $u_n \to u$. According to theorem7, we know that in $L^{2^*/(2^*-1)}(\Omega)$ in $g(\cdot, u_n) \to g(\cdot, u)$. Therefore

$$\begin{aligned}\left|\int (g(x,u_n)-g(x,u))(u_n-u)\right| &\le |g(\cdot,u_n)-g(\cdot,u)|_{2^*/(2^*-1)} \cdot |u_n-u|_{2^*} \\ &\le C \cdot |g(\cdot,u_n)-g(\cdot,u)|_{2^*/(2^*-1)} \to 0.\end{aligned}$$

Notice again $\langle f_+'(u_n) - f_+'(u), u_n - u\rangle \to 0$, yes

$$\begin{aligned}\int |\nabla(u_n-u)|^2 &= \langle f_+'(u_n) - f_+'(u), u_n - u\rangle \\ &\quad + \int (g(x,u_n)-g(x,u))(u_n-u) \to 0.\end{aligned}$$

That is, $u_n \to u$ in $H_0^1(\Omega)$. □

Remark 15. A

(1) Considering the process of verifying the (PS) condition above, after deriving that the $(PS)_c$ sequence $\{u_n\}$ is bounded, the Sobolev compact embedding and the subcritical growth condition (g_1) immediately yield that $\{u_n\}$ has a convergent subsequence. To ensure that $\{u_n\}$ is bounded, some conditions need to be added to the nonlinear term $g(x,t)$ near ∞, such as (g_2) here.

(2) From (g_2), it can be seen by integration that there exists $C > 0$ such that

$$G(x,t) \ge C(|t|^\mu - 1), \qquad \forall (x,t) \in \Omega \times \mathbb{R}.$$

Furthermore, from (g_2), we have

$$\lambda = \lim_{|t|\to\infty} \frac{g(x,t)}{t} = +\infty. \tag{5.14}$$

When (5.14) holds, (5.9) is called superlinear. If $\lambda \in (0, +\infty)$, then (5.9) is called asymptotically linear. When $\lambda = 0$, (5.9) is called sublinear.

Example 16. Let $\lambda_\ell < \lambda_{\ell+1}$, $p \in C(\Omega \times \mathbb{R}, \mathbb{R})$ satisfy

(p_1) exists $A \in \mathbb{R}$, $|p(x,t)| \le A$.

(p_2^+) $\lim_{|t|\to\infty} P(x,t) = +\infty$, where $P(x,t) = \int_0^t p(x,s)\,\mathrm{d}s$.

Then the Dirichlet problem

$$-\Delta u = \lambda_\ell u + p(x,u), \qquad u|_{\partial\Omega} = 0 \tag{5.15}$$

There is a weak solution $u \in H_0^1(\Omega)$.

Proof. Let's find C^1- functional $f : H_0^1(\Omega) \to \mathbb{R}$,

$$f(u) = \frac{1}{2}\int |\nabla u|^2 - \frac{\lambda_\ell}{2}\int |u|^2 - \int P(x,u)$$

critical point. Note $X^- = \operatorname{span}\{\phi_1, \cdots, \phi_\ell\}$, $X^+ = (X^-)^\perp$. Then

$$\text{When} u \in X^-, \int |\nabla u|^2 \le \lambda_\ell \int |u|^2,$$

$$\text{When} u \in X^+, \int |\nabla u|^2 \ge \lambda_{\ell+1} \int |u|^2.$$

Let $u_n \in X^-$, $\|u_n\| \to \infty$. Let $v_n = \|u_n\|^{-1}u_n$, then $\|v_n\| = 1$. Because $\dim X^- < \infty$, the existence of $v \in X^- \setminus 0$ makes $\|v_n - v\| \to 0$. Set

$$\Theta = \{x \in \Omega |\, v(x) \neq 0\}$$

The measure of is positive, and for $x \in \Theta$ there is $|u_n(x)| \to \infty$. by (p_2^+) and Fatou Lemma knowledge

$$\int_\Theta P(x, u_n) \to +\infty.$$

From (p_2^+) we can also get $R > 0$, when $|Whent| \geq R$, $P(x,t) \geq 0$. Therefore

$$-P(x,t) \leq M := \sup_{(x,t)\in\Omega\times[-R,R]} |P(x,t)|.$$

then

$$\begin{aligned}\int P(x,u_n) &= \int_\Theta P(x,u_n) + \int_{\Omega\setminus\Theta} P(x,u_n)\\ &\geq \int_\Theta P(x,u_n) - M|\Omega\setminus\Theta| \to +\infty. \end{aligned} \tag{5.16}$$

Therefore

$$\begin{aligned} f(u_n) &= \frac{1}{2}\int |\nabla u_n|^2 - \frac{\lambda_\ell}{2}\int |u_n|^2 - \int P(x,u_n)\\ &\leq -\int P(x,u_n) \to -\infty. \end{aligned}$$

Right now

$$\lim_{u\in X^-, \|u\|\to\infty} f(u) = -\infty.$$

Let $\varepsilon \in (0, \lambda_{\ell+1} - \lambda_\ell)$. From

$$\lim_{|t|\to\infty} \frac{P(x,t)}{t^2} = \lim_{|t|\to\infty} \frac{p(x,t)}{t} = 0$$

we know that there exists $C > 0$ such that

$$P(x,t) \leq \frac{\varepsilon}{2}t^2 + C.$$

Thus, when $u \in X^+$,

$$\begin{aligned} f(u) &= \frac{1}{2}\int |\nabla u|^2 - \frac{\lambda_\ell}{2}\int |u|^2 - \int P(x,u)\\ &\geq \left(\frac{1}{2} - \frac{\lambda_\ell + \varepsilon}{2\lambda_{\ell+1}}\right)\int |\nabla u|^2 - C|\Omega| \geq -C|\Omega|. \end{aligned}$$

To apply the saddle point theorem, we still need to verify the (PS) condition. Let $\{u_n\} \subset H_0^1(\Omega)$ be such that $f(u_n) \to c$, $f'(u_n) \to 0$. We only need to show that $\{u_n\}$ is bounded in $H_0^1(\Omega)$.

If $\{u_n\}$ is unbounded, without loss of generality, assume $|u_n| \to \infty$. Let $u_n = u_n^- + u_n^+$, Among them $u_n^\pm \in X^\pm$. Then for large n,

$$\begin{aligned} \|u_n^+\| &\geq \langle f'(u_n), u_n^+\rangle\\ &= \int \nabla u_n \nabla u_n^+ - \lambda_\ell \int u_n u_n^+ - \int p(x,u_n)u_n^+ \end{aligned}$$

$$= \int |\nabla u_n^+|^2 - \lambda_\ell \int |u_n^+|^2 - \int p(x,u_n)u_n^+$$
$$\geq \left(1 - \frac{\lambda_\ell}{\lambda_{\ell+1}}\right) \int |\nabla u_n^+|^2 - A \int |u_n^+|$$
$$\geq \left(1 - \frac{\lambda_\ell}{\lambda_{\ell+1}}\right) \|u_n^+\| - C\|u_n^+\|.$$

Therefore $\{u_n^+\}$ is bounded in $H_0^1(\Omega)$, there exists $C > 0$ such that

$$\int |\nabla u_n|^2 - \lambda_\ell \int |u_n|^2 \leq \int |\nabla u_n^+|^2 - \lambda_\ell \int |u_n^+|^2 \leq C. \tag{5.17}$$

Remember $v_n = \|u_n\|^{-1}u_n$, then there is $v \in X^- \setminus 0$ This makes $\|v_n - v\| \to 0$. Similarly to (5.16), it can be proved that

$$\int P(x,u_n) \to +\infty.$$

Therefore, from (5.17), we get

$$f(u_n) = \frac{1}{2}\int |\nabla u_n|^2 - \frac{\lambda_\ell}{2}\int |u_n|^2 - \int P(x,u_n) \to -\infty,$$

which contradicts $f(u_n) \to c$. □

Similarly, it can be proved that when p satisfies (p_1) and

(p_2^-) $\lim_{|t|\to\infty} P(x,t) = -\infty$

When , (5.15) has a solution.

Exercise 16. Let $\lambda_\ell < \lambda_{\ell+1}$, $g \in C(\Omega \times \mathbb{R}, \mathbb{R})$,

$$\lim_{|t|\to\infty} \frac{g(x,t)}{t} = \lambda \in (\lambda_\ell, \lambda_{\ell+1}),$$

Then (5.9) has a solution.

Example 17. Let $g \in C(\Omega \times \mathbb{R}, \mathbb{R})$ satisfies (g_1), (g_2) and

(g_0') $\lim_{t\to 0} \frac{2G(x,t)}{t^2} = \lambda \in [\lambda_\ell, \lambda_{\ell+1})$,

(g_3) $G(x,t) \geq \frac{\lambda_\ell}{2}t^2$, $\forall (x,t) \in \Omega \times \mathbb{R}$.

Then (5.9) has a non-zero solution.

Proof. We only need to find the critical points of the C^1-functional $f : H_0^1(\Omega) \to \mathbb{R}$,

$$f(u) = \frac{1}{2}\int |\nabla u|^2 - \int G(x,u)$$

From the proof of Example 15, we know that f satisfies the (PS) condition.

Let $X^- = \text{span}\{\phi_1, \ldots, \phi_\ell\}$, $X^+ = (X^-)^\perp$. From (g_2), there exists $C_1 > 0$ such that

$$G(x,t) \geq C_1(|t|^\mu - 1).$$

Let $\varepsilon \in (\lambda, \lambda_{\ell+1})$. From (g_1) and (g_0'), there exists $C_2 > 0$ such that

$$G(x,t) \leq \frac{\varepsilon}{2}|t|^2 + C_2|t|^\theta.$$

When $u \in X^+$,

$$f(u) = \frac{1}{2}\int |\nabla u|^2 - \int G(x,u) \geq \frac{1}{2}\int |\nabla u|^2 - \frac{\varepsilon}{2}\int |u|^2 - C_2\int |u|^\theta$$

$$\geq \left(\frac{1}{2} - \frac{\varepsilon}{2\lambda_{\ell+1}}\right) \int |\nabla u|^2 - C_2 \int |u|^\theta$$

$$\geq \left(\frac{1}{2} - \frac{\varepsilon}{2\lambda_{\ell+1}}\right) \|u\|^2 - C\|u\|^\theta.$$

Since $\theta > 2$, it is desirable $\rho > 0$ to fully satisfy

$$\alpha = \inf_{S_\rho} f > 0,$$

Where $S_\rho = \{u \in X^+ | \, \|u\| = \rho\}$.

From (g_2), when $u \in X^-$

$$f(u) = \frac{1}{2}\int |\nabla u|^2 - \int G(x,u) \leq \frac{1}{2}\int |\nabla u|^2 - \frac{\lambda_\ell}{2}\int |u|^2 \leq 0.$$

For $u \in X^- \oplus \mathbb{R}\phi_{\ell+1}$,

$$f(u) = \frac{1}{2}\int |\nabla u|^2 - \int G(x,u) \leq \frac{1}{2}\int |\nabla u|^2 - C_1 \int |u|^\mu + C_1|\Omega|$$

$$= \frac{1}{2}\|u\|^2 - C_1|u|_\mu^\mu + C_1|\Omega|.$$

Since $\mu > 2$, and all norms are equivalent on the finite-dimensional space $X^- \oplus \mathbb{R}\phi_{\ell+1}$, for sufficiently large R, we have

$$\sup_{\partial Q_R} f \leq 0,$$

where $Q_R = \{u = u^- + t\phi_\ell | \, u^- \in X^-, t \geq 0, \|u\| \leq R\}$.

By the generalized mountain pass lemma, f has a critical value $c \geq \alpha > 0$, and thus a non-zero critical point. □

5.4. Multiple Critical Points of Even Functionals. Functionals with symmetry have a richer theory of critical points. We restrict ourselves to considering even functionals, which are symmetric under the action of the $\mathbb{Z}_2$ group.

Lemma 11. *Let U be a bounded symmetric open neighborhood of* 0 *in $\mathbb{R}^n$, $m < n$. Then any odd continuous mapping $\psi : \partial U \to \mathbb{R}^m$ has a zero point.*

Proof. Suppose otherwise, $0 \notin \psi(\partial U)$. Let $r = \operatorname{dist}(0, \psi(\partial U)) > 0$. Let $\tilde{\psi} : \overline{U} \to \mathbb{R}^m$ be the odd continuous extension of ψ. For $y \in B_r(0) \setminus \mathbb{R}^m$, by Borsuk's theorem,

$$\deg(\tilde{\psi}, U, y) = \deg(\tilde{\psi}, U, 0) \neq 0.$$

Therefore, $y \in \tilde{\psi}(U) \subset \mathbb{R}^m$, a contradiction. □

Lemma 12. *Let Y, Z be subspaces of a Banach space X, $X = Y + Z$, $Y \cap Z \neq \{0\}$,* $\dim Y < \infty$. *Let $\rho > r > 0$. Let*

$$B = \{u \in Y | \, \|u\| \leq \rho\}, \qquad N = \{u \in Z | \, \|u\| = r\}.$$

If $\gamma : B \to X$ Let $\gamma : B \to Y$ be an odd continuous map such that $\gamma|_{\partial B} = \mathbf{1}_{\partial B}$, then $\gamma(B) \cap N \neq \emptyset$.

Proof. Let $U = \{u \in B | \, \|\gamma(u)\| < r\}$. Since $\rho > r$ and $\gamma(0) = 0$, U is a bounded symmetric open neighborhood of 0 in Y. Let V be a proper subspace of Y such that $X = V \oplus Z$, and let $P : X \to V$ be the projection. By Lemma 11, the odd continuous map

$$\psi : \partial U \to V, \qquad u \longmapsto P\gamma(u)$$

has a zero $u^* \in \partial U$. It is easy to see that $\gamma(u^*) \in \gamma(B) \cap N$. □

Now let $\{X_j\}_{j=1}^{\infty}$ be a sequence of d-dimensional subspaces of the Banach space X, such that $X = \overline{\bigoplus_{j\geq 1} X_j}$. Let

$$Y_k = \bigoplus_{j=1}^{k} X_j, \qquad Z_k = \overline{\bigoplus_{j\geq k} X_j}. \tag{5.18}$$

Theorem 48 (Bartsch 1992)**.** *If $f \in C^1(X,\mathbb{R})$ satisfies the (PS) condition and is even. Also assume that for $k \geq k_0$, there exist $\rho_k > r_k > 0$ make*

(i) $b_k = \inf_{u\in Z_k, \|u\|=r_k} f(u) \to +\infty$.

(ii) $a_k = \sup_{u\in Y_k, \|u\|=\rho_k} f(u) \leq 0$.

Then f has a list of unbounded critical values ??$\{c_k\}$.

Remark 16. remember

$$B_k = \{u \in Y_k \,|\, \|u\| \leq \rho_k\}, \qquad N_k = \{u \in Z_k \,|\, \|u\| = r_k\},$$

but

$$c_k = \inf_{\gamma\in\mathcal{A}_k} \sup_{u\in B_k} f(\gamma(u)).$$

where $\mathcal{A}_k = \left\{\gamma : B_k \to X \,|\, \gamma \text{odd continuous}, \gamma|_{\partial B_k} = \mathbf{1}_{\partial B_k}\right\}$.

Proof. For every sufficiently large k, we have $b_k > a_k$. By Lemma 12, for all $\gamma \in \mathcal{A}_k$, we have $\gamma(B_k) \cap N_k \neq \emptyset$. Thus $c_k \geq b_k$, and therefore $c_k \to +\infty$.

If c_k is not a critical value of f, then there exists $\varepsilon > 0$ such that $f^{-1}[c_k-\varepsilon, c_k+\varepsilon]\cap K = \emptyset$. Since $c_k > a_k$, we can also require $c_k - \varepsilon > a_k$, i.e., $\partial B_k \subset f_{c_k-\varepsilon}$. By the deformation lemma, there exists a continuous map $\eta : f_{c_k+\varepsilon} \to f_{c_k+\varepsilon}$ such that

$$\eta(f_{c_k+\varepsilon}) \subset f_{c_k-\varepsilon}, \text{ and } \eta|_{f_{c_k-\varepsilon}} = \mathbf{1}_{f_{c_k-\varepsilon}}.$$

Choose $\gamma \in \mathcal{A}_k$ such that

$$\sup_{u\in B_k} f(\gamma(u)) < c_k + \varepsilon,$$

Let $\zeta = \eta \circ \gamma$. Then $\zeta : B_k \to X$ is continuous, and when $u \in \partial B_k \subset f_{c_k-\varepsilon}$, $\zeta(u) = u$. Thus $\zeta \in \mathcal{A}_k$. But

$$\sup_{u\in B_k} f(\zeta(u)) \leq c_k - \varepsilon,$$

which contradicts the definition of c_k. □

Example 18 (Ambrosetti-Rabinowitz 1973)**.** Let $g \in C(\Omega \times \mathbb{R}, \mathbb{R})$ satisfy $(g_1), (g_2)$ and $g(x,-u) = -g(x,u)$. Then (5.9) has a sequence of solutions $\{u_n\}$ such that $f(u_n) \to +\infty$.

Proof. We apply Theorem 48 to the C^1-functional $f : H_0^1(\Omega) \to \mathbb{R}$,

$$f(u) = \frac{1}{2}\int |\nabla u|^2 - \int G(x,u)$$

Let $X_j = \operatorname{span}\{\phi_j\}$, and construct Y_k, Z_k as in (5.18).

Let

$$\beta_k = \sup_{u\in Z_k, \|u\|=1} |u|_\theta.$$

For $u \in Z_k$, $\|u\| = 1$, we have

$$\int |\nabla u|^2 \geq \lambda_k \int |u|^2, \text{ that is, } |u|_2 \leq \frac{1}{\sqrt{\lambda_k}}\|u\| = \frac{1}{\sqrt{\lambda_k}}.$$

By the interpolation inequality, for some $\tau \in (0,1)$, we have

$$|u|_\theta \le |u|_2^\tau \cdot |u|_{2^*}^{1-\tau} \le C|u|_2^\tau \le C\left(\frac{1}{\sqrt{\lambda_k}}\right)^\tau.$$

Since $\lambda_k \to +\infty$, we have $\beta_k \to 0$.

By (g_1), there exists $c > 0$ such that

$$|G(x,t)| \le c(1+|t|^\theta).$$

Take $r_k = (c\theta\beta_k^\theta)^{1/(2-\theta)}$, then for $u \in Z_k$, $\|u\| = r_k$, yes

$$\begin{aligned} f(u) &= \frac{1}{2}\int |\nabla u|^2 - \int G(x,u) \ge \frac{1}{2}\|u\|^2 - c|u|_\theta^\theta - c|\Omega| \\ &\ge \frac{1}{2}\|u\|^2 - c\beta_k^\theta \|u\|^\theta - c|\Omega| \\ &= \left(\frac{1}{2} - \frac{1}{\theta}\right)(c\theta\beta_k^\theta)^{2/(2-\theta)} - c|\Omega|. \end{aligned}$$

Because $\beta_k \to 0$, $\theta > 2$, we get from the above formula

$$b_k = \inf_{u \in Z_k, \|u\| = r_k} f(u) \to +\infty.$$

From (g_2), there exists $C > 0$ so that

$$G(x,t) \ge C(|t|^\mu - 1).$$

For $\forall u \in Y_k$,

$$f(u) = \frac{1}{2}\int |\nabla u|^2 - \int G(x,u) \le \frac{1}{2}\|u\|^2 - C|u|_\mu^\mu + C|\Omega|.$$

Since all norms are equivalent on the finite-dimensional space Y_k, there exists $\rho_k > r_k$ such that

$$a_k = \sup_{u \in Y_k, \|u\| = \rho_k} f(u) \le 0.$$

Also, f satisfies the (PS) condition. By Theorem 48, f has a sequence of critical points $\{u_n\}$ such that $f(u_n) \to \infty$. □

Remark 17. The proof that $a_k \le 0$ above strongly depends on condition (g_2). However, many superlinear functions, such as

$$g(x,t) = t\log(1+|t|),$$

do not satisfy (g_2). Work on removing condition (g_2) can be found in [**?**].